Algebra 2

Contents

1. Order and primitive roots 3
2. The index 6
3. Binomial congruences and quadratic residues 7
4. The Legendre symbol and quadratic reciprocity 9
5. The Jacobi symbol 12
6. Polynomial congruences and Hensel's lemma 13
7. Arithmetic functions 15
8. p-adic numbers 18
9. The p-adic metric 20
10. Worked exercises 23
11. Geometry of $\mathbb{Q}_p$ 26
12. Sequences and series in $\mathbb{Q}_p$ 26
13. Topology on $\mathbb{Q}_p$ 27
14. Finite fields 28
15. Worked exercises on p-adic numbers 30
16. Trace and norm 35
17. Field extensions 36
    - 17.1. The Frobenius automorphism 36
    - 17.2. Separable extensions 36
    - 17.3. Embeddings and primitive elements 39
    - 17.4. Normal extensions 40
18. Galois theory 41
19. Worked exercises on fields 45
20. Cyclotomic fields 49
21. Ruler and compass constructions 51
22. Worked exercises on Galois groups 54
23. Solvable groups 57

24. Solvability by radicals 58
25. Review exercises 62

1. Order and primitive roots

Remark 1.1 (Recall). Let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct primes p_i and $\alpha_i \in \mathbb{N}$. Then $\varphi(n) = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right)$, and φ is multiplicative. If $m \in \mathbb{N}$, $a \in \mathbb{Z}$ and $(a, m) = 1$, then $a^{\varphi(m)} \equiv 1 \pmod{m}$ (Euler–Fermat).

Definition 1.2 (Order). Let $m \in \mathbb{N}$ and $a \in \mathbb{Z}$ with $(a, m) = 1$. The *order* of a modulo m, denoted $O_m(a)$, is the least $k \in \mathbb{N}$ with $a^k \equiv 1 \pmod{m}$. It exists by Remark 1.1.

Proposition 1.3. Let $m \in \mathbb{N}$, $a \in \mathbb{Z}$ with $(a, m) = 1$, and $k, s, t \in \mathbb{Z}_{\geq 0}$. Then:

1) $a^k \equiv 1 \pmod{m}$ if and only if $O_m(a) \mid k$. In particular $O_m(a) \mid \varphi(m)$.

2) $a^s \equiv a^t \pmod{m}$ if and only if $s \equiv t \pmod{O_m(a)}$.

Proof. Write $d = O_m(a)$.

1) If $d \mid k$, then $k = de$ with $e \in \mathbb{Z}_{\geq 0}$, and $a^k = \left(a^d\right)^e \equiv 1 \pmod{m}$. Conversely, let $a^k \equiv 1 \pmod{m}$ and write $k = qd + r$ with $q \in \mathbb{Z}_{\geq 0}$ and $0 \leq r < d$. Then $1 \equiv a^k = \left(a^d\right)^q a^r \equiv a^r \pmod{m}$. Since $r < d$ and d is the least positive exponent with this property, $r = 0$, so $d \mid k$. The particular case follows from Remark 1.1.

2) By symmetry assume $s \geq t$. Since $(a, m) = 1$, a^t is invertible modulo m. Hence $a^s \equiv a^t \pmod{m}$ holds if and only if $a^{s-t} \equiv 1 \pmod{m}$, which by 1) holds if and only if $d \mid s - t$, that is, $s \equiv t \pmod{d}$. ■

Example 1.4. Compute $O_{23}(5)$.

Solution. By Proposition 1.3, $O_{23}(5) \mid \varphi(23) = 22$, so $O_{23}(5) \in \{1, 2, 11, 22\}$. Here $5 \not\equiv 1$ and $5^2 = 25 \equiv 2 \not\equiv 1 \pmod{23}$. For 5^{11} use $11 = (1011)_2 = 8 + 2 + 1$ and repeated squaring: $5^2 \equiv 2$, $5^4 \equiv 4$, $5^8 \equiv 16 \pmod{23}$. Thus

$$5^{11} = 5^8 \cdot 5^2 \cdot 5 \equiv 16 \cdot 2 \cdot 5 = 160 \equiv -1 \pmod{23},$$

since $160 = 7 \cdot 23 - 1$. Hence $5^{11} \not\equiv 1$ and $O_{23}(5) = 22$. ◊

Definition 1.5 (Primitive root). Let $m \in \mathbb{N}$. An integer $a \in \mathbb{Z}$ with $(a, m) = 1$ is a *primitive root* modulo m if $O_m(a) = \varphi(m)$.

Lemma 1.6. Let $m \in \mathbb{N}$, $a \in \mathbb{Z}$ with $(a, m) = 1$, $d = O_m(a)$ and $j \in \mathbb{Z}_{\geq 0}$. Then $O_m(a^j) = d/(j, d)$.

Proof. Let $e = (j, d)$. For $k \in \mathbb{N}$, Proposition 1.3 gives $\left(a^j\right)^k \equiv 1 \pmod{m}$ if and only if $d \mid jk$, that is, $d/e \mid (j/e)k$. Since $(d/e, j/e) = 1$, this holds if and only if $d/e \mid k$. The least such k is d/e. ■

Remark 1.7 (Recall). Let p be a prime and $f \in \mathbb{Z}[x]$ a polynomial of degree $n \geq 1$ whose leading coefficient is not divisible by p. Then $f(x) \equiv 0 \pmod{p}$ has at most n pairwise incongruent solutions (Lagrange).

Theorem 1.8. Let $m \geq 2$. There exists a primitive root modulo m if and only if $m = 2$, $m = 4$, $m = p^\alpha$ or $m = 2p^\alpha$, where p is an odd prime and $\alpha \in \mathbb{N}$.

Proof. **The cases $m = 2, 4$.** $O_2(1) = 1 = \varphi(2)$. Since $3 \not\equiv 1$ and $3^2 = 9 \equiv 1 \pmod 4$, $O_4(3) = 2 = \varphi(4)$.

The case $m = p$, p an odd prime. For $d \mid p-1$ let $\psi(d)$ be the number of residues a modulo p with $p \nmid a$ and $O_p(a) = d$. By Proposition 1.3 every such order divides $p-1$, so $\sum_{d \mid p-1} \psi(d) = p-1$. Let $d \mid p-1$ with $\psi(d) \geq 1$, and let a have order d. By Proposition 1.3 the residues $1, a, \ldots, a^{d-1}$ are pairwise incongruent, and they are roots of $x^d - 1$. By Remark 1.7, $x^d - 1$ has no other roots modulo p, so every residue of order d is congruent to some a^j with $0 \leq j < d$. By Lemma 1.6, a^j has order d if and only if $(j, d) = 1$. Hence $\psi(d) = \varphi(d)$ whenever $\psi(d) \geq 1$, and so $\psi(d) \leq \varphi(d)$ for every $d \mid p-1$. Since $\sum_{d \mid p-1} \varphi(d) = p-1$ by Theorem 7.9, equality holds for every d. In particular $\psi(p-1) = \varphi(p-1) \geq 1$.

The case $m = p^\alpha$, p an odd prime, $\alpha \geq 2$. Let g be a primitive root modulo p. We may assume $g^{p-1} \not\equiv 1 \pmod{p^2}$. Indeed, $g + p$ is also a primitive root modulo p, and by the binomial theorem

$$(g+p)^{p-1} \equiv g^{p-1} + (p-1)pg^{p-2} \equiv g^{p-1} - pg^{p-2} \pmod{p^2}.$$

If $g^{p-1} \equiv 1 \pmod{p^2}$, then $(g+p)^{p-1} \equiv 1 - pg^{p-2} \not\equiv 1 \pmod{p^2}$, because $p \nmid g$; in that case replace g by $g + p$.

So $g^{p-1} = 1 + pu$ with $p \nmid u$. We claim that for every $k \in \mathbb{N}$ there is $u_k \in \mathbb{Z}$ with $p \nmid u_k$ and $g^{(p-1)p^{k-1}} = 1 + p^k u_k$. For $k = 1$ take $u_1 = u$. If the claim holds for k, then

$$g^{(p-1)p^k} = \left(1 + p^k u_k\right)^p = 1 + p^{k+1} u_k + \sum_{i=2}^{p} \binom{p}{i} p^{ik} u_k^i.$$

For $2 \leq i \leq p-1$ the prime p divides $\binom{p}{i}$, so the term is divisible by p^{2k+1}, and $2k + 1 \geq k + 2$. The term $i = p$ is divisible by p^{pk}, and $pk \geq 3k \geq k + 2$. Hence the sum equals $p^{k+2} w$ with $w \in \mathbb{Z}$, and $u_{k+1} = u_k + pw$ is not divisible by p.

Now let $d = O_{p^\alpha}(g)$. From $g^d \equiv 1 \pmod p$ and Proposition 1.3 modulo p we get $p - 1 \mid d$. Also $d \mid \varphi(p^\alpha) = (p-1)p^{\alpha-1}$. Hence $d = (p-1)p^i$ with $0 \leq i \leq \alpha - 1$. If $i \leq \alpha - 2$, then $d \mid (p-1)p^{\alpha-2}$ and so $g^{(p-1)p^{\alpha-2}} \equiv 1 \pmod{p^\alpha}$. By the claim with $k = \alpha - 1$, however, $g^{(p-1)p^{\alpha-2}} = 1 + p^{\alpha-1} u_{\alpha-1}$ with $p \nmid u_{\alpha-1}$, which is not congruent to 1 modulo p^α. Hence $d = \varphi(p^\alpha)$.

The case $m = 2p^\alpha$, p an odd prime, $\alpha \geq 1$. Let g be a primitive root modulo p^α. Replacing g by $g + p^\alpha$ if necessary, g is odd; it remains a primitive root modulo p^α, and $(g, 2p^\alpha) = 1$. If $g^d \equiv 1 \pmod{2p^\alpha}$, then $g^d \equiv 1 \pmod{p^\alpha}$, so $\varphi(p^\alpha) \mid d$. Since $\varphi(2p^\alpha) = \varphi(p^\alpha)$, Proposition 1.3 gives $O_{2p^\alpha}(g) = \varphi(2p^\alpha)$.

No primitive root otherwise. Let $m \geq 2$ be of none of the listed forms, and write $m = 2^\beta n$ with n odd. If $n = 1$, then $\beta \geq 3$. If n has two distinct prime divisors, write $n = p^\alpha n'$ with $p \nmid n'$ and $n' \geq 3$, and put $m_1 = 2^\beta p^\alpha$, $m_2 = n'$. If $n = p^\alpha$, then $\beta \geq 2$; put $m_1 = 2^\beta$, $m_2 = p^\alpha$. So either $m = 2^\beta$ with $\beta \geq 3$, or $m = m_1 m_2$ with $(m_1, m_2) = 1$ and $m_1, m_2 \geq 3$.

In the first case, $a^{2^{k-2}} \equiv 1 \pmod{2^k}$ for every odd a and every $k \geq 3$. For $k = 3$, $a^2 - 1 = (a-1)(a+1)$ is a product of two consecutive even numbers, hence divisible by 8. If $a^{2^{k-2}} = 1 + 2^k t$, then $a^{2^{k-1}} = 1 + 2^{k+1}t + 2^{2k}t^2 \equiv 1 \pmod{2^{k+1}}$. Hence $O_{2^\beta}(a) \leq 2^{\beta-2} < 2^{\beta-1} = \varphi(2^\beta)$.

In the second case, note that $\varphi(n)$ is even for $n \geq 3$: if an odd prime p divides n, then $p - 1 \mid \varphi(n)$ by Remark 1.1; otherwise $n = 2^\gamma$ with $\gamma \geq 2$ and $\varphi(n) = 2^{\gamma-1}$. Let L be the least common multiple of $\varphi(m_1)$ and $\varphi(m_2)$. Both are even, so $L \leq \varphi(m_1)\varphi(m_2)/2 = \varphi(m)/2$. For $(a, m) = 1$, Euler–Fermat gives $a^L \equiv 1$ modulo m_1 and modulo m_2, hence modulo $m = m_1 m_2$. So $O_m(a) \leq L < \varphi(m)$. ■

Remark 1.9. There is no primitive root modulo 12. Indeed, $\varphi(12) = 4$, and each of $1, 5, 7, 11$ has square congruent to 1 modulo 12, so every order is at most 2.

Theorem 1.10. Let $m \geq 2$ and suppose that g is a primitive root modulo m. Then the primitive roots modulo m are, up to congruence, exactly the powers g^j with $1 \leq j \leq \varphi(m)$ and $(j, \varphi(m)) = 1$. Hence there are $\varphi(\varphi(m))$ of them. In particular, for a prime p there are $\varphi(\varphi(p)) = \varphi(p-1)$ primitive roots modulo p.

Proof. A primitive root a satisfies $(a, m) = 1$. By Theorem 1.17 and $g^{\varphi(m)} \equiv 1 = g^0 \pmod{m}$, a is congruent to exactly one g^j with $1 \leq j \leq \varphi(m)$. By Lemma 1.6, $O_m(g^j) = \varphi(m)/(j, \varphi(m))$, which equals $\varphi(m)$ if and only if $(j, \varphi(m)) = 1$. The number of such j is $\varphi(\varphi(m))$. ■

Theorem 1.11 (cited)**.** There is a constant $c > 0$ such that $\varphi(n) > cn/\log\log n$ for every $n \geq 3$.

Procedure 1.12 (Checking primitive roots). Let p be a prime and $g \in \mathbb{Z}$ with $p \nmid g$.

1) Compute the prime divisors $q_1, \ldots, q_s$ of $p - 1$.
2) Check that $g^{(p-1)/q_i} \not\equiv 1 \pmod{p}$ for every $1 \leq i \leq s$.

Then g is a primitive root modulo p if and only if every check in step 2 succeeds.

Proof. If g is a primitive root, then $O_p(g) = p - 1$ does not divide $(p-1)/q_i < p - 1$, so $g^{(p-1)/q_i} \not\equiv 1$ by Proposition 1.3. Conversely, suppose $d = O_p(g) < p - 1$. Since $d \mid p - 1$, the quotient $(p-1)/d > 1$ has a prime divisor q_i, and then $d \mid (p-1)/q_i$, so $g^{(p-1)/q_i} \equiv 1 \pmod{p}$. ■

Theorem 1.13 (Burgess) (cited)**.** For a prime p let $n(p)$ be the least primitive root modulo p in $\mathbb{N}$. For every $\varepsilon > 0$ there is $c(\varepsilon) > 0$ such that $n(p) < c(\varepsilon) p^{1/4+\varepsilon}$ for all primes p.

Theorem 1.14 (Shoup) (cited)**.** Assume the generalized Riemann hypothesis for Dirichlet L-functions. Then there is an absolute constant $c > 0$ such that for every prime $p \geq 3$

$$n(p) \leq c\omega(p-1)^4(\log\omega(p-1) + 1)^4(\log p)^2,$$

where $\omega(k)$ is the number of distinct prime divisors of k. In particular $n(p) = O((\log p)^6)$.

Theorem 1.15 (Prime number theorem) (cited)**.** Let $\pi(x)$ be the number of primes not exceeding x. There is $c > 0$ such that

$$\pi(x) = \int_2^x \frac{dt}{\log t} + O\left(xe^{-c\sqrt{\log x}}\right) \quad (x \to \infty).$$

If the Riemann hypothesis holds, the error term is $O(\sqrt{x}\log x)$.

Definition 1.16 (Reduced residue system). Let $m \in \mathbb{N}$. Integers $a_1, ..., a_{\varphi(m)}$ form a *reduced residue system* modulo m if $(a_i, m) = 1$ for every i and $a_i \not\equiv a_j \pmod m$ for all $i \neq j$. Such a system contains exactly one element of each residue class coprime to m, since there are $\varphi(m)$ such classes.

Theorem 1.17. Let $m \in \mathbb{N}$ and $g \in \mathbb{Z}$ with $(g, m) = 1$. Then g is a primitive root modulo m if and only if $1, g, g^2, ..., g^{\varphi(m)-1}$ form a reduced residue system modulo m.

Proof. Every power of g is coprime to m. Suppose g is a primitive root and $g^i \equiv g^j \pmod m$ with $0 \leq j \leq i \leq \varphi(m) - 1$. By Proposition 1.3, $i \equiv j \pmod{\varphi(m)}$, and $0 \leq i - j < \varphi(m)$ forces $i = j$. So the $\varphi(m)$ powers are pairwise incongruent.

Conversely, if the powers form a reduced residue system, then $g^j \not\equiv g^0 = 1 \pmod m$ for $1 \leq j \leq \varphi(m) - 1$, because they are pairwise incongruent. Hence $O_m(g) \geq \varphi(m)$, and $O_m(g) \mid \varphi(m)$ by Proposition 1.3, so $O_m(g) = \varphi(m)$. ∎

2. The index

Definition 2.1 (Index, discrete logarithm). Let p be a prime, g a primitive root modulo p, and $a \in \mathbb{Z}$ with $p \nmid a$. The *index* or *discrete logarithm* of a to the base g modulo p is the unique k with $1 \leq k \leq p - 1$ and $g^k \equiv a \pmod p$. It is denoted $\operatorname{ind}_g a$. Existence and uniqueness follow from Theorem 1.17, since $g^{p-1} \equiv 1 = g^0 \pmod p$.

Proposition 2.2. Let p be a prime and g a primitive root modulo p. For $i, j \in \mathbb{Z}_{\geq 0}$, $g^i \equiv g^j \pmod p$ if and only if $i \equiv j \pmod{p-1}$. For $a, b \in \mathbb{Z}$ with $p \nmid ab$, $a \equiv b \pmod p$ if and only if $\operatorname{ind}_g a \equiv \operatorname{ind}_g b \pmod{p-1}$.

Proof. The first statement is Proposition 1.3 2) with $O_p(g) = \varphi(p) = p - 1$. For the second, $a \equiv b \pmod p$ holds if and only if $g^{\operatorname{ind}_g a} \equiv g^{\operatorname{ind}_g b} \pmod p$, and the first statement applies. ∎

Theorem 2.3. Let p be a prime, g and h primitive roots modulo p, $a, b \in \mathbb{Z}$ with $p \nmid ab$, and $n \in \mathbb{Z}_{\geq 0}$. Then, modulo $p - 1$:

1) $\operatorname{ind}_g(ab) \equiv \operatorname{ind}_g a + \operatorname{ind}_g b$,

2) $\operatorname{ind}_g(a^n) \equiv n \operatorname{ind}_g a$,

3) $\operatorname{ind}_h a \equiv \operatorname{ind}_h g \cdot \operatorname{ind}_g a$.

Proof. 1) $g^{\operatorname{ind}_g(ab)} \equiv ab$ and $g^{\operatorname{ind}_g a + \operatorname{ind}_g b} = g^{\operatorname{ind}_g a} g^{\operatorname{ind}_g b} \equiv ab \pmod p$; apply Proposition 2.2.

2) $g^{\operatorname{ind}_g(a^n)} \equiv a^n$ and $g^{n \operatorname{ind}_g a} = \left(g^{\operatorname{ind}_g a}\right)^n \equiv a^n \pmod p$; apply Proposition 2.2.

3) $h^{\operatorname{ind}_h a} \equiv a \equiv g^{\operatorname{ind}_g a} \equiv \left(h^{\operatorname{ind}_h g}\right)^{\operatorname{ind}_g a} = h^{\operatorname{ind}_h g \cdot \operatorname{ind}_g a} \pmod p$; apply Proposition 2.2 with base h. ■

Remark 2.4. Indices are computed with an index table: list g^n modulo p for $n = 1, \ldots, p-1$ and read the table in reverse.

Example 2.5. Find a primitive root g modulo 17 and compute $\operatorname{ind}_g 11$.

Solution. Here $\varphi(17) = 16$, whose only prime divisor is 2, so by Procedure 1.12 an integer g with $17 \nmid g$ is a primitive root if and only if $g^8 \not\equiv 1 \pmod{17}$. Since $2^8 = 256 = 15 \cdot 17 + 1 \equiv 1$, the integer 2 is not a primitive root. Since $3^4 = 81 \equiv -4$ and $3^8 \equiv 16 \equiv -1 \pmod{17}$, the integer 3 is a primitive root. The powers of 3 modulo 17 are:

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
3^n	3	9	10	13	5	15	11	16	14	8	7	4	12	2	6	1

Reading the table in reverse, $\operatorname{ind}_3 11 = 7$. ◊

Remark 2.6 (Discrete logarithm problem). Given a prime p, a primitive root g modulo p and an integer k with $1 \le k \le p-1$, find x with $1 \le x \le p-1$ and $g^x \equiv k \pmod p$. No classical algorithm solving this problem in time polynomial in $\log p$ is known.

Remark 2.7. The index generalizes to any finite cyclic group $G = \langle g \rangle$ of order N: for $b \in G$, $\operatorname{ind}_g b$ is the unique k with $1 \le k \le N$ and $g^k = b$. The discrete logarithm problem is posed in the same way.

3. Binomial congruences and quadratic residues

Definition 3.1 (Quadratic residue). Let $p > 2$ be a prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. Then a is a *quadratic residue* modulo p if $x^2 \equiv a \pmod p$ has a solution, and a *quadratic non-residue* otherwise.

Let p be a prime, $a \in \mathbb{Z}$ with $(a, p) = 1$, and $k \in \mathbb{N}$. We consider the congruence $x^k \equiv a \pmod p$.

Remark 3.2 (Recall). Let $n \in \mathbb{N}$ and $a, b \in \mathbb{Z}$. The congruence $ay \equiv b \pmod n$ is solvable if and only if $(a, n) \mid b$, and then it has exactly (a, n) pairwise incongruent solutions modulo n.

Theorem 3.3. Let p be a prime, $a \in \mathbb{Z}$ with $(a, p) = 1$, and $k \in \mathbb{N}$. The congruence $x^k \equiv a \pmod p$ is solvable if and only if

$$a^{(p-1)/(k, p-1)} \equiv 1 \pmod p.$$

In that case the number of pairwise incongruent solutions is $(k, p-1)$. If g is a primitive root modulo p, the condition is equivalent to $(k, p-1) \mid \operatorname{ind}_g a$.

Proof. Let g be a primitive root modulo p (Theorem 1.8) and $e = (k, p-1)$. Every solution x satisfies $p \nmid x$, since $p \nmid a$. By the definition of the index and Proposition 2.2, $x \mapsto \operatorname{ind}_g x$ is a bijection from the residue classes modulo p not divisible by p onto the residue classes modulo $p-1$. We seek solutions in the form $x \equiv g^{\operatorname{ind}_g x} \pmod p$. Then

$$x^k \equiv \left(g^{\operatorname{ind}_g x}\right)^k = g^{k \operatorname{ind}_g x} \quad \text{and} \quad a \equiv g^{\operatorname{ind}_g a} \pmod p,$$

so by Proposition 2.2, $x^k \equiv a \pmod p$ if and only if $k \operatorname{ind}_g x \equiv \operatorname{ind}_g a \pmod{p-1}$. Put $y = \operatorname{ind}_g x$. The solutions x modulo p correspond bijectively to the solutions y modulo $p-1$ of the linear congruence

$$ky \equiv \operatorname{ind}_g a \pmod{p-1}.$$

By Remark 3.2 it is solvable if and only if $e \mid \operatorname{ind}_g a$, and then it has exactly e solutions.

It remains to show that $e \mid \operatorname{ind}_g a$ if and only if $a^{(p-1)/e} \equiv 1 \pmod p$. We have

$$a^{(p-1)/e} \equiv \left(g^{\operatorname{ind}_g a}\right)^{(p-1)/e} = g^{\operatorname{ind}_g a \cdot (p-1)/e} \pmod p.$$

By Proposition 1.3 with $O_p(g) = p-1$, this is congruent to 1 if and only if $p-1 \mid \operatorname{ind}_g a \cdot (p-1)/e$, that is, if and only if $e \mid \operatorname{ind}_g a$. ■

Example 3.4. Solve $5x^{22} \equiv 6 \pmod{13}$.

Solution. Let $y = x^{22}$. Then $5y \equiv 6 \pmod{13}$ with $(5, 13) = 1$. Multiplying by 5 gives $25y \equiv 30 \pmod{13}$, and $25 \equiv -1$, $30 \equiv 4$, so $-y \equiv 4$ and $y \equiv -4 \equiv 9 \pmod{13}$. It remains to solve $x^{22} \equiv 9 \pmod{13}$.

Here $\varphi(13) = 12 = 2^2 \cdot 3$. Since $2^4 = 16 \equiv 3$ and $2^6 = 64 \equiv -1 \pmod{13}$, Procedure 1.12 shows that 2 is a primitive root modulo 13, that is, $O_{13}(2) = 12$. Taking indices to the base 2 and using Theorem 2.3,

$$22 \operatorname{ind}_2 x \equiv \operatorname{ind}_2 9 \pmod{12}.$$

The powers of 2 modulo 13 are:

n	1	2	3	4	5	6	7	8	9	10	11	12
2^n	2	4	8	3	6	12	11	9	5	10	7	1

Hence $\operatorname{ind}_2 9 = 8$. Let $z = \operatorname{ind}_2 x$. Then $22z \equiv 8 \pmod{12}$, and $(22, 12) = 2 \mid 8$, so there are two solutions modulo 12. Dividing by 2 gives $11z \equiv 4 \pmod 6$, that is, $-z \equiv 4$, so $z \equiv -4 \equiv 2 \pmod 6$. Thus $z \equiv 2$ or $z \equiv 8 \pmod{12}$. From $\operatorname{ind}_2 x \equiv 2$ we get $x \equiv 2^2 = 4$, and from $\operatorname{ind}_2 x \equiv 8$ we get $x \equiv 2^8 \equiv 9 \pmod{13}$. The solutions are $x \equiv 4$ and $x \equiv 9 \pmod{13}$. ◊

Corollary 3.5. Let p be an odd prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. Then a is a quadratic residue modulo p if and only if $a^{(p-1)/2} \equiv 1 \pmod p$, and in that case $x^2 \equiv a \pmod p$ has exactly 2

pairwise incongruent solutions. Among the $p-1$ residue classes coprime to p there are $(p-1)/2$ quadratic residues and $(p-1)/2$ quadratic non-residues.

Proof. Since p is odd, $(2, p-1) = 2$, and the first two statements are Theorem 3.3 with $k = 2$. The map $x \mapsto x^2$ sends the $p-1$ residue classes coprime to p onto the quadratic residues, and each quadratic residue has exactly 2 preimages. Hence there are $(p-1)/2$ quadratic residues, and the remaining $(p-1)/2$ classes are non-residues. ∎

Proposition 3.6. Let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$. Then $a^{(p-1)/2} \equiv \pm 1 \pmod{p}$, and a is a quadratic residue modulo p if and only if $a^{(p-1)/2} \equiv 1 \pmod{p}$.

Proof. Let $b = a^{(p-1)/2}$. By Fermat's theorem $b^2 = a^{p-1} \equiv 1 \pmod{p}$, so $p \mid (b-1)(b+1)$. Since p is prime, $b \equiv 1$ or $b \equiv -1 \pmod{p}$. The criterion is Corollary 3.5. ∎

4. The Legendre symbol and quadratic reciprocity

Definition 4.1 (Legendre symbol). Let $p > 2$ be a prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. The *Legendre symbol* is

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a quadratic residue modulo } p \\ -1 & \text{if } a \text{ is a quadratic non-residue modulo } p. \end{cases}$$

Corollary 4.2 (Euler's criterion). Let $p > 2$ be a prime and $a \in \mathbb{Z}$ with $(a, p) = 1$. Then $\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}$.

Proof. By Proposition 3.6, $a^{(p-1)/2} \equiv 1 = \left(\frac{a}{p}\right)$ if a is a quadratic residue, and otherwise $a^{(p-1)/2} \equiv -1 = \left(\frac{a}{p}\right)$. ∎

Proposition 4.3. Let $p > 2$ be a prime and $a, b \in \mathbb{Z}$ with $(a, p) = (b, p) = 1$. Then:

1) If $a \equiv b \pmod{p}$, then $\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$.
2) $\left(\frac{a}{p}\right)\left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$.
3) $\left(\frac{-1}{p}\right) = 1$ if $p \equiv 1 \pmod{4}$, and $\left(\frac{-1}{p}\right) = -1$ if $p \equiv -1 \pmod{4}$.
4) $\left(\frac{2}{p}\right) = 1$ if $p \equiv \pm 1 \pmod{8}$, and $\left(\frac{2}{p}\right) = -1$ if $p \equiv \pm 3 \pmod{8}$.

Proof. 1) If $x^2 \equiv a \pmod{p}$, then $x^2 \equiv b \pmod{p}$, and conversely.

2) By Corollary 4.2, $\left(\frac{ab}{p}\right) \equiv (ab)^{(p-1)/2} = a^{(p-1)/2} b^{(p-1)/2} \equiv \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod{p}$. Both sides lie in $\{1, -1\}$ and $p > 2$, so the congruence is an equality.

3) By Corollary 4.2, $\left(\frac{-1}{p}\right) \equiv (-1)^{(p-1)/2} \pmod{p}$, and as in 2) this is an equality. The exponent $(p-1)/2$ is even if and only if $p \equiv 1 \pmod{4}$.

4) is proved after Lemma 4.4. ∎

Lemma 4.4 (Gauss). Let p be an odd prime, $a \in \mathbb{Z}$ with $p \nmid a$, and $S = \{1, 2, \ldots, (p-1)/2\}$. For $j \in S$ let t_j be the least positive residue of ja modulo p, and let μ be the number of $j \in S$ with $t_j > p/2$. Then $\left(\frac{a}{p}\right) = (-1)^\mu$.

Proof. For $j \in S$ put $s_j = t_j$ if $t_j < p/2$ and $s_j = p - t_j$ otherwise. Then $s_j \in S$ and $s_j \equiv \pm ja \pmod p$, with the minus sign for exactly μ values of j. If $s_i = s_j$, then $ia \equiv \pm ja$, so $i \equiv \pm j \pmod p$ because $p \nmid a$. Since $2 \le i + j \le p - 1$, the minus sign is impossible, so $i = j$. Hence $j \mapsto s_j$ is a permutation of S, and

$$\prod_{j \in S} ja \equiv (-1)^\mu \prod_{j \in S} s_j = (-1)^\mu \left(\frac{p-1}{2}\right)! \pmod p.$$

The left side is $a^{(p-1)/2}((p-1)/2)!$. Cancelling the unit $((p-1)/2)!$ gives $a^{(p-1)/2} \equiv (-1)^\mu$, and Corollary 4.2 gives $\left(\frac{a}{p}\right) \equiv (-1)^\mu \pmod p$. Both sides lie in $\{1, -1\}$ and $p > 2$, so they are equal. ■

Proof of Proposition 4.3 4). Take $a = 2$ in Lemma 4.4. For $j \in S$ we have $2j \le p - 1$, so $t_j = 2j$, and μ is the number of $j \in S$ with $j > p/4$, namely $\mu = (p-1)/2 - \lfloor p/4 \rfloor$. Writing $p = 8l + r$ with $r \in \{1, 3, 5, 7\}$:

$$r = 1 : \mu = 4l - 2l = 2l, \quad r = 3 : \mu = (4l + 1) - 2l = 2l + 1,$$
$$r = 5 : \mu = (4l + 2) - (2l + 1) = 2l + 1, \quad r = 7 : \mu = (4l + 3) - (2l + 1) = 2l + 2.$$

So μ is even for $p \equiv \pm 1 \pmod 8$ and odd for $p \equiv \pm 3 \pmod 8$. ■

Lemma 4.5. Let p be an odd prime, a an odd integer with $p \nmid a$, and μ as in Lemma 4.4. Then

$$\mu \equiv \sum_{j=1}^{(p-1)/2} \left\lfloor \frac{ja}{p} \right\rfloor \pmod 2.$$

Proof. With the notation of Lemma 4.4, $ja = p\lfloor ja/p \rfloor + t_j$. Let $u_1, \ldots, u_\lambda$ be the t_j below $p/2$ and $v_1, \ldots, v_\mu$ those above. By the proof of Lemma 4.4, the numbers $u_1, \ldots, u_\lambda, p - v_1, \ldots, p - v_\mu$ are $1, \ldots, (p-1)/2$ in some order. Summing over $j \in S$,

$$a \sum_{j \in S} j = p \sum_{j \in S} \left\lfloor \frac{ja}{p} \right\rfloor + \sum_i u_i + \sum_i v_i, \quad \sum_{j \in S} j = \sum_i u_i + \mu p - \sum_i v_i.$$

Subtracting, $(a-1)\sum_{j \in S} j = p\left(\sum_{j \in S} \lfloor ja/p \rfloor - \mu\right) + 2\sum_i v_i$. The left side is even because a is odd, and p is odd, so $\sum_{j \in S} \lfloor ja/p \rfloor - \mu$ is even. ■

Theorem 4.6 (Quadratic reciprocity law). Let p and q be distinct odd primes. Then

$$\left(\frac{p}{q}\right) = \begin{cases} -\left(\frac{q}{p}\right) & \text{if } p \equiv q \equiv -1 \pmod 4 \\ \left(\frac{q}{p}\right) & \text{otherwise.} \end{cases}$$

Equivalently, $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\cdot\frac{q-1}{2}}$.

Proof. The two forms are equivalent: both symbols are ± 1, and $\frac{p-1}{2} \cdot \frac{q-1}{2}$ is odd if and only if both factors are odd, that is, if and only if $p \equiv q \equiv -1 \pmod 4$.

By Lemma 4.4 and Lemma 4.5, $\left(\frac{q}{p}\right) = (-1)^A$ and $\left(\frac{p}{q}\right) = (-1)^B$ with

$$A = \sum_{j=1}^{(p-1)/2} \left\lfloor \frac{jq}{p} \right\rfloor, \quad B = \sum_{k=1}^{(q-1)/2} \left\lfloor \frac{kp}{q} \right\rfloor.$$

Consider the lattice points (j,k) with $1 \le j \le (p-1)/2$ and $1 \le k \le (q-1)/2$; there are $\frac{p-1}{2} \cdot \frac{q-1}{2}$ of them. None satisfies $qj = pk$, since $p \nmid qj$. For fixed j, the points with $k < qj/p$ are those with $1 \le k \le \lfloor qj/p \rfloor$, and $\lfloor qj/p \rfloor \le (q-1)/2$ because $qj/p < q/2$. So A counts the points with $qj > pk$, and symmetrically B counts those with $qj < pk$. Hence $A + B = \frac{p-1}{2} \cdot \frac{q-1}{2}$. ■

Example 4.7. Compute $\left(\frac{12}{31}\right)$.

Solution.

$$\left(\frac{12}{31}\right) = \left(\frac{2^2 \cdot 3}{31}\right) = \left(\frac{2}{31}\right)^2 \left(\frac{3}{31}\right) = \left(\frac{3}{31}\right) = -\left(\frac{31}{3}\right) = -\left(\frac{1}{3}\right) = -1.$$

The fourth equality is Theorem 4.6 with $3 \equiv 31 \equiv -1 \pmod 4$, and the fifth uses $31 \equiv 1 \pmod 3$. Hence $x^2 \equiv 12 \pmod{31}$ has no solution. ◊

Example 4.8. Compute $\left(\frac{1783}{7523}\right)$, where 1783 and 7523 are primes.

Solution.

$$\left(\frac{1783}{7523}\right) = -\left(\frac{7523}{1783}\right) = -\left(\frac{391}{1783}\right) = \left(\frac{1783}{391}\right) = \left(\frac{219}{391}\right) = \left(\frac{3}{391}\right)\left(\frac{73}{391}\right).$$

The first equality is Theorem 4.6 with $1783 \equiv 7523 \equiv -1 \pmod 4$, and the second uses $7523 = 4 \cdot 1783 + 391$. From here on $391 = 17 \cdot 23$ appears as a denominator, so the symbols with denominator 391 are Jacobi symbols. The third equality is Theorem 5.4 with $391 \equiv 1783 \equiv -1 \pmod 4$, the fourth is Proposition 5.3 1) with $1783 = 4 \cdot 391 + 219$, and the fifth is Proposition 5.3 2) with $219 = 3 \cdot 73$.

$$\left(\frac{3}{391}\right)\left(\frac{73}{391}\right) = -\left(\frac{391}{3}\right)\left(\frac{391}{73}\right) = -\left(\frac{1}{3}\right)\left(\frac{26}{73}\right) = -\left(\frac{2}{73}\right)\left(\frac{13}{73}\right) = -\left(\frac{13}{73}\right)$$
$$= -\left(\frac{73}{13}\right) = -\left(\frac{8}{13}\right) = -\left(\frac{2}{13}\right)^3 = -(-1)^3 = 1.$$

The first equality is Theorem 5.4 with $3 \equiv 391 \equiv -1 \pmod 4$ and $73 \equiv 1 \pmod 4$. The remaining symbols have prime denominators. We used $391 = 130 \cdot 3 + 1$, $391 = 5 \cdot 73 + 26$, $\left(\frac{2}{73}\right) = 1$ since $73 \equiv 1 \pmod 8$, Theorem 4.6 with $13 \equiv 1 \pmod 4$, $73 = 5 \cdot 13 + 8$, and $\left(\frac{2}{13}\right) = -1$ since $13 \equiv -3 \pmod 8$. ◊

5. The Jacobi symbol

Definition 5.1 (Jacobi symbol). Let $n \in \mathbb{N}$ be odd with $n > 1$, let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct primes p_i, and let $a \in \mathbb{Z}$ with $(a, n) = 1$. The *Jacobi symbol* is

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right)^{\alpha_1} \cdots \left(\frac{a}{p_r}\right)^{\alpha_r},$$

where the factors are Legendre symbols. For n prime it coincides with the Legendre symbol, which justifies the notation.

Lemma 5.2. For odd integers u and v,

$$\frac{uv-1}{2} \equiv \frac{u-1}{2} + \frac{v-1}{2} \pmod 2, \qquad \frac{u^2v^2-1}{8} \equiv \frac{u^2-1}{8} + \frac{v^2-1}{8} \pmod 2.$$

Proof. The differences are $(u-1)(v-1)/2$ and $(u^2-1)(v^2-1)/8$. In the first, $4 \mid (u-1)(v-1)$. In the second, 8 divides each of u^2-1 and v^2-1, since each is a product of two consecutive even numbers, so $64 \mid (u^2-1)(v^2-1)$. ∎

Proposition 5.3. Let $m, n \in \mathbb{N}$ be odd with $m, n > 1$, and let $a, b \in \mathbb{Z}$.

1) If $(a, n) = 1$ and $a \equiv b \pmod n$, then $\left(\frac{a}{n}\right) = \left(\frac{b}{n}\right)$.
2) If $(ab, n) = 1$, then $\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right)$.
3) $\left(\frac{-1}{n}\right) = 1$ if $n \equiv 1 \pmod 4$, and $\left(\frac{-1}{n}\right) = -1$ if $n \equiv -1 \pmod 4$.
4) $\left(\frac{2}{n}\right) = 1$ if $n \equiv \pm 1 \pmod 8$, and $\left(\frac{2}{n}\right) = -1$ if $n \equiv \pm 3 \pmod 8$.
5) If $(a, mn) = 1$, then $\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right)\left(\frac{a}{n}\right)$.

Proof. 1) and 2) follow factor by factor from Proposition 4.3 1) and 2), since $a \equiv b \pmod n$ implies $a \equiv b \pmod{p_i}$. 5) holds because the prime factorization of mn is obtained by adding the exponents in those of m and n.

For 3) and 4), Proposition 4.3 gives $\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2}$ and $\left(\frac{2}{p}\right) = (-1)^{(p^2-1)/8}$ for odd primes p; for the second, $p \equiv \pm 1 \pmod 8$ gives $p^2 \equiv 1 \pmod{16}$ and $p \equiv \pm 3 \pmod 8$ gives $p^2 \equiv 9 \pmod{16}$. By Lemma 5.2, $n \mapsto (-1)^{(n-1)/2}$ and $n \mapsto (-1)^{(n^2-1)/8}$ are multiplicative on odd integers. Hence $\left(\frac{-1}{n}\right) = (-1)^{(n-1)/2}$ and $\left(\frac{2}{n}\right) = (-1)^{(n^2-1)/8}$, and the same computations modulo 4 and 16 give the stated cases. ∎

Theorem 5.4 (Reciprocity for the Jacobi symbol). Let $m, n \in \mathbb{N}$ be odd with $m, n > 1$ and $(m, n) = 1$. Then

$$\left(\frac{m}{n}\right) = \begin{cases} -\left(\frac{n}{m}\right) & \text{if } n \equiv m \equiv -1 \pmod 4 \\ \left(\frac{n}{m}\right) & \text{otherwise.} \end{cases}$$

Proof. As in Theorem 4.6, the claim is equivalent to $\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = (-1)^{\frac{m-1}{2}\cdot\frac{n-1}{2}}$. Write $m = p_1 \cdots p_s$ and $n = q_1 \cdots q_t$ as products of not necessarily distinct primes; $p_i \neq q_j$ since $(m,n) = 1$. By Proposition 5.3 2) and 5),

$$\left(\frac{m}{n}\right)\left(\frac{n}{m}\right) = \prod_{i,j}\left(\frac{p_i}{q_j}\right)\left(\frac{q_j}{p_i}\right) = \prod_{i,j}(-1)^{P_i Q_j} = (-1)^{\left(\sum_i P_i\right)\left(\sum_j Q_j\right)},$$

where $P_i = (p_i - 1)/2$, $Q_j = (q_j - 1)/2$, and the middle equality is Theorem 4.6. By Lemma 5.2, $\sum_i P_i \equiv (m-1)/2$ and $\sum_j Q_j \equiv (n-1)/2 \pmod 2$. ∎

Example 5.5. Compute $\left(\frac{31}{95}\right)$.

Solution. $\left(\frac{31}{95}\right) = -\left(\frac{95}{31}\right) = -\left(\frac{2}{31}\right) = -1$, by Theorem 5.4 with $31 \equiv 95 \equiv -1 \pmod 4$, then $95 = 3 \cdot 31 + 2$, and $\left(\frac{2}{31}\right) = 1$ since $31 \equiv -1 \pmod 8$. ◊

Example 5.6. Compute $\left(\frac{60}{101}\right)$.

Solution.

$$\left(\frac{60}{101}\right) = \left(\frac{2^2 \cdot 15}{101}\right) = \left(\frac{2}{101}\right)^2\left(\frac{15}{101}\right) = \left(\frac{15}{101}\right) = \left(\frac{101}{15}\right)$$
$$= \left(\frac{11}{15}\right) = -\left(\frac{15}{11}\right) = -\left(\frac{4}{11}\right) = -\left(\frac{2}{11}\right)^2 = -1.$$

The fourth equality is Theorem 5.4 with $101 \equiv 1 \pmod 4$, the fifth uses $101 = 6 \cdot 15 + 11$, and the sixth is Theorem 5.4 with $11 \equiv 15 \equiv -1 \pmod 4$. ◊

Remark 5.7. If $x^2 \equiv a \pmod n$ is solvable, then a is a quadratic residue modulo every prime divisor of n, so $\left(\frac{a}{n}\right) = 1$. Hence $\left(\frac{a}{n}\right) = -1$ implies that $x^2 \equiv a \pmod n$ has no solution. The converse fails: $\left(\frac{2}{15}\right) = \left(\frac{2}{3}\right)\left(\frac{2}{5}\right) = (-1)(-1) = 1$, but $x^2 \equiv 2 \pmod 3$ has no solution, so neither has $x^2 \equiv 2 \pmod{15}$.

6. Polynomial congruences and Hensel's lemma

Definition 6.1 (Polynomial congruence). Let $f(x) = a_n x^n + \cdots + a_1 x + a_0$ with $a_i \in \mathbb{Z}$ and $a_n \neq 0$, and let $m \in \mathbb{N}$. A *polynomial congruence* is a congruence $f(x) \equiv 0 \pmod m$.

Let p be a prime. Starting from a solution of $f(x) \equiv 0 \pmod p$, we lift it to solutions modulo p^k. Here f' denotes the formal derivative of f, which again lies in $\mathbb{Z}[x]$.

Theorem 6.2 (Hensel). Let $f \in \mathbb{Z}[x]$, p a prime, and $c \in \mathbb{Z}$ with $f(c) \equiv 0 \pmod p$ and $f'(c) \not\equiv 0 \pmod p$. Then for every $k \in \mathbb{N}$ the congruence $f(x) \equiv 0 \pmod{p^k}$ has exactly one solution $x \equiv c_k \pmod{p^k}$ with $c_k \equiv c \pmod p$.

Lemma 6.3. Let $f \in \mathbb{Z}[x]$, p a prime, $a, t \in \mathbb{Z}$ and $j \in \mathbb{N}$. Then

$$f(a+tp^j) \equiv f(a) + tp^j f'(a) \pmod{p^{j+1}}.$$

Proof. For a polynomial of degree n the Taylor expansion is exact:

$$f(a+tp^j) = f(a) + tp^j f'(a) + t^2 p^{2j}\frac{f''(a)}{2!} + \cdots + t^n p^{nj}\frac{f^{(n)}(a)}{n!}.$$

Each $f^{(r)}(a)/r!$ is an integer, because $(x^s)^{(r)} = s(s-1)\cdots(s-r+1)x^{s-r}$ and $s(s-1)\cdots(s-r+1)/r! = \binom{s}{r}$. Every term with $r \geq 2$ is divisible by p^{2j}, and $2j \geq j+1$ since $j \geq 1$. ∎

Proof of Theorem 6.2. We proceed by induction on k. For $k=1$, $c_1 = c$ is a solution, and every solution congruent to c modulo p is congruent to c_1.

Let $k \geq 2$ and assume that $f(x) \equiv 0 \pmod{p^{k-1}}$ has exactly one solution $x \equiv c_{k-1} \pmod{p^{k-1}}$ with $c_{k-1} \equiv c \pmod{p}$. Let x be a solution of $f(x) \equiv 0 \pmod{p^k}$ with $x \equiv c \pmod{p}$. Then $f(x) \equiv 0 \pmod{p^{k-1}}$, so $x \equiv c_{k-1} \pmod{p^{k-1}}$ by the uniqueness in the induction hypothesis, that is, $x = c_{k-1} + tp^{k-1}$ with $t \in \mathbb{Z}$. By Lemma 6.3 with $j = k-1 \geq 1$,

$$f\big(c_{k-1} + tp^{k-1}\big) \equiv f(c_{k-1}) + tp^{k-1} f'(c_{k-1}) \pmod{p^k}.$$

By the induction hypothesis $p^{k-1} \mid f(c_{k-1})$, and $f'(c_{k-1}) \equiv f'(c) \pmod{p}$ since $c_{k-1} \equiv c \pmod{p}$ and $f' \in \mathbb{Z}[x]$. Hence $f\big(c_{k-1} + tp^{k-1}\big) \equiv 0 \pmod{p^k}$ if and only if

$$\frac{f(c_{k-1})}{p^{k-1}} + tf'(c) \equiv 0 \pmod{p}.$$

This is a linear congruence modulo p in t, and $f'(c) \not\equiv 0 \pmod{p}$, so by Remark 3.2 it has exactly one solution $t \equiv t_0 \pmod{p}$. Writing $t = t_0 + sp$ with $s \in \mathbb{Z}$,

$$x = c_{k-1} + (t_0 + sp)p^{k-1} = c_{k-1} + t_0 p^{k-1} + sp^k,$$

so every such solution satisfies $x \equiv c_{k-1} + t_0 p^{k-1} \pmod{p^k}$. Conversely, $c_k = c_{k-1} + t_0 p^{k-1}$ is a solution, and $c_k \equiv c_{k-1} \equiv c \pmod{p}$. Hence the solution is unique modulo p^k. ∎

Example 6.4. Solve $x^3 + 2x - 22 \equiv 0 \pmod{125}$.

Solution. Let $f(x) = x^3 + 2x - 22$, so $f'(x) = 3x^2 + 2$ and $125 = 5^3$.

Modulo 5. The values of $f(0), \ldots, f(4)$ modulo 5 are $3, 1, 0, 1, 0$, so $x \equiv 2$ or $x \equiv 4 \equiv -1 \pmod{5}$.

Case 1: $x \equiv 2 \pmod{5}$. Here $f'(2) = 14 \equiv -1 \not\equiv 0 \pmod{5}$, so by Theorem 6.2 there is exactly one solution modulo 125 in this class. Write $x = 5t + 2$ with $t \in \mathbb{Z}$. Modulo 25,

$$(5t+2)^3 + 2(5t+2) - 22 \equiv 60t + 8 + 10t + 4 - 22 = 70t - 10 \pmod{25},$$

and $70t - 10 \equiv 0 \pmod{25}$ if and only if $-5t \equiv 10 \pmod{25}$, that is, $t \equiv -2 \pmod{5}$. So $t = 5s - 2$ with $s \in \mathbb{Z}$, and $x = 5(5s-2) + 2 = 25s - 8$. Modulo 125, Lemma 6.3 gives

$$f(25s - 8) \equiv f(-8) + 25s f'(-8) = -550 + 25s \cdot 194 \equiv -50 + 25s \cdot 194 \pmod{125},$$

which vanishes modulo 125 if and only if $-2 + 194s \equiv 0 \pmod{5}$, that is, $s \equiv -2 \pmod{5}$. So $s = 5r - 2$ with $r \in \mathbb{Z}$, and $x = 25(5r-2) - 8 = 125r - 58$. Hence $x \equiv -58 \equiv 67 \pmod{125}$.

Case 2: $x \equiv -1 \pmod 5$. Here $f'(-1) = 5 \equiv 0 \pmod 5$, and Theorem 6.2 does not apply. By the proof of Theorem 6.2, if $c_{k-1} \equiv -1 \pmod 5$ solves the congruence modulo 5^{k-1}, then $c_{k-1} + t5^{k-1}$ solves it modulo 5^k if and only if $f(c_{k-1})/5^{k-1} + tf'(-1) \equiv 0 \pmod 5$, that is, if and only if $5^k \mid f(c_{k-1})$. So either every lift is a solution or none is.

Since $f(-1) = -25 \equiv 0 \pmod{25}$, every $x \equiv -1 \pmod 5$ solves $f(x) \equiv 0 \pmod{25}$, namely $x \equiv 4, 9, 14, 19, 24 \pmod{25}$. For each such c and $j \in \mathbb{Z}$, Lemma 6.3 and $5 \mid f'(c)$ give $f(c + 25j) \equiv f(c) \pmod{125}$. Now $f(4) = 50$, $f(9) = 725 \equiv 100$, $f(24) = 13850 \equiv 100$, while $f(14) = 2750 = 22 \cdot 125$ and $f(19) = 6875 = 55 \cdot 125$. So the solutions in this class are $x \equiv 14 + 25j$ and $x \equiv 19 + 25j \pmod{125}$ with $0 \le j \le 4$.

Altogether the congruence has 11 solutions modulo 125: $x \equiv 67$, and $x \equiv 14 + 25j$, $19 + 25j$ for $0 \le j \le 4$. ◊

7. Arithmetic functions

Definition 7.1 (Arithmetic function). An *arithmetic function* is a function $f : \mathbb{N} \to \mathbb{C}$. It is *multiplicative* if $f(ab) = f(a)f(b)$ for all $a, b \in \mathbb{N}$ with $(a, b) = 1$, and *additive* if $f(ab) = f(a) + f(b)$ for all $a, b \in \mathbb{N}$ with $(a, b) = 1$.

Example 7.2. Let $d(n)$, also written $\tau(n)$, be the number of positive divisors of n. If $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct primes p_i, then $d(n) = \prod_{i=1}^{r} (1 + \alpha_i)$, because the divisors of n are exactly the products $p_1^{\beta_1} \cdots p_r^{\beta_r}$ with $0 \le \beta_i \le \alpha_i$. Hence d is multiplicative: for $(m, n) = 1$ the prime divisors of m and n are disjoint, and $d(mn) = d(m)d(n)$.

Example 7.3. Let $\sigma(n)$ be the sum of the positive divisors of n; for instance $\sigma(6) = 1 + 2 + 3 + 6 = 12$. For $n = \prod_{i=1}^{r} p_i^{\alpha_i}$ as above, summing over the divisors described in the previous example gives

$$\sigma(n) = \prod_{i=1}^{r} \sum_{j=0}^{\alpha_i} p_i^j.$$

As for d, this formula shows that σ is multiplicative.

Example 7.4. Euler's function φ is multiplicative (Remark 1.1). Let $\omega(n)$ be the number of distinct prime divisors of n; then ω is additive. Let $\Omega(n)$ be the number of prime divisors of n counted with multiplicity, so $\Omega(n) = \sum_{i=1}^{r} \alpha_i$ for n as above; then Ω is additive.

Proposition 7.5.

1) If f is multiplicative and not identically 0, then $f(1) = 1$.

2) If f is additive, then $f(1) = 0$.

Proof. 1) Choose a with $f(a) \neq 0$. Since $(1, a) = 1$, $f(a) = f(1 \cdot a) = f(1)f(a)$, so $f(1) = 1$.

2) Since $(1, 1) = 1$, $f(1) = f(1 \cdot 1) = f(1) + f(1)$, so $f(1) = 0$. ■

Definition 7.6 (Summation function). Let $f : \mathbb{N} \to \mathbb{C}$ be an arithmetic function. Its *summation function* is $f^+(n) = F(n) = \sum_{d|n} f(d)$, where d runs over the positive divisors of n.

Lemma 7.7. Let $a, b \in \mathbb{N}$ with $(a, b) = 1$ and let $d \mid ab$. Then there are unique $d_1, d_2 \in \mathbb{N}$ with $d_1 \mid a$, $d_2 \mid b$ and $d = d_1 d_2$. Hence $(d_1, d_2) \mapsto d_1 d_2$ is a bijection from the pairs of positive divisors of a and b onto the positive divisors of ab.

Proof. Put $d_1 = (d, a)$ and $d_2 = (d, b)$. They are coprime and both divide d, so $d_1 d_2 \mid d$. Conversely, let q^e be the exact power of a prime q dividing d. Since q divides at most one of a, b, say a, and $q^e \mid ab$, we get $q^e \mid a$ and so $q^e \mid d_1$. Hence $d \mid d_1 d_2$, and $d = d_1 d_2$. If also $d = e_1 e_2$ with $e_1 \mid a$ and $e_2 \mid b$, then $e_1 \mid (d, a) = d_1$ and $e_2 \mid d_2$, and $e_1 e_2 = d_1 d_2$ forces $e_1 = d_1$, $e_2 = d_2$. Every product $d_1 d_2$ with $d_1 \mid a$, $d_2 \mid b$ divides ab, which gives the bijection. ∎

Theorem 7.8. If $f : \mathbb{N} \to \mathbb{C}$ is multiplicative, then $F(n) = \sum_{d|n} f(d)$ is multiplicative.

Proof. Let $(m, n) = 1$. By Lemma 7.7, and since $(d_1, d_2) = 1$ for $d_1 \mid m$, $d_2 \mid n$,

$$F(mn) = \sum_{d|mn} f(d) = \sum_{d_1|m,\ d_2|n} f(d_1 d_2) = \sum_{d_1|m,\ d_2|n} f(d_1) f(d_2) = \sum_{d_1|m} f(d_1) \sum_{d_2|n} f(d_2) = F(m) F(n).$$

∎

Theorem 7.9. For every $n \in \mathbb{N}$, $\varphi^+(n) = \sum_{d|n} \varphi(d) = n$.

Proof. For $n = 1$ both sides equal 1. For a prime power $n = p^\alpha$ with $\alpha \geq 1$,

$$F(p^\alpha) = \sum_{d|p^\alpha} \varphi(d) = \sum_{i=0}^{\alpha} \varphi(p^i) = 1 + (p - 1) + (p^2 - p) + \cdots + (p^\alpha - p^{\alpha-1}) = p^\alpha.$$

For $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$, F is multiplicative by Theorem 7.8, so $F(n) = F(p_1^{\alpha_1}) \cdots F(p_r^{\alpha_r}) = p_1^{\alpha_1} \cdots p_r^{\alpha_r} = n$. ∎

Definition 7.10 (Möbius function). The *Möbius function* is

$$\mu(n) = \begin{cases} 1 & \text{if } n = 1 \\ (-1)^r & \text{if } n = p_1 \cdots p_r \text{ with distinct primes } p_i \\ 0 & \text{otherwise.} \end{cases}$$

So $\mu(n) \neq 0$ if and only if n is square-free. For example, $\mu(10) = (-1)^2 = 1$.

Proposition 7.11. The function μ is multiplicative, and

$$\mu^+(n) = \sum_{d|n} \mu(d) = \begin{cases} 1 \text{ if } n = 1 \\ 0 \text{ otherwise.} \end{cases}$$

Proof. Let $(m,n)=1$. If m or n is not square-free, neither is mn, and both sides of $\mu(mn)=\mu(m)\mu(n)$ vanish. Otherwise m and n are products of r and s distinct primes, the primes of m and n are distinct, and $\mu(mn)=(-1)^{r+s}=\mu(m)\mu(n)$.

For $n=1$, $\mu^+(1)=\mu(1)=1$. For $n=p^\alpha$ with $\alpha\geq 1$, $\mu^+(p^\alpha)=\mu(1)+\mu(p)+\cdots+\mu(p^\alpha)=1+(-1)=0$. For $n=p_1^{\alpha_1}\cdots p_r^{\alpha_r}$ with $r\geq 1$, μ^+ is multiplicative by Theorem 7.8, so $\mu^+(n)=\mu^+(p_1^{\alpha_1})\cdots\mu^+(p_r^{\alpha_r})=0$. ∎

Theorem 7.12 (Möbius inversion formula). Let $f,F:\mathbb{N}\to\mathbb{C}$ be arithmetic functions with $F(n)=\sum_{d\mid n}f(d)$ for every $n\in\mathbb{N}$. Then $f(n)=\sum_{d\mid n}\mu(d)F(n/d)$ for every $n\in\mathbb{N}$.

Proof. The pairs (d,c) of positive integers with $d\mid n$ and $c\mid n/d$ are exactly those with $cd\mid n$, that is, with $c\mid n$ and $d\mid n/c$. Hence, using Proposition 7.11,

$$\sum_{d\mid n}\mu(d)F\left(\frac{n}{d}\right)=\sum_{d\mid n}\mu(d)\sum_{c\mid n/d}f(c)=\sum_{c\mid n}f(c)\sum_{d\mid n/c}\mu(d)=f(n),$$

since the inner sum is 1 for $c=n$ and 0 otherwise. ∎

Theorem 7.13. Let $f,F:\mathbb{N}\to\mathbb{C}$ with $F(n)=\sum_{d\mid n}f(d)$ for every n. If F is multiplicative, then f is multiplicative.

Proof. Let $(m,n)=1$. By Theorem 7.12 and Lemma 7.7, writing $d=d_1d_2$ with $d_1\mid m$, $d_2\mid n$,

$$\begin{aligned}f(mn)&=\sum_{d\mid mn}\mu(d)F\left(\frac{mn}{d}\right)=\sum_{d_1\mid m,\ d_2\mid n}\mu(d_1d_2)F\left(\frac{m}{d_1}\cdot\frac{n}{d_2}\right)\\&=\sum_{d_1\mid m}\mu(d_1)F\left(\frac{m}{d_1}\right)\sum_{d_2\mid n}\mu(d_2)F\left(\frac{n}{d_2}\right)=f(m)f(n).\end{aligned}$$

The third equality uses that μ and F are multiplicative, with $(d_1,d_2)=1$ and $(m/d_1,n/d_2)=1$. ∎

Example 7.14. Let $F(n)=n$. Compute f with $F=f^+$.

Solution. By Theorem 7.9, $F=\varphi^+$, so Theorem 7.12 gives

$$\varphi(n)=\sum_{d\mid n}\mu(d)F\left(\frac{n}{d}\right)=n\sum_{d\mid n}\frac{\mu(d)}{d}.$$

Both F and φ are multiplicative. For $n=p_1^{\alpha_1}\cdots p_r^{\alpha_r}$, only square-free d contribute, that is, products of distinct p_i, so

$$\varphi(n) = n - \frac{n}{p_1} - \cdots - \frac{n}{p_r} + \frac{n}{p_1 p_2} + \cdots + \frac{n}{p_{r-1} p_r} - \frac{n}{p_1 p_2 p_3} - \cdots + (-1)^r \frac{n}{p_1 \cdots p_r} = n \prod_{i=1}^{r} \left(1 - \frac{1}{p_i}\right).$$

◊

8. p-adic numbers

Let p be a prime. Every $n \in \mathbb{Z}_{\geq 0}$ has a unique expansion $n = a_0 + a_1 p + \cdots + a_k p^k$ with $0 \leq a_i < p$.

Definition 8.1 (p-adic integer). A *p-adic integer* is a formal series $\alpha = a_0 + a_1 p + a_2 p^2 + \cdots$ with digits $a_i \in \{0, 1, \ldots, p-1\}$. The set of p-adic integers is denoted $\mathbb{Z}_p$. For $k \in \mathbb{N}$ the *truncation* of α is $\alpha_k = \sum_{i=0}^{k-1} a_i p^i$.

Remark 8.2. Let $(y_k)_{k \geq 1}$ be integers with $y_{k+1} \equiv y_k \pmod{p^k}$ for every k. Then there is exactly one $\alpha \in \mathbb{Z}_p$ with $\alpha_k \equiv y_k \pmod{p^k}$ for every k. Indeed, let x_k be the least non-negative residue of y_k modulo p^k. Then $x_{k+1} \equiv x_k \pmod{p^k}$ and $0 \leq x_k < p^k$ give $x_{k+1} = x_k + a_k p^k$ with $0 \leq a_k < p$, and $\alpha = \sum_k a_k p^k$ is the required series. Uniqueness holds because $0 \leq \alpha_k < p^k$ determines α_k from its residue, and the truncations determine the digits.

Definition 8.3 (Operations on $\mathbb{Z}_p$). For $\alpha, \beta \in \mathbb{Z}_p$, the sum $\alpha + \beta$ and the product $\alpha\beta$ are the p-adic integers with

$$(\alpha + \beta)_k \equiv \alpha_k + \beta_k \pmod{p^k}, \quad (\alpha\beta)_k \equiv \alpha_k \beta_k \pmod{p^k} \quad (k \in \mathbb{N}).$$

They exist and are unique by Remark 8.2, since the sequences $\alpha_k + \beta_k$ and $\alpha_k \beta_k$ satisfy its compatibility condition. Each ring axiom holds modulo every p^k, so $\mathbb{Z}_p$ is a commutative ring with zero $0 + 0p + \cdots$ and identity $1 + 0p + \cdots$.

Example 8.4. $-1 = (p-1) + (p-1)p + (p-1)p^2 + \cdots$ in $\mathbb{Z}_p$, because the truncations of the right side are $(p-1)\left(1 + p + \cdots + p^{k-1}\right) = p^k - 1 \equiv -1 \pmod{p^k}$.

Example 8.5. Let $\alpha, \beta \in \mathbb{Z}_3$ with $\alpha = 2 + 1 \cdot 3 + \cdots$ and $\beta = 1 + 2 \cdot 3 + \cdots$. Then $(\alpha + \beta)_1 \equiv 2 + 1 \equiv 0 \pmod{3}$. Further $\alpha_2 = 5$ and $\beta_2 = 7$, so $(\alpha + \beta)_2 \equiv 12 \equiv 3 \pmod{3^2}$. Hence $\alpha + \beta = 0 + 1 \cdot 3 + \cdots$.

Remark 8.6. $\mathbb{Z} \subset \mathbb{Z}_p$: the integer n is identified with the p-adic integer whose truncations satisfy $\alpha_k \equiv n \pmod{p^k}$. For $n \geq 0$ this is the base-p expansion of n followed by zero digits. The map is an injective ring homomorphism, since distinct integers differ modulo p^k for large k.

Example 8.7. The equation $2x + 3 = 0$ has no solution in $\mathbb{Z}$, but it has one in $\mathbb{Z}_3$. Formally,

$$-\frac{3}{2} = \frac{3}{-2} = \frac{3}{1-3} = 3 \cdot \frac{1}{1-3} = 3(1 + 3 + 3^2 + \cdots),$$

which suggests $\gamma = 0 + 1 \cdot 3 + 1 \cdot 3^2 + \cdots$. Indeed, $\gamma_k = 3 + 3^2 + \cdots + 3^{k-1} = \left(3^k - 3\right)/2$, so $2\gamma_k + 3 = 3^k \equiv 0 \pmod{3^k}$ for every k, and $2\gamma + 3 = 0$ in $\mathbb{Z}_3$.

Remark 8.8. Let $x = a/b$ with $a, b \in \mathbb{Z}$ and $p \nmid b$. Then there is exactly one $\xi \in \mathbb{Z}_p$ with $b\xi = a$, and we write $\xi = a/b$. Indeed, b is invertible modulo every p^k; if b_k^{-1} denotes an inverse modulo p^k, then $y_k = ab_k^{-1}$ satisfies $y_{k+1} \equiv y_k \pmod{p^k}$, and the ξ of Remark 8.2 satisfies $(b\xi)_k \equiv a \pmod{p^k}$ for all k. If $b\xi = b\xi'$, then $b(\xi - \xi')_k \equiv 0 \pmod{p^k}$, so $(\xi - \xi')_k = 0$ for all k and $\xi = \xi'$.

Definition 8.9 (p-adic number). A *p-adic number* is a formal series $x = \sum_{k=-N}^{\infty} a_k p^k$ with $N \in \mathbb{Z}_{\geq 0}$ and digits $a_k \in \{0, \ldots, p-1\}$; series that differ only by leading zero digits are identified. The set of p-adic numbers is denoted $\mathbb{Q}_p$, and $\mathbb{Z}_p \subset \mathbb{Q}_p$ consists of the series with $a_k = 0$ for $k < 0$.

Proposition 8.10. For $\alpha \in \mathbb{Z}_p$ and $N \in \mathbb{Z}_{\geq 0}$ let $p^{-N}\alpha$ denote the series whose digit at p^{k-N} is the digit of α at p^k. Every $x \in \mathbb{Q}_p$ has this form, and $p^{-N}\alpha = p^{-N-1}(p\alpha)$. Define

$$p^{-N}\alpha + p^{-N}\beta = p^{-N}(\alpha + \beta), \quad (p^{-N}\alpha)(p^{-N}\beta) = p^{-2N}(\alpha\beta).$$

Then $\mathbb{Q}_p$ is a field containing $\mathbb{Z}_p$ as a subring. An element $u \in \mathbb{Z}_p$ is a unit of $\mathbb{Z}_p$ if and only if its digit a_0 is nonzero, and every $x \in \mathbb{Q}_p$ with $x \neq 0$ can be written uniquely as $x = p^v u$ with $v \in \mathbb{Z}$ and u a unit of $\mathbb{Z}_p$.

Proof. Multiplication by p in $\mathbb{Z}_p$ shifts the digits up by one, since $(p\alpha)_{k+1} = p\alpha_k$. This gives $p^{-N}\alpha = p^{-N-1}(p\alpha)$, so the definitions do not depend on the choice of a common N, and the ring axioms follow from those of $\mathbb{Z}_p$.

If u has $a_0 \neq 0$, then every u_k is coprime to p. Let w_k be an inverse of u_k modulo p^k; then $w_{k+1} \equiv w_k \pmod{p^k}$, and the resulting $w \in \mathbb{Z}_p$ satisfies $(uw)_k \equiv 1$ for all k, so $uw = 1$. Conversely, $uw = 1$ gives $u_1 w_1 \equiv 1 \pmod{p}$, so $a_0 = u_1 \neq 0$.

For $x \neq 0$ let v be the least index of a nonzero digit. Shifting the digits gives $x = p^v u$ with $u \in \mathbb{Z}_p$ whose digit a_0 is nonzero, so u is a unit, and v and u are determined by x. Then $x^{-1} = p^{-v}u^{-1}$, so $\mathbb{Q}_p$ is a field. ∎

Remark 8.11. $\mathbb{Q} \subset \mathbb{Q}_p$: every nonzero rational number is $p^v a/b$ with $v \in \mathbb{Z}$ and $p \nmid ab$, and $a/b \in \mathbb{Z}_p$ by Remark 8.8. This gives an injective field homomorphism $\mathbb{Q} \to \mathbb{Q}_p$.

Example 8.12. Is $x^2 - 2 = 0$ solvable in $\mathbb{Z}_7$?

Solution. Let $\alpha = a_0 + a_1 \cdot 7 + a_2 \cdot 7^2 + \cdots$ be a solution. Then $a_0^2 \equiv 2 \pmod{7}$, so $a_0 = 3$ or $a_0 = 4$, that is, $\alpha_1 = 3$ or $\alpha_1 = 4$.

Let $\alpha_1 = 3$. We seek $\alpha_2 = a_0 + a_1 \cdot 7$ modulo 49:

$$\alpha_2^2 - 2 = (a_0 + a_1 \cdot 7)^2 - 2 = a_0^2 - 2 + 2 \cdot 7 a_0 a_1 + a_1^2 7^2 \equiv 0 \pmod{7^2}$$

if and only if $a_0^2 - 2 + 14 a_0 a_1 \equiv 0 \pmod{7^2}$, that is, $3^2 - 2 + 14 \cdot 3 a_1 \equiv 0 \pmod{49}$, which gives $1 + 6a_1 \equiv 0 \pmod 7$ and $a_1 = 1$. Continuing recursively, $\alpha = 3 + 1 \cdot 7 + 2 \cdot 7^2 + 6 \cdot 7^3 + \cdots$. The other solution is $\alpha = 4 + 5 \cdot 7 + 4 \cdot 7^2 + \cdots$.

Both solutions exist: $f(x) = x^2 - 2$ has $f'(3) = 6 \not\equiv 0$ and $f'(4) = 8 \not\equiv 0 \pmod 7$, so Theorem 6.2 gives compatible solutions c_k modulo 7^k, and by Remark 8.2 they define $\alpha \in \mathbb{Z}_7$ with $(\alpha^2 - 2)_k = 0$ for every k, that is, $\alpha^2 = 2$. ◊

Remark 8.13. Write $\mathbb{Q}_\infty = \mathbb{R}$, and let $p \le \infty$ range over the primes and ∞. If a polynomial equation with rational coefficients has a solution in $\mathbb{Q}$, then it has a solution in $\mathbb{Q}_p$ for every $p \le \infty$, since $\mathbb{Q} \subset \mathbb{Q}_p$.

Theorem 8.14 (Hasse–Minkowski) (cited)**.** Let $f(x_1, \ldots, x_n) \in \mathbb{Q}[x_1, \ldots, x_n]$ be a quadratic form. The equation $f(x_1, \ldots, x_n) = 0$ has a nontrivial solution in $\mathbb{Q}$ if and only if it has a nontrivial solution in $\mathbb{Q}_p$ for every $p \le \infty$.

9. The p-adic metric

Definition 9.1 (Absolute value). Let F be a field. A map $|\cdot| : F \to \mathbb{R}_{\ge 0}$ is an *absolute value* if for all $x, y \in F$:

1) $|x| = 0$ if and only if $x = 0$,
2) $|xy| = |x|\,|y|$,
3) $|x + y| \le |x| + |y|$.

It is *non-archimedean* if moreover $|x + y| \le \max\{|x|, |y|\}$ for all $x, y \in F$.

Remark 9.2. From $|1| = |1|^2$ and $|1| \neq 0$ we get $|1| = 1$, and from $|-1|^2 = |1|$ we get $|-1| = 1$. Hence an absolute value induces a metric $d(x, y) = |x - y|$ on F.

Definition 9.3 (Ultrametric space). A metric space whose metric is induced by a non-archimedean absolute value is called *ultrametric*. Its metric satisfies $d(x, z) \le \max\{d(x, y), d(y, z)\}$.

Definition 9.4 (p-adic valuation). The *p-adic valuation* is the function $v_p : \mathbb{Q} \to \mathbb{Z} \cup \{\infty\}$ defined as follows. For $0 \neq x \in \mathbb{Z}$, $v_p(x)$ is the unique non-negative integer with $x = p^{v_p(x)} x'$ and $p \nmid x'$. For $x \in \mathbb{Q} \setminus \mathbb{Z}$, write $x = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$, and put $v_p(x) = v_p(a) - v_p(b)$. Finally $v_p(0) = \infty$.

Proposition 9.5. The valuation v_p is well defined, and for all $x, y \in \mathbb{Q}$:

1) $v_p(xy) = v_p(x) + v_p(y)$,

2) $v_p(x+y) \geq \min\{v_p(x), v_p(y)\}$.

Proof. On nonzero integers, $v_p(ab) = v_p(a) + v_p(b)$ by unique factorization. If $a/b = a'/b'$, then $ab' = a'b$, so $v_p(a) + v_p(b') = v_p(a') + v_p(b)$, and the definition does not depend on the representation. Property 1 follows from the integer case. For 2 we may assume $x, y \neq 0$; write $x = p^v a/b$ and $y = p^w c/d$ with $p \nmid abcd$ and $v \leq w$. Then $x + y = p^v(ad + p^{w-v}cb)/(bd)$, and $p \nmid bd$, so $v_p(x+y) \geq v$. ∎

Definition 9.6 (p-adic absolute value). The *p-adic absolute value* $|\cdot|_p : \mathbb{Q} \to \mathbb{R}_{\geq 0}$ is $|x|_p = p^{-v_p(x)}$ for $x \neq 0$ and $|0|_p = 0$.

Theorem 9.7. $|\cdot|_p$ is a non-archimedean absolute value on $\mathbb{Q}$.

Proof. $|x|_p = 0$ only for $x = 0$. By Proposition 9.5, $|xy|_p = p^{-v_p(x)-v_p(y)} = |x|_p\,|y|_p$ and $|x+y|_p \leq p^{-\min\{v_p(x),v_p(y)\}} = \max\{|x|_p, |y|_p\}$, which also implies the triangle inequality. ∎

Remark 9.8. $\lim_{n\to\infty} p^n = 0$ in the p-adic metric. Indeed, let $\varepsilon > 0$ and choose $N \in \mathbb{N}$ with $1/N < \varepsilon$. Since $p^N > N$, we have $1/p^N < \varepsilon$, so for $n \geq N$, $|p^n|_p = 1/p^n < \varepsilon$.

Definition 9.9 (Equivalent absolute values). Two absolute values on a field are *equivalent* if they induce the same topology. The *trivial* absolute value is $|x| = 1$ for $x \neq 0$, $|0| = 0$. The *standard* absolute value on $\mathbb{Q}$ is denoted $|\cdot|_\infty$.

Theorem 9.10 (Ostrowski). Every nontrivial absolute value on $\mathbb{Q}$ is equivalent to $|\cdot|_\infty$ or to $|\cdot|_p$ for some prime p.

Proof. Let $|\cdot|$ be a nontrivial absolute value on $\mathbb{Q}$. It suffices to find $c > 0$ with $|x| = |x|_\infty^c$ for all x, or $|x| = |x|_p^c$ for all x: then the open balls of $|\cdot|$ of radius r are the balls of the other absolute value of radius $r^{1/c}$, so the topologies agree. Since $|-1| = 1$ and $|\cdot|$ is multiplicative, it suffices to determine $|n|$ for $n \in \mathbb{N}$. For $n \in \mathbb{N}$ the triangle inequality gives $|n| \leq n$.

Case 1: $|n| \leq 1$ for every $n \in \mathbb{N}$. If $|n| = 1$ for all $n \in \mathbb{N}$, then $|\cdot|$ is trivial by multiplicativity. So some $n \in \mathbb{N}$ has $|n| < 1$; let n_0 be the least one. Then $n_0 > 1$, and n_0 is a prime p: if $n_0 = ab$ with $1 < a, b < n_0$, then $|a| = |b| = 1$ and $|n_0| = 1$. Suppose a prime $q \neq p$ also has $|q| < 1$. Choose $M \in \mathbb{N}$ with $|p|^M < 1/2$ and $|q|^M < 1/2$. Since $(p^M, q^M) = 1$, there are $u, w \in \mathbb{Z}$ with $up^M + wq^M = 1$, and then $1 = |1| \leq |u|\,|p|^M + |w|\,|q|^M < 1/2 + 1/2 = 1$, because $|u|, |w| \leq 1$. This is a contradiction, so $|q| = 1$ for every prime $q \neq p$. Put $c = -\log|p| \,/ \log p > 0$, so $|p| = p^{-c}$. For $n = p^v n'$ with $p \nmid n'$ we get $|n| = p^{-cv} = |n|_p^c$.

Case 2: $|n_1| > 1$ for some $n_1 \in \mathbb{N}$. We first show that for all integers $m, n > 1$,

$$|m| \leq \max\{1, |n|\}^{\log m / \log n}.$$

Let $k \in \mathbb{N}$ and write $m^k = \sum_{i=0}^{s} d_i n^i$ in base n, with $0 \leq d_i < n$ and $s \leq k \log m / \log n$. Then $|d_i| \leq d_i < n$, and

$$|m|^k \le \sum_{i=0}^{s} |d_i|\ |n|^i \le (s+1)n \max\{1,|n|\}^s \le \left(1 + k\frac{\log m}{\log n}\right) n \max\{1,|n|\}^{k \log m/\log n}.$$

Taking k-th roots and letting $k \to \infty$ gives the inequality. Applying it with $m = n_1$ shows $1 < |n_1| \le \max\{1,|n|\}^{\log n_1/\log n}$, so $|n| > 1$ for every $n > 1$. Hence $|m|^{1/\log m} \le |n|^{1/\log n}$ for all $m, n > 1$, and by symmetry equality holds. So $|n|^{1/\log n} = e^c$ for a constant c, with $c > 0$ since $|n| > 1$. Then $|n| = n^c = |n|_\infty^c$ for $n > 1$, and also for $n = 1$. ∎

Remark 9.11 (Recall). A *completion* of a metric space X is a complete metric space $\hat{X}$ together with an isometric embedding $X \to \hat{X}$ whose image is dense. A completion exists and is unique up to an isometry that is the identity on X.

Theorem 9.12 (Construction of $\mathbb{Q}_p$ and $\mathbb{Z}_p$). For $0 \ne x \in \mathbb{Q}_p$ let $v(x)$ be the least index of a nonzero digit of x, and put $|x|_p = p^{-v(x)}$, $|0|_p = 0$. Then:

1) $|\cdot|_p$ is a non-archimedean absolute value on $\mathbb{Q}_p$ that extends the p-adic absolute value of $\mathbb{Q}$;
2) $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \le 1\}$;
3) $\mathbb{Q}$ is dense in $\mathbb{Q}_p$;
4) $\mathbb{Q}_p$ is complete.

Hence $\mathbb{Q}_p$ is the completion of $\mathbb{Q}$ with respect to $|\cdot|_p$. The field $\mathbb{Q}$ itself is not complete under the p-adic metric.

Proof. We use one fact throughout: for $\alpha, \beta \in \mathbb{Z}_p$ and $k \in \mathbb{N}$, $|\alpha - \beta|_p \le p^{-k}$ holds if and only if $\alpha_k = \beta_k$. Indeed, $(\alpha - \beta)_k \equiv \alpha_k - \beta_k \pmod{p^k}$ and $0 \le \alpha_k, \beta_k < p^k$, and $|\alpha - \beta|_p \le p^{-k}$ means that the first k digits of $\alpha - \beta$ vanish.

1) In the notation of Proposition 8.10, $x = p^v u$ with u a unit gives $v(x) = v$. For $x = p^v u$ and $y = p^w u'$, $xy = p^{v+w} uu'$ with uu' a unit, so $|xy|_p = |x|_p\ |y|_p$. For the ultrametric inequality, write $x = p^{-N}\alpha$ and $y = p^{-N}\beta$ with $\alpha, \beta \in \mathbb{Z}_p$. If the first k digits of α and β vanish, then $(\alpha + \beta)_k \equiv \alpha_k + \beta_k = 0$, so those of $\alpha + \beta$ vanish; hence $v(x + y) \ge \min\{v(x), v(y)\}$. For rational $x = p^v a/b$ with $p \nmid ab$, the element $a/b \in \mathbb{Z}_p$ has first digit $ab^{-1} \not\equiv 0 \pmod{p}$, so it is a unit and $v(x) = v = v_p(x)$.

2) $x \in \mathbb{Z}_p$ if and only if x has no nonzero digit of negative index, that is, $v(x) \ge 0$.

3) Let $x = \sum_{k \ge -N} a_k p^k$ and let $s_n = \sum_{k=-N}^{n-1} a_k p^k \in \mathbb{Q}$. The first nonzero digit of $x - s_n$ has index at least n, so $|x - s_n|_p \le p^{-n} \to 0$.

4) Let (x_n) be a Cauchy sequence in $\mathbb{Q}_p$. It is bounded, so there is N with $|x_n|_p \le p^N$ for all n; then $y_n = p^N x_n \in \mathbb{Z}_p$ by 2), and (y_n) is Cauchy. Choose $n_1 < n_2 < \cdots$ with $|y_n - y_m|_p \le p^{-k}$ for $n, m \ge n_k$. By the fact above, $(y_n)_k = \left(y_{n_k}\right)_k =: z_k$ for $n \ge n_k$. Then $z_{k+1} = \left(y_{n_{k+1}}\right)_{k+1} \equiv \left(y_{n_{k+1}}\right)_k = z_k \pmod{p^k}$, so by Remark 8.2 there is $\eta \in \mathbb{Z}_p$ with $\eta_k = z_k$. For $n \ge n_k$ we get $(y_n)_k = \eta_k$, that is, $|y_n - \eta|_p \le p^{-k}$. So $y_n \to \eta$ and $x_n \to p^{-N}\eta$.

By 1), 3) and 4), $\mathbb{Q}_p$ with $|\cdot|_p$ is a completion of $\mathbb{Q}$. Finally, $\mathbb{Q}$ is not complete: for $p = 7$, the truncations of the element α with $\alpha^2 = 2$ constructed in the example on $x^2 - 2$ form a Cauchy sequence in $\mathbb{Q}$ by 3). Its limit in $\mathbb{Q}_7$ is α, which is not rational, since 2 is not the square of a rational number. ∎

10. Worked exercises

Example 10.1. Show that 2 is a primitive root modulo 11 and solve $7x^8 \equiv 5 \pmod{11}$.

Solution. $O_{11}(2) \mid \varphi(11) = 10$, so $O_{11}(2) \in \{1, 2, 5, 10\}$. Since $2^1 = 2$, $2^2 = 4$ and $2^5 = 32 \equiv -1 \pmod{11}$, $O_{11}(2) = 10$, and 2 is a primitive root modulo 11. Taking indices, Theorem 2.3 gives $\operatorname{ind}_2(7x^8) \equiv \operatorname{ind}_2 5 \pmod{10}$, that is, $\operatorname{ind}_2 7 + 8 \operatorname{ind}_2 x \equiv \operatorname{ind}_2 5 \pmod{10}$. The tables of powers and indices are:

a	1	2	3	4	5	6	7	8	9	10
2^a	2	4	8	5	10	9	7	3	6	1
$\operatorname{ind}_2 a$	10	1	8	2	4	9	7	3	6	5

So $7 + 8 \operatorname{ind}_2 x \equiv 4 \pmod{10}$, that is, $8 \operatorname{ind}_2 x \equiv 7 \pmod{10}$. Since $(8, 10) = 2 \nmid 7$, there is no solution (Remark 3.2). ◊

Example 10.2. Solve $x^3 - x^2 - 1 \equiv 0 \pmod{27}$.

Solution. Here $27 = 3^3$. Let $f(x) = x^3 - x^2 - 1$. Modulo 3, $f(0) \equiv f(1) \equiv -1$ and $f(2) = 3 \equiv 0$, so $x \equiv -1 \pmod{3}$ is the only solution. Since $f'(-1) = 5 \not\equiv 0 \pmod{3}$, Theorem 6.2 gives exactly one solution modulo 27.

Write $x = 3b - 1$. Modulo 9,

$$(3b)^3 - 3(3b)^2 + 3(3b) - 1 - (3b)^2 + 2 \cdot 3b - 1 - 1 \equiv 6b - 3 \pmod{9},$$

and $6b - 3 \equiv 0 \pmod{9}$ if and only if $2b - 1 \equiv 0 \pmod{3}$, that is, $b \equiv 2 \pmod{3}$. So $b = 3c + 2$ and $x = 3(3c + 2) - 1 = 9c + 5$. Modulo 27,

$$(9c)^3 + 3 \cdot 5(9c)^2 + 3 \cdot 5^2(9c) + 5^3 - (9c)^2 - 2 \cdot 5(9c) - 5^2 - 1 \equiv 585c + 99 \equiv 18c + 18 \pmod{27},$$

and $18c + 18 \equiv 0 \pmod{27}$ if and only if $c \equiv -1 \pmod{3}$. Hence $x \equiv 9 \cdot (-1) + 5 = -4 \pmod{27}$. Equivalently, the solution is found digit by digit in the form $x = a + 3b + 3^2c$. ◊

Example 10.3. Let $p > 2$ be a prime. Show that $(p - 1)! \equiv -1 \pmod{p}$.

Solution. Let g be a primitive root modulo p. Both $1, 2, \ldots, p - 1$ and, by Theorem 1.17, $1, g, g^2, \ldots, g^{p-2}$ form reduced residue systems modulo p, so

$$(p - 1)! \equiv g^{1+2+\cdots+(p-2)} = g^{(p-2)(p-1)/2} = \left(g^{(p-1)/2}\right)^{p-2} \equiv (-1)^{p-2} = -1 \pmod{p}.$$

Here $g^{(p-1)/2} \equiv -1$ by Proposition 3.6, since $g^{(p-1)/2} \not\equiv 1$ because $O_p(g) = p - 1$; and $p - 2$ is odd. ◊

Example 10.4. Let $p > 2$ be a prime and g a primitive root modulo p. Show that g is a quadratic non-residue modulo p.

Solution. If g were a quadratic residue, then $g^{(p-1)/2} \equiv 1 \pmod{p}$ by Proposition 3.6, contradicting $O_p(g) = p-1$. ◊

Example 10.5. Decide whether $x^2 \equiv 66 \pmod{101}$, $x^2 \equiv 7! \pmod{83}$ and $x^2 \equiv 30 \pmod{77}$ are solvable.

Solution. Using $101 \equiv 1 \pmod{4}$, $101 \equiv -3 \pmod{8}$, $101 = 33 \cdot 3 + 2 = 9 \cdot 11 + 2$:

$$\left(\frac{66}{101}\right) = \left(\frac{2 \cdot 3 \cdot 11}{101}\right) = \left(\frac{2}{101}\right)\left(\frac{3}{101}\right)\left(\frac{11}{101}\right)$$
$$= -\left(\frac{101}{3}\right)\left(\frac{101}{11}\right) = -\left(\frac{2}{3}\right)\left(\frac{2}{11}\right) = -(-1)(-1) = -1,$$

so $x^2 \equiv 66 \pmod{101}$ has no solution.

Next, $7! = 1 \cdot 2 \cdot \cdots \cdot 7$ and $\left(\frac{4}{83}\right) = 1$, $\left(\frac{6}{83}\right) = \left(\frac{2}{83}\right)\left(\frac{3}{83}\right)$, so

$$\left(\frac{7!}{83}\right) = \left(\frac{1}{83}\right)\left(\frac{2}{83}\right)\cdots\left(\frac{7}{83}\right) = \left(\frac{5}{83}\right)\left(\frac{7}{83}\right)$$
$$= \left(\frac{83}{5}\right) \cdot \left(-\left(\frac{83}{7}\right)\right) = -\left(\frac{3}{5}\right)\left(\frac{6}{7}\right) = -\left(\frac{3}{5}\right)\left(\frac{-1}{7}\right) = -(-1)(-1) = -1,$$

using $5 \equiv 1 \pmod{4}$, $7 \equiv 83 \equiv -1 \pmod{4}$, $83 = 16 \cdot 5 + 3 = 11 \cdot 7 + 6$, $\left(\frac{3}{5}\right) = -1$ and $\left(\frac{-1}{7}\right) = -1$. So $x^2 \equiv 7! \pmod{83}$ has no solution.

Finally $77 = 7 \cdot 11$, and we check the two primes separately. Modulo 7,

$$\left(\frac{30}{7}\right) = \left(\frac{2 \cdot 3 \cdot 5}{7}\right) = \left(\frac{2}{7}\right)\left(\frac{3}{7}\right)\left(\frac{5}{7}\right)$$
$$= 1 \cdot \left(-\left(\frac{7}{3}\right)\right)\left(\frac{7}{5}\right) = -\left(\frac{1}{3}\right)\left(\frac{2}{5}\right) = -1 \cdot (-1) = 1,$$

so $x^2 \equiv 30 \pmod{7}$ is solvable. Modulo 11, $\left(\frac{30}{11}\right) = \left(\frac{8}{11}\right) = \left(\frac{2}{11}\right)^3 = (-1)^3 = -1$, so $x^2 \equiv 30 \pmod{11}$ is not solvable. A solution modulo 77 would be a solution modulo 11, so $x^2 \equiv 30 \pmod{77}$ has no solution. ◊

Example 10.6. Compute $\left(\frac{1234567}{225}\right)$ and $\left(\frac{589}{1999}\right)$.

Solution. Since 1234567 has digit sum 28 and last digit 7, $(1234567, 15) = 1$. By Proposition 5.3 5),

$$\left(\frac{1234567}{225}\right) = \left(\frac{1234567}{15^2}\right) = \left(\frac{1234567}{15}\right)^2 = 1.$$

Using $589 \equiv 1 \pmod{4}$, $1999 = 3 \cdot 589 + 232$, $232 = 2^3 \cdot 29$, $589 \equiv 5 \pmod{8}$, $29 \equiv 1 \pmod{4}$ and $589 = 20 \cdot 29 + 9$,

$$\left(\frac{589}{1999}\right) = \left(\frac{1999}{589}\right) = \left(\frac{232}{589}\right) = \left(\frac{2^3 \cdot 29}{589}\right) = \left(\frac{2}{589}\right)^3 \left(\frac{29}{589}\right) = (-1)^3 \left(\frac{29}{589}\right)$$
$$= -\left(\frac{589}{29}\right) = -\left(\frac{9}{29}\right) = -\left(\frac{3}{29}\right)^2 = -1.$$

◊

Definition 10.7. An arithmetic function f is *totally multiplicative* if $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$. The *von Mangoldt function* is $\Lambda(n) = \log p$ if $n = p^k$ for a prime p and $k \in \mathbb{N}$, and $\Lambda(n) = 0$ otherwise. The *Liouville function* is $\lambda(n) = (-1)^{\Omega(n)}$. For a statement P, $\mathbb{1}\{P\}$ is 1 if P holds and 0 otherwise.

Example 10.8. Show that $\sum_{d|n} \Lambda(d) = \log n$ for $n \in \mathbb{N}$.

Solution. Let $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$. Only the prime-power divisors p_i^j with $1 \le j \le \alpha_i$ contribute, so

$$\sum_{d|n} \Lambda(d) = \sum_{p^\alpha | n} \Lambda(p^\alpha) = \sum_{i=1}^{r} \sum_{j=1}^{\alpha_i} \Lambda\left(p_i^j\right) = \sum_{i=1}^{r} \sum_{j=1}^{\alpha_i} \log p_i$$
$$= \sum_{i=1}^{r} \alpha_i \log p_i = \sum_{i=1}^{r} \log\left(p_i^{\alpha_i}\right) = \log\left(p_1^{\alpha_1} \cdots p_r^{\alpha_r}\right) = \log n.$$

For $n = 1$ both sides are 0. ◊

Example 10.9. Show that $\Lambda(n) = -\sum_{d|n} \mu(d) \log d$.

Solution. By the previous example and Theorem 7.12,

$$\Lambda(n) = \sum_{d|n} \mu(d) \log\left(\frac{n}{d}\right) = \sum_{d|n} \mu(d)(\log n - \log d)$$
$$= \log n \sum_{d|n} \mu(d) - \sum_{d|n} \mu(d) \log d$$
$$= \log n \cdot \mathbb{1}\{n = 1\} - \sum_{d|n} \mu(d) \log d = -\sum_{d|n} \mu(d) \log d,$$

using Proposition 7.11 and $\log 1 = 0$. ◊

Example 10.10. Show that $\sum_{d|n} \lambda(d) = \mathbb{1}\{n \text{ is a square}\}$.

Solution. Since $\Omega(mn) = \Omega(m) + \Omega(n)$ for all $m, n \in \mathbb{N}$, $\lambda(mn) = (-1)^{\Omega(m)+\Omega(n)} = \lambda(m)\lambda(n)$, so λ is totally multiplicative, in particular multiplicative. By Theorem 7.8 its summation function F is multiplicative. For a prime power,

$$F(p^\alpha) = \sum_{d|p^\alpha} \lambda(d) = \lambda(1) + \lambda(p) + \cdots + \lambda(p^\alpha) = 1 - 1 + (-1)^2 + (-1)^3 + \cdots + (-1)^\alpha = \mathbb{1}\{2 \mid \alpha\}.$$

Hence $F(n) = F(p_1^{\alpha_1}) \cdots F(p_r^{\alpha_r}) = \mathbb{1}\{\text{every } \alpha_i \text{ is even}\} = \mathbb{1}\{n \text{ is a square}\}$. ◊

11. Geometry of $\mathbb{Q}_p$

Lemma 11.1. Let F be a field with a non-archimedean absolute value $|\cdot|$, and let $x, y \in F$ with $|x| \neq |y|$. Then $|x+y| = \max\{|x|, |y|\}$.

Proof. Assume $|x| > |y|$. Since $|\cdot|$ is non-archimedean, $|x+y| \leq \max\{|x|, |y|\} = |x|$. Also $x = (x+y) - y$, so $|x| \leq \max\{|x+y|, |-y|\} = \max\{|x+y|, |y|\}$. Since $|x| > |y|$, the maximum is $|x+y|$, and $|x| \leq |x+y|$. Hence $|x+y| = |x|$. ∎

Definition 11.2. In a metric space (X, d), a *triangle* is a triple of distinct points a, b, c, with side lengths $d(a,b)$, $d(a,c)$, $d(b,c)$. It is *isosceles* if two of its side lengths are equal. Three distinct points are *collinear* if, after relabelling, $d(a,c) = d(a,b) + d(b,c)$, and they form a *right triangle* if, after relabelling, $d(a,c)^2 = d(a,b)^2 + d(b,c)^2$. We write $d_p(x,y) = |x-y|_p$ on $\mathbb{Q}_p$.

Theorem 11.3. Every triangle in $\mathbb{Q}_p$ is isosceles. More precisely, its two longest sides have equal length.

Proof. Let $a, b, c \in \mathbb{Q}_p$ be distinct. If $d_p(a,b) = d_p(b,c)$, the triangle is isosceles. Otherwise $|a-b|_p \neq |b-c|_p$, and since $(a-b) + (b-c) = a-c$, Lemma 11.1 gives $|a-c|_p = \max\{|a-b|_p, |b-c|_p\}$. So $|a-c|_p$ equals $|a-b|_p$ or $|b-c|_p$, namely the larger of the two. In both cases the two longest sides are equal. ∎

Theorem 11.4. No three distinct points of $\mathbb{Q}_p$ are collinear.

Proof. Suppose $a, b, c \in \mathbb{Q}_p$ are distinct and $d_p(a,c) = d_p(a,b) + d_p(b,c)$. Since $d_p(a,b), d_p(b,c) > 0$, we get $d_p(a,c) > d_p(a,b), d_p(b,c)$. So the side ac is strictly the longest, which contradicts Theorem 11.3. ∎

Theorem 11.5. There are no right triangles in $\mathbb{Q}_p$.

Proof. Suppose $a, b, c \in \mathbb{Q}_p$ form a right triangle with $d_p(a,c)^2 = d_p(a,b)^2 + d_p(b,c)^2$. Again $d_p(a,c) > d_p(a,b), d_p(b,c)$. Since the triangle is isosceles, $d_p(a,b) = d_p(b,c)$, so ac is the base, it is the longest side, and $d_p(a,c) = \sqrt{2}\, d_p(a,b)$. By Theorem 11.3, however, the base of a triangle in $\mathbb{Q}_p$ that is not equilateral is its shortest side, a contradiction. ∎

12. Sequences and series in $\mathbb{Q}_p$

Theorem 12.1. A sequence (a_n) in $\mathbb{Q}_p$ is a Cauchy sequence if and only if for every $\varepsilon > 0$ there is $N \in \mathbb{N}$ with $|a_{n+1} - a_n|_p < \varepsilon$ for all $n \geq N$.

Proof. If (a_n) is Cauchy and $\varepsilon > 0$, there is $N \in \mathbb{N}$ with $|a_n - a_m|_p < \varepsilon$ for all $n, m \geq N$. For $n \geq N$ also $n+1 \geq N$, so $|a_{n+1} - a_n|_p < \varepsilon$.

Conversely, let $\varepsilon > 0$ and choose N with $|a_{n+1} - a_n|_p < \varepsilon$ for all $n \geq N$. Let $m, n \geq N$. If $n = m$, then $|a_n - a_m|_p = 0 < \varepsilon$. Otherwise assume $n > m$ and write $n = m + l$ with $l \in \mathbb{N}$. Then

$$|a_n - a_m|_p = |(a_{m+l} - a_{m+l-1}) + (a_{m+l-1} - a_{m+l-2}) + \cdots + (a_{m+1} - a_m)|_p.$$

Since the p-adic absolute value is non-archimedean,

$$|a_n - a_m|_p \le \max\{|a_{m+i} - a_{m+i-1}|_p : 1 \le i \le l\}.$$

The maximum is attained at some $1 \le i_0 \le l$, and $m + i_0 - 1 \ge N$, so $|a_n - a_m|_p \le |a_{m+i_0} - a_{m+i_0-1}|_p < \varepsilon$. ∎

Corollary 12.2. Let $(a_n)_{n\ge 0}$ be a sequence in $\mathbb{Q}_p$. Then $\sum_{n=0}^{\infty} a_n$ converges if and only if $\lim_{n\to\infty} a_n = 0$.

Proof. Let $S_n = \sum_{i=0}^{n} a_i$ be the partial sums. If the series converges, then (S_n) converges and hence is Cauchy. Let $\varepsilon > 0$; there is $N \in \mathbb{N}$ with $|S_{n+1} - S_n|_p < \varepsilon$ for all $n \ge N$. Since $|S_{n+1} - S_n|_p = |a_{n+1}|_p$, we get $|a_n|_p < \varepsilon$ for all $n > N$.

Conversely, let $\lim_{n\to\infty} a_n = 0$ and $\varepsilon > 0$. There is $N \in \mathbb{N}$ with $|a_n|_p < \varepsilon$ for all $n > N$, hence $|S_{n+1} - S_n|_p = |a_{n+1}|_p < \varepsilon$ for all $n \ge N$. By Theorem 12.1, (S_n) is Cauchy, and it converges because $\mathbb{Q}_p$ is complete (Theorem 9.12). ∎

Proposition 12.3. Every series $\sum_{n=n_0}^{\infty} a_n p^n$ with $n_0 \in \mathbb{Z}$ and $a_n \in \{0, 1, \ldots, p-1\}$ converges in $\mathbb{Q}_p$, and its sum is the p-adic number with digits a_n.

Proof. $|a_n p^n|_p \le p^{-n} \to 0$, so the series converges by Corollary 12.2. Its partial sums are the rational numbers s_n of part 3) of the proof of Theorem 9.12, which converge to the p-adic number with digits a_n. ∎

13. Topology on $\mathbb{Q}_p$

The p-adic metric induces a topology on $\mathbb{Q}_p$. For $a \in \mathbb{Q}_p$ and $r > 0$ write $B(a, r) = \{x \in \mathbb{Q}_p : |x - a|_p < r\}$ for the open ball and $\overline{B}(a, r) = \{x \in \mathbb{Q}_p : |x - a|_p \le r\}$ for the closed ball. As in every metric space, open balls are open and closed balls are closed. The values of $|\cdot|_p$ on $\mathbb{Q}_p$ are 0 and the powers p^k, $k \in \mathbb{Z}$.

Remark 13.1 (cited)**.** $\mathbb{Q}_p$ admits no ordering compatible with its field operations.

Theorem 13.2. In $\mathbb{Q}_p$ every open ball is closed, and every closed ball of positive radius is open.

Proof. Let $B(a, r)$ be an open ball and let n be the largest integer with $r \le p^{-n}$, so $p^{-(n+1)} < r \le p^{-n}$. No p-adic number has absolute value strictly between $p^{-(n+1)}$ and p^{-n}, hence

$$B(a, r) = \{x \in \mathbb{Q}_p : |x - a|_p < p^{-n}\} = \{x \in \mathbb{Q}_p : |x - a|_p \le p^{-(n+1)}\} = \overline{B}(a, p^{-(n+1)}),$$

which is closed.

Let $\overline{B}(a, r)$ be a closed ball with $r > 0$ and $x \in \overline{B}(a, r)$. For $y \in B(x, r)$, $|y - a|_p \le \max\{|y - x|_p, |x - a|_p\} \le r$, so $B(x, r) \subseteq \overline{B}(a, r)$. Hence $\overline{B}(a, r)$ is open. ∎

Corollary 13.3. $\mathbb{Z}_p$ is both closed and open in $\mathbb{Q}_p$.

Proof. By Theorem 9.12, $\mathbb{Z}_p = \{x \in \mathbb{Q}_p : |x|_p \leq 1\} = \overline{B}(0,1)$ is a closed ball, hence closed. By the theorem, it is open as well. ∎

Corollary 13.4. $\mathbb{Q}_p$ is totally disconnected, that is, its only connected subsets are the empty set and the singletons.

Proof. Let $A \subseteq \mathbb{Q}_p$ contain distinct points a, b and put $r = |a-b|_p$. The ball $U = B(a,r)$ is open and closed, contains a and not b. Hence $A = (A \cap U) \cup (A \setminus U)$ is a decomposition into two disjoint nonempty sets that are open in A, and A is not connected. ∎

Theorem 13.5. $\mathbb{Z}_p$ is compact.

Proof. $\mathbb{Z}_p$ is a metric space, so it suffices to show that every sequence (x_n) in $\mathbb{Z}_p$ has a convergent subsequence. Infinitely many x_n share the same first digit; among these, infinitely many share the same second digit; continuing, we obtain infinite index sets $I_1 \supseteq I_2 \supseteq \cdots$ such that all x_n with $n \in I_k$ have the same truncation $(x_n)_k$. Choose $n_1 < n_2 < \cdots$ with $n_k \in I_k$. For $j, l \geq k$ the truncations $\left(x_{n_j}\right)_k$ and $\left(x_{n_l}\right)_k$ agree, so $|x_{n_j} - x_{n_l}|_p \leq p^{-k}$, and $\left(x_{n_k}\right)$ is Cauchy. It converges in $\mathbb{Q}_p$ by Theorem 9.12, and its limit lies in $\mathbb{Z}_p$, which is closed. ∎

14. Finite fields

Definition 14.1 (Finite field). A *finite field* is a field F with $|F| < \infty$. The *characteristic* $\operatorname{char} F$ is the least $n \in \mathbb{N}$ with $n \cdot 1 = 0$ in F. A finite field with q elements is denoted $\mathbb{F}_q$, and $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ for a prime p.

Theorem 14.2. Let F be a finite field. Then $\operatorname{char} F$ is a prime p, and $|F| = p^n$ for some $n \in \mathbb{N}$. The multiplicative group $F^* = F^\times = F \setminus \{0\}$ is cyclic, $F^\times = \langle g \rangle$; a generator g is called a *primitive element* of F. Consequently $a^q = a$ for every $a \in \mathbb{F}_q$.

Proof. Since F is finite, the elements $n \cdot 1$ are not pairwise distinct, so $\operatorname{char} F$ exists. If $\operatorname{char} F = ab$ with $1 < a, b$, then $(a \cdot 1)(b \cdot 1) = 0$ with both factors nonzero, which is impossible in a field; so $\operatorname{char} F = p$ is prime. Then $\{n \cdot 1 : n \in \mathbb{Z}\}$ is a subfield isomorphic to $\mathbb{F}_p$, and F is a finite-dimensional vector space over it, of dimension n say, so $|F| = p^n$.

Let $|F| = q$. The proof of Theorem 1.8 in the case $m = p$ applies to $F^\times$, which has order $q-1$: every element has an order dividing $q-1$, a polynomial $x^d - 1$ over the field F has at most d roots, and $\sum_{d \mid q-1} \varphi(d) = q-1$ by Theorem 7.9. Hence $F^\times$ has $\varphi(q-1) \geq 1$ elements of order $q-1$, and it is cyclic. Finally $a^{q-1} = 1$ for $a \neq 0$, so $a^q = a$ for every $a \in F$. ∎

Lemma 14.3. Let p be a prime. Then $p \mid \binom{p}{j}$ for all $0 < j < p$.

Proof. We have $\binom{p}{j} = p!/((p-j)!j!)$, and

$$\frac{p}{j}\binom{p-1}{j-1} = \frac{p}{j}\cdot\frac{(p-1)!}{(j-1)!(p-j)!} = \frac{p!}{(p-j)!j!} = \binom{p}{j}.$$

Hence $j\binom{p}{j} = p\binom{p-1}{j-1}$, so $p \mid j\binom{p}{j}$. Since $(p,j)=1$, $p \mid \binom{p}{j}$. ∎

Theorem 14.4. Let $\mathbb{F}_q$ be a finite field with $\operatorname{char} \mathbb{F}_q = p$. Then $\varphi : \mathbb{F}_q \to \mathbb{F}_q$, $\varphi(a) = a^p$, is a ring homomorphism.

Proof. Let $a, b \in \mathbb{F}_q$; $\mathbb{F}_q$ is commutative. Then $\varphi(ab) = (ab)^p = a^p b^p = \varphi(a)\varphi(b)$, and by the binomial theorem and the lemma,

$$\varphi(a+b) = (a+b)^p = a^p + \sum_{j=1}^{p-1}\binom{p}{j}a^j b^{p-j} + b^p = a^p + b^p = \varphi(a) + \varphi(b).$$

From $\varphi(a-b) + \varphi(b) = \varphi(a)$ we get $\varphi(a-b) = a^p - b^p = \varphi(a) - \varphi(b)$. ∎

Remark 14.5 (Recall). Let $f \in \mathbb{F}_q[x]$ be irreducible of degree m. Then $\mathbb{F}_q[x]/(f)$ is a field containing a copy of $\mathbb{F}_q$, with q^m elements, in which the class of x is a root of f. If $K \supseteq \mathbb{F}_q$ is a field and $\alpha \in K$ is a root of f, then $\mathbb{F}_q(\alpha) = \{g(\alpha) : g \in \mathbb{F}_q[x], \deg g < m\}$ has exactly q^m elements. We write $\mathbb{F}_{q^m}$ for a field with q^m elements that contains $\mathbb{F}_q$.

Lemma 14.6. Let $f \in \mathbb{F}_q[x]$ be irreducible with $\deg f = m$. Then f has a root α in some $\mathbb{F}_{q^m}$, and for every such root the roots of f in $\mathbb{F}_{q^m}$ are exactly the m pairwise distinct elements $\alpha, \alpha^q, \alpha^{q^2}, \ldots, \alpha^{q^{m-1}}$.

Proof. A root exists in $\mathbb{F}_q[x]/(f)$ by Remark 14.5. Let $\beta \in \mathbb{F}_{q^m}$ be a root of f. We claim $f(\beta^q) = 0$. Let $f(x) = \sum_{i=0}^{m} a_i x^i$. Since $a_i \in \mathbb{F}_q$, $a_i^q = a_i$ by Theorem 14.2, and $x \mapsto x^q$ is a ring homomorphism, being an iterate of the map in Theorem 14.4 because q is a power of p. Hence

$$\begin{aligned} f(\beta^q) &= a_0 + a_1\beta^q + a_2(\beta^2)^q + \cdots + a_m(\beta^m)^q \\ &= a_0^q + (a_1\beta)^q + \cdots + (a_m\beta^m)^q \\ &= (a_0 + a_1\beta + \cdots + a_m\beta^m)^q = f(\beta)^q = 0. \end{aligned}$$

It follows that $\alpha, \alpha^q, (\alpha^q)^q, \ldots, \alpha^{q^{m-1}}$ are roots of f.

Suppose $\alpha^{q^i} = \alpha^{q^j}$ with $0 \le i < j \le m-1$. Raising to the power q^{m-j} gives $\alpha^{q^{m-j+i}} = \alpha^{q^m} = \alpha$, by Theorem 14.2 applied to $\mathbb{F}_{q^m}$. So α is a root of $x^{q^k} - x$ with $k = m - j + i$ and $1 \le k < m$. The set $L = \{\beta \in \mathbb{F}_{q^m} : \beta^{q^k} = \beta\}$ is a subfield, because $\beta \mapsto \beta^{q^k}$ is a ring homomorphism and $(\beta^{-1})^{q^k} = \left(\beta^{q^k}\right)^{-1}$. It contains $\mathbb{F}_q$ and α, hence $\mathbb{F}_q(\alpha)$, which has q^m elements by Remark 14.5. But L consists of roots of $x^{q^k} - x$, so $|L| \le q^k < q^m$, a contradiction, since $f(\alpha) = 0$, $\deg f = m$ and f is irreducible. Hence the m elements are pairwise distinct. A polynomial of degree m has at most m roots in a field, so these are all the roots of f in $\mathbb{F}_{q^m}$, and f splits completely over $\mathbb{F}_{q^m}$. ∎

Definition 14.7 (Conjugates). Let $\mathbb{F}_q \subseteq \mathbb{F}_{q^m}$ and $\alpha \in \mathbb{F}_{q^m}$. The elements $\alpha, \alpha^q, \alpha^{q^2}, \ldots, \alpha^{q^{m-1}}$ are called the *conjugates* of α with respect to $\mathbb{F}_q$.

Example 14.8. For fields $K \subseteq L$, the *degree* $[L : K]$ is the dimension of L as a vector space over K, and the *minimal polynomial* m_α of $\alpha \in L$ is the monic polynomial of least degree over K with root α. For $\mathbb{R} \subset \mathbb{C} = \mathbb{R}(i)$ we have $[\mathbb{C} : \mathbb{R}] = 2$, $m_i(x) = x^2 + 1$ and $\mathbb{C} \cong \mathbb{R}[x]/(x^2 + 1)$. The roots of $x^2 + 1$ are i and $-i$. Complex conjugation $\overline{a + bi} = a - bi$ satisfies $\bar{i} = -i$, so it interchanges the roots of the irreducible polynomial $x^2 + 1$.

Definition 14.9 (Automorphism over $\mathbb{F}_q$). An *automorphism of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$* is an isomorphism of $\mathbb{F}_{q^m}$ onto itself that leaves every element of $\mathbb{F}_q$ invariant.

Remark 14.10. Such a map is also called an *$\mathbb{F}_q$-automorphism*.

Example 14.11. For $\mathbb{R} \subset \mathbb{C}$, conjugation $z \mapsto \overline{z}$ fixes the real numbers, so it is an $\mathbb{R}$-automorphism of $\mathbb{C}$.

15. Worked exercises on p-adic numbers

Lemma 15.1. Let $x \in \mathbb{Q}_p$ with $|x|_p < 1$. Then $\sum_{n=0}^{\infty} x^n$ converges in $\mathbb{Q}_p$ and its sum is $1/(1 - x)$.

Proof. By Lemma 11.1, $|1 - x|_p = \max\{1, |x|_p\} = 1$, so $1 - x \neq 0$. The partial sums are $S_N = \sum_{n=0}^{N} x^n = (1 - x^{N+1})/(1 - x)$, hence

$$|S_N - \frac{1}{1-x}|_p = \frac{|x|_p^{N+1}}{|1-x|_p} = |x|_p^{N+1} \to 0.$$

■

Remark 15.2. For $x \in \mathbb{Q}_p$ we write $v_p(x) = v(x)$, with v as in Theorem 9.12. By part 1) of that theorem this extends the p-adic valuation of $\mathbb{Q}$, and $|x|_p = p^{-v_p(x)}$ for all $x \neq 0$.

Example 15.3. Find the p-adic expansions of $2/3$ in $\mathbb{Q}_2$, of $-1/6$ in $\mathbb{Q}_7$ and of $1/6$ in $\mathbb{Q}_3$.

Solution. **$2/3$ in $\mathbb{Q}_2$.** Since $|-2|_2 = 1/2 < 1$, Lemma 15.1 gives $\frac{1}{3} = \frac{1}{1-(-2)} = \sum_{n=0}^{\infty} (-2)^n$. This is a convergent series but not an expansion, since the digits must be 0 and 1. Similarly

$$-1 = \frac{1}{1-2} = \sum_{n=0}^{\infty} 2^n, \quad -\frac{1}{3} = \frac{1}{1-4} = \sum_{n=0}^{\infty} 4^n = 1 + 2^2 + 2^4 + \cdots,$$

and the latter is an expansion. Hence

$$\frac{2}{3} = -2 \cdot \left(-\frac{1}{3}\right) = -2 \sum_{n=0}^{\infty} 4^n = -(2 + 2^3 + 2^5 + \cdots).$$

By Example 15.5 with $p = 2$, lowest digit $a_1 = 1$ and $a_n = 1$ exactly for odd n, the negation has digits $2 - a_1 = 1$ at 2^1 and $1 - a_n$ at 2^n for $n \geq 2$:

$$\frac{2}{3} = 2 + 2^2 + 2^4 + 2^6 + \cdots.$$

Alternatively, $2/3 \in \mathbb{Z}_2$ by Remark 8.8, and its truncations x_k satisfy $3x_k \equiv 2 \pmod{2^k}$, so they are found by solving congruences. If $3x \equiv 2 \pmod 2$, then $x \equiv 0 \pmod 2$ and $x = 2y$. Then $3(2y) \equiv 2 \pmod 4$ gives $2y \equiv 2 \pmod 4$, so $y \equiv 1 \pmod 2$, $y = 2z + 1$ and $x = 4z + 2$. Then $3(4z + 2) \equiv 2 \pmod 8$ gives $4z + 6 \equiv 2 \pmod 8$, that is, $2z + 3 \equiv 1 \pmod 4$, so $z \equiv 1 \pmod 2$, $z = 2w + 1$ and

$$x = 4(2w + 1) + 2 = 8w + 6.$$

So $x_3 = 6 = 0 + 1 \cdot 2 + 1 \cdot 2^2$, in agreement with the first three digits above.

$-1/6$ in $\mathbb{Q}_7$. By Lemma 15.1, $-\frac{1}{6} = \frac{1}{1-7} = \sum_{n=0}^{\infty} 7^n = 1 + 7 + 7^2 + \cdots$, which is an expansion. Alternatively, the truncations satisfy $6x_k \equiv -1 \pmod{7^k}$. Modulo 7, $6x \equiv -1$ gives $x \equiv 1 \pmod 7$, so $x = 7y + 1$. Then $6(7y + 1) \equiv -1 \pmod{49}$ gives $42y + 6 \equiv -1 \pmod{49}$, that is, $6y + 1 \equiv 0 \pmod 7$, so $y \equiv 1 \pmod 7$ and $x_2 = 1 + 1 \cdot 7$.

$1/6$ in $\mathbb{Q}_3$. We have $\frac{1}{6} = \frac{1}{2} \cdot \frac{1}{3}$. The formula $\frac{1}{2} = \frac{1}{1-(-1)}$ does not help: $|-1|_3 = 1$, so the terms of $\sum (-1)^n$ do not tend to 0 and the series diverges by Corollary 12.2. Instead,

$$\frac{1}{2} = \frac{2}{4} = \frac{2}{1-(-3)} = 2\sum_{n=0}^{\infty} (-3)^n.$$

The digits must be $0, 1, 2$, and the terms with odd n have coefficient -2. Using $-2 \cdot 3^{2k+1} = (1 - 3) \cdot 3^{2k+1} = 3^{2k+1} - 3^{2k+2}$,

$$2(-3)^{2k+1} + 2(-3)^{2k+2} = 3^{2k+1} - 3^{2k+2} + 2 \cdot 3^{2k+2} = 3^{2k+1} + 3^{2k+2}.$$

Grouping the terms in consecutive pairs after the first replaces the sequence of partial sums by a subsequence, which has the same limit. Hence

$$\frac{1}{2} = 2 + \sum_{k=0}^{\infty} (3^{2k+1} + 3^{2k+2}) = 2 + 3 + 3^2 + 3^3 + \cdots,$$

and multiplying by 3^{-1} shifts the digits:

$$\frac{1}{6} = 2 \cdot 3^{-1} + 1 + 3 + 3^2 + \cdots.$$

◊

Example 15.4. Let $a = a_0 + a_1 p + a_2 p^2 + \cdots \in \mathbb{Z}_p$. Show that a is a unit of $\mathbb{Z}_p$ if and only if $a_0 \neq 0$, and that this holds if and only if $|a|_p = 1$.

Solution. The first equivalence is part of Proposition 8.10. For $a \in \mathbb{Z}_p$ we have $v_p(a) \geq 0$, and $v_p(a) = 0$ holds if and only if $a_0 \neq 0$. Hence $a_0 \neq 0$ if and only if $|a|_p = p^0 = 1$. ◊

Example 15.5. Let $0 \neq a \in \mathbb{Q}_p$ with expansion $a = \sum_{n=-m}^{\infty} a_n p^n$, where $m \in \mathbb{Z}$ and $a_{-m} \neq 0$. Find the expansion of $-a$.

Solution. Write $-a = \sum_{n=-m}^{\infty} b_n p^n$; its lowest index is at least $-m$ because $v_p(-a) = v_p(a)$. Then $a + (-a) = 0$. Multiplying by p^m,

$$\alpha = p^m a = a_{-m} + a_{-m+1}p + \cdots + a_0 p^m + a_1 p^{m+1} + \cdots, \quad \beta = p^m(-a) = b_{-m} + b_{-m+1}p + \cdots,$$

with $\alpha, \beta \in \mathbb{Z}_p$ and $\alpha + \beta = 0$. So the truncations satisfy $\alpha_k + \beta_k \equiv 0 \pmod{p^k}$ for every $k \in \mathbb{N}$.

For $k = 1$: $a_{-m} + b_{-m} \equiv 0 \pmod{p}$, and $1 \le a_{-m} + b_{-m} \le 2p - 1$, so $a_{-m} + b_{-m} = p$ and $b_{-m} = p - a_{-m}$. Hence $\alpha_1 + \beta_1 = p$.

Continuing the process, suppose $\alpha_k + \beta_k = p^k$. Then

$$\alpha_{k+1} + \beta_{k+1} = p^k + (a_{k-m} + b_{k-m})p^k \equiv 0 \pmod{p^{k+1}},$$

so $1 + a_{k-m} + b_{k-m} \equiv 0 \pmod{p}$. Since $1 \le 1 + a_{k-m} + b_{k-m} \le 2p - 1$, this gives $1 + a_{k-m} + b_{k-m} = p$, that is, $b_{k-m} = p - 1 - a_{k-m}$, and $\alpha_{k+1} + \beta_{k+1} = p^{k+1}$. By induction,

$$-a = (p - a_{-m})p^{-m} + (p - 1 - a_{-m+1})p^{-m+1} + \cdots + (p - 1 - a_0) + (p - 1 - a_1)p + \cdots.$$

◊

Example 15.6 (Legendre's formula). Let $n \in \mathbb{N}$ and let p be a prime. Show that

$$v_p(n!) = \sum_{i=1}^{\infty} \left\lfloor \frac{n}{p^i} \right\rfloor,$$

where the sum is finite because its terms vanish for $p^i > n$. Consequently $n! = \prod_{p \le n} p^{\alpha_p}$ with $\alpha_p = \sum_{i \ge 1} \lfloor n/p^i \rfloor$.

Solution. Since $n! = 1 \cdot 2 \cdot \cdots \cdot n$, Proposition 9.5 gives $v_p(n!) = \sum_{j=1}^{n} v_p(j)$, and $v_p(j)$ is the number of $i \ge 1$ with $p^i \mid j$. Exchanging the order of summation,

$$v_p(n!) = \sum_{j=1}^{n} \sum_{i \ge 1,\ p^i \mid j} 1 = \sum_{i \ge 1} \#\{1 \le j \le n : p^i \mid j\}.$$

The factors divisible by p are $p, 2p, \ldots, tp$, where $tp \le n < (t+1)p$, so there are $t = \lfloor n/p \rfloor$ of them. In the same way, exactly $\lfloor n/p^2 \rfloor$ factors are divisible by p^2, and $\lfloor n/p^i \rfloor$ by p^i. The prime divisors of $n!$ are the primes $p \le n$, which gives the factorization. ◊

Example 15.7. Show that $x^2 = 6$ has a solution in $\mathbb{Q}_5$.

Solution. $x^2 \equiv 6 \pmod{5}$ has the solution $x \equiv 1 \pmod{5}$. We prove by induction that there are integers x_n, $n \ge 1$, with $x_n^2 \equiv 6 \pmod{5^n}$, $x_1 = 1$ and $x_{n+1} \equiv x_n \pmod{5^n}$. Given x_n with $x_n \equiv 1 \pmod{5}$, write $x_n^2 = 5^n s + 6$ and $x_{n+1} = x_n + t5^n$ with $t \in \mathbb{Z}$. Then

$$x_{n+1}^2 = x_n^2 + 2x_n t 5^n + t^2 5^{2n} \equiv 6 \pmod{5^{n+1}}$$

holds if and only if

$$2x_n t + \frac{x_n^2 - 6}{5^n} \equiv 0 \pmod{5}.$$

Since $(2x_n, 5) = 1$, this linear congruence has a solution t (Remark 3.2), which is the lifting step of Theorem 6.2. Then $x_{n+1} \equiv x_n \equiv 1 \pmod 5$.

Since $|x_{n+1} - x_n|_5 \le 5^{-n}$, the sequence (x_n) is Cauchy by Theorem 12.1 and converges to some $x \in \mathbb{Q}_5$ by Theorem 9.12; $x \in \mathbb{Z}_5$ since $\mathbb{Z}_5$ is closed. Finally $|x + x_n|_5 \le 1$, so

$$|x^2 - 6|_5 \le \max\{|x - x_n|_5 \ |x + x_n|_5, |x_n^2 - 6|_5\} \le \max\{|x - x_n|_5, 5^{-n}\} \to 0,$$

and $x^2 = 6$. ◊

Example 15.8. Consider $\mathbb{Q}$ with the 5-adic metric and the triangle with vertices $A = 2/15$, $B = 1/5$, $C = 7/15$. Compute its side lengths.

Solution. Recall that $v_p(a/b) = v_p(a) - v_p(b)$ and $|x|_p = p^{-v_p(x)}$ for $x \neq 0$. Here $v_5(-1) = 0$ and $v_5(15) = 1$. Hence

$$|A - B|_5 = |\frac{2}{15} - \frac{3}{15}|_5 = |-\frac{1}{15}|_5 = \frac{1}{5^{-1}} = 5,$$
$$|B - C|_5 = |-\frac{4}{15}|_5 = 5, \quad |C - A|_5 = |\frac{5}{15}|_5 = |\frac{1}{3}|_5 = 1.$$

In agreement with Theorem 11.3, the two longest sides are equal. ◊

Example 15.9. Characterize the units of $\mathbb{Z}_p$ in terms of v_p. Prove that $0 \neq x \in \mathbb{Q}_p$ is a square in $\mathbb{Q}_p$ if and only if $x = p^{2n}y^2$ with $n \in \mathbb{Z}$ and y a unit of $\mathbb{Z}_p$.

Solution. By Theorem 9.12 2), $x \in \mathbb{Q}_p$ lies in $\mathbb{Z}_p$ if and only if $v_p(x) \ge 0$. By Example 15.4, an element of $\mathbb{Z}_p$ is a unit if and only if $v_p = 0$. Hence $\mathbb{Z}_p^\times = \{x \in \mathbb{Q}_p : v_p(x) = 0\}$.

If $x = p^{2n}y^2$, then $p^{2n} = (p^n)^2$ and $x = (p^n y)^2$ with $p^n y \in \mathbb{Q}_p$. Conversely, let $x = z^2$ with $z \in \mathbb{Q}_p$; then $z \neq 0$. By Proposition 8.10, $z = p^n y$ with $n = v_p(z) \in \mathbb{Z}$ and $y \in \mathbb{Z}_p^\times$. Hence $x = z^2 = (p^n y)^2 = p^{2n}y^2$, and $y^2 = y \cdot y \in \mathbb{Z}_p^\times$. ◊

Example 15.10. Prove that $(x^2 - 2)(x^2 - 17)(x^2 - 34) = 0$ has a root in $\mathbb{Q}_p$ for every $p \le \infty$, but no root in $\mathbb{Q}$.

Solution. None of $2, 17, 34$ is the square of a rational number, since $v_q(z^2) = 2v_q(z)$ is even for every prime q and every $0 \neq z \in \mathbb{Q}$, whereas $v_2(2) = v_{17}(17) = v_2(34) = 1$. So there is no rational root.

$p = \infty$. $\mathbb{Q}_\infty = \mathbb{R}$ and $\sqrt{2} \in \mathbb{R}$ is a root.

p odd, $p \neq 17$. Then $p \nmid 2 \cdot 17 \cdot 34$. If $\left(\frac{2}{p}\right) = 1$ or $\left(\frac{17}{p}\right) = 1$, one of $2, 17$ is a quadratic residue modulo p. Otherwise $\left(\frac{2}{p}\right) = \left(\frac{17}{p}\right) = -1$, and by Proposition 4.3 2)

$$\left(\frac{34}{p}\right) = \left(\frac{2}{p}\right)\left(\frac{17}{p}\right) = (-1)(-1) = 1,$$

so there is x with $x^2 \equiv 34 \pmod p$. In each case some $a \in \{2, 17, 34\}$ and $c \in \mathbb{Z}$ satisfy $c^2 \equiv a \pmod p$, and $p \nmid c$. For $f(x) = x^2 - a$ we have $f'(c) = 2c \not\equiv 0 \pmod p$, so as in the solution of Example 8.12, Theorem 6.2 yields $\alpha \in \mathbb{Z}_p$ with $\alpha^2 = a$.

$p = 17$. $6^2 = 36 \equiv 2 \pmod{17}$ and $f'(6) = 12 \not\equiv 0 \pmod{17}$ for $f(x) = x^2 - 2$, so 2 is a square in $\mathbb{Q}_{17}$ in the same way.

$p = 2$. We show that 17 is a square in $\mathbb{Q}_2$; Theorem 6.2 does not apply, since $f'(x) = 2x \equiv 0 \pmod 2$. We construct odd integers x_k, $k \geq 3$, with $x_k^2 \equiv 17 \pmod{2^k}$ and $x_{k+1} \equiv x_k \pmod{2^{k-1}}$. Take $x_3 = 1$. Given x_k, write $x_k^2 - 17 = 2^k s$ and put $x_{k+1} = x_k + t2^{k-1}$ with $t \in \{0, 1\}$. Then

$$x_{k+1}^2 = x_k^2 + tx_k 2^k + t^2 2^{2k-2} \equiv 17 + 2^k(s + tx_k) \pmod{2^{k+1}},$$

since $2k - 2 \geq k + 1$ for $k \geq 3$. As x_k is odd, $t \equiv s \pmod 2$ makes $s + tx_k$ even, and $x_{k+1}^2 \equiv 17 \pmod{2^{k+1}}$. Now $|x_{k+1} - x_k|_2 \leq 2^{-(k-1)}$, so (x_k) converges to some $x \in \mathbb{Z}_2$, and the argument at the end of the solution of Example 15.7 gives $x^2 = 17$. ◊

Example 15.11. Let $f \in \mathbb{Z}[x]$. Is it true that f is irreducible in $\mathbb{Q}[x]$ if and only if f is irreducible in $\mathbb{Q}_p[x]$ for every prime p?

Solution. One implication holds. If f is reducible over $\mathbb{Q}$, that is, $f = gh$ with $g, h \in \mathbb{Q}[x]$ of positive degree, then this is a factorization in $\mathbb{Q}_p[x]$, so f is reducible over $\mathbb{Q}_p$. Hence irreducibility over every $\mathbb{Q}_p$ implies irreducibility over $\mathbb{Q}$.

The other implication fails. $x^2 + 1$ is irreducible over $\mathbb{Q}$, since -1 is not a rational square. It is reducible over $\mathbb{Q}_5$: $2^2 \equiv -1 \pmod 5$ and $f'(2) = 4 \not\equiv 0 \pmod 5$ for $f(x) = x^2 + 1$, so Theorem 6.2 gives $\iota \in \mathbb{Z}_5$ with $\iota^2 = -1$, and $x^2 + 1 = (x - \iota)(x + \iota)$.

A stronger example: $x^4 + 1$ is irreducible over $\mathbb{Q}$ but reducible over $\mathbb{Q}_p$ for every odd prime p.

Irreducibility over $\mathbb{Q}$. It has no rational root, since $x^4 \geq 0 > -1$ for real x. In $\mathbb{R}[x]$,

$$x^4 + 1 = (x^2 + 1)^2 - 2x^2 = \left(x^2 - \sqrt{2}x + 1\right)\left(x^2 + \sqrt{2}x + 1\right),$$

and both factors are irreducible over $\mathbb{R}$ (discriminant $2 - 4 < 0$). By unique factorization in $\mathbb{R}[x]$, a monic quadratic factor of $x^4 + 1$ over $\mathbb{Q}$ would equal one of them, but neither has rational coefficients.

Reducibility over $\mathbb{Q}_p$, p odd. Recall from Proposition 4.3 that

$$\left(\frac{-1}{p}\right) = \begin{cases} 1 & \text{if } p \equiv 1 \pmod 4 \\ -1 & \text{if } p \equiv 3 \pmod 4 \end{cases}, \quad \left(\frac{2}{p}\right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod 8 \\ -1 & \text{if } p \equiv \pm 3 \pmod 8 \end{cases}.$$

Since $\left(\frac{-1}{p}\right)\left(\frac{2}{p}\right) = \left(\frac{-2}{p}\right)$, at least one of $-1, 2, -2$ is a quadratic residue modulo p; in particular $\left(\frac{-2}{p}\right) = 1$ if and only if $p \equiv 1, 3 \pmod 8$. As in the previous example, Theorem 6.2 lifts a square root modulo p of -1, 2 or -2 to $\mathbb{Q}_p$, since p is odd.

- If $\left(\frac{2}{p}\right) = 1$, there is $a \in \mathbb{Z}_p$ with $a^2 = 2$, and $x^4 + 1 = (x^2 + 1)^2 - 2x^2 = (x^2 + 1 - ax)(x^2 + 1 + ax)$.
- If $\left(\frac{-2}{p}\right) = 1$, there is $b \in \mathbb{Z}_p$ with $b^2 = -2$, and

$$x^4+1=(x^2-1)^2+2x^2=(x^2-1)^2-(-2)x^2=(x^2-1-bx)(x^2-1+bx).$$

- If $\left(\frac{-1}{p}\right)=1$, that is, $p\equiv 1 \pmod 4$, there is $c\in\mathbb{Z}_p$ with $c^2=-1$, and $x^4+1=x^4-(-1)=(x^2-c)(x^2+c)$.

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Remark 15.12 (cited). The prime 2 is a genuine exception: x^4+1 is irreducible over $\mathbb{Q}_2$, although $x^4+1\equiv(x+1)^4 \pmod 2$. Likewise x^2+1 is irreducible over $\mathbb{Q}_2$, although $x^2+1\equiv(x+1)^2 \pmod 2$.

16. Trace and norm

Throughout, q is a prime power, $m\in\mathbb{N}$, and $\mathbb{F}_q\subseteq\mathbb{F}_{q^m}$ as in Remark 14.5.

Definition 16.1 (Trace). Let $\alpha\in\mathbb{F}_{q^m}$. The *relative trace* of α over $\mathbb{F}_q$ is

$$\operatorname{tr}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)=\alpha+\alpha^q+\cdots+\alpha^{q^{m-1}}.$$

If $\mathbb{F}_q=\mathbb{F}_p$ with p prime, then $\operatorname{tr}_{\mathbb{F}_{p^m}\mid\mathbb{F}_p}$ is called the *trace*.

Definition 16.2 (Norm). Let $L=\mathbb{F}_{q^m}$, $K=\mathbb{F}_q$ and $\alpha\in L$. The *relative norm* of α over K is

$$\mathrm{N}_{L\mid K}(\alpha)=\alpha\cdot\alpha^q\cdot\cdots\cdot\alpha^{q^{m-1}}.$$

Lemma 16.3. Let $\alpha\in\mathbb{F}_{q^m}$. Then $\operatorname{tr}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)\in\mathbb{F}_q$ and $\mathrm{N}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)\in\mathbb{F}_q$. More precisely, if $m_\alpha(x)=\sum_{i=0}^{r}a_ix^i$ is the minimal polynomial of α over $\mathbb{F}_q$, then $r\mid m$ and

$$\operatorname{tr}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)=-\frac{m}{r}\cdot a_{r-1},\quad \mathrm{N}_{\mathbb{F}_{q^m}\mid\mathbb{F}_q}(\alpha)=(-1)^m a_0^{m/r}.$$

Proof. α is a root of $x^{q^m}-x$ by Theorem 14.2, so it has a minimal polynomial $m_\alpha\in\mathbb{F}_q[x]$, which is monic and irreducible, of degree $r=[\mathbb{F}_q(\alpha):\mathbb{F}_q]$; so $a_r=1$. By Remark 14.5, $\mathbb{F}_q(\alpha)$ is a field with q^r elements containing $\mathbb{F}_q$. Since $\mathbb{F}_{q^m}$ is a vector space over $\mathbb{F}_q(\alpha)$ of some dimension s, we get $q^m=(q^r)^s$, so $r\mid m$ and $m=rs$.

By Lemma 14.6 applied to m_α and the field $\mathbb{F}_q(\alpha)$, $m_\alpha(x)=\prod_{i=0}^{r-1}\left(x-\alpha^{q^i}\right)$. By Theorem 14.2 applied to $\mathbb{F}_q(\alpha)$, $\alpha^{q^r}=\alpha$, hence $\alpha^{q^{jr}}=\alpha$ and $\alpha^{q^{i+jr}}=\left(\alpha^{q^{jr}}\right)^{q^i}=\alpha^{q^i}$ for all $i,j\geq 0$. Splitting $0\leq i\leq m-1$ into m/r blocks of length r,

$$\prod_{i=0}^{m-1}\left(x-\alpha^{q^i}\right)=\prod_{i=0}^{r-1}\left(x-\alpha^{q^i}\right)\cdot\prod_{i=0}^{r-1}\left(x-\alpha^{q^{i+r}}\right)\cdot\cdots\cdot\prod_{i=0}^{r-1}\left(x-\alpha^{q^{i+r(m/r-1)}}\right)=m_\alpha(x)^{m/r}.$$

So the coefficients of the left side lie in $\mathbb{F}_q$. Its coefficient of x^{m-1} is $-\left(\alpha+\alpha^q+\cdots+\alpha^{q^{m-1}}\right)=-\operatorname{tr}(\alpha)$, and its constant term is $(-1)^m\alpha\cdot\alpha^q\cdot\cdots\cdot\alpha^{q^{m-1}}=(-1)^m\,\mathrm{N}(\alpha)$. On the right, the coefficient of x^{m-1} in $\left(x^r+a_{r-1}x^{r-1}+\cdots+a_0\right)^{m/r}$ is $(m/r)a_{r-1}$, and the constant term is $a_0^{m/r}$. Comparing coefficients gives both formulas. ■

17. Field extensions

17.1. The Frobenius automorphism

Definition 17.1 (Frobenius automorphism). The map $\Phi : \mathbb{F}_{q^m} \to \mathbb{F}_{q^m}$, $\Phi(\alpha) = \alpha^q$, is the *Frobenius automorphism* of $\mathbb{F}_{q^m}$ over $\mathbb{F}_q$.

Theorem 17.2. The $\mathbb{F}_q$-automorphisms of $\mathbb{F}_{q^m}$ are exactly $\Phi, \Phi^2, ..., \Phi^m = \mathrm{id}$, and these are pairwise distinct.

Proof. Φ is an iterate of $a \mapsto a^p$, where $p = \mathrm{char}\ \mathbb{F}_q$, since q is a power of p; so it is a ring homomorphism by Theorem 14.4. It fixes $\mathbb{F}_q$, since $a^q = a$ for $a \in \mathbb{F}_q$ by Theorem 14.2. It is injective: $\Phi(\alpha) = \Phi(\beta)$ if and only if $\Phi(\alpha - \beta) = 0$, if and only if $(\alpha - \beta)^q = 0$, if and only if $\alpha = \beta$. An injective map of the finite set $\mathbb{F}_{q^m}$ into itself is surjective. So Φ is an $\mathbb{F}_q$-automorphism, and so are its powers. By Theorem 14.2 applied to $\mathbb{F}_{q^m}$, $\Phi^m(\alpha) = \alpha^{q^m} = \alpha$, so $\Phi^m = \mathrm{id}$.

Let β be a generator of $\mathbb{F}_{q^m}^\times$ (Theorem 14.2). Then $\mathbb{F}_q(\beta) = \mathbb{F}_{q^m}$, so by Remark 14.5 the minimal polynomial $f(x) = x^m + a_{m-1}x^{m-1} + \cdots + a_0 \in \mathbb{F}_q[x]$ of β has degree m. By Lemma 14.6 its roots in $\mathbb{F}_{q^m}$ are the pairwise distinct elements $\beta, \beta^q, ..., \beta^{q^{m-1}}$. Hence $\Phi^k(\beta) = \beta^{q^k}$, $1 \le k \le m$, are pairwise distinct, and so are $\Phi, ..., \Phi^m$.

Now let ψ be an $\mathbb{F}_q$-automorphism of $\mathbb{F}_{q^m}$. Then

$$0 = \psi(0) = \psi(f(\beta)) = \psi(\beta^m + a_{m-1}\beta^{m-1} + \cdots + a_0).$$

Since ψ is an $\mathbb{F}_q$-automorphism, it fixes the coefficients, hence

$$\psi(f(\beta)) = \psi(\beta)^m + a_{m-1}\psi(\beta)^{m-1} + \cdots + a_1\psi(\beta) + a_0 = f(\psi(\beta)).$$

So $\psi(\beta)$ is a root of f, that is, $\psi(\beta) = \beta^{q^k} = \Phi^k(\beta)$ for some $1 \le k \le m$. Two $\mathbb{F}_q$-automorphisms that agree on β agree on every $g(\beta)$ with $g \in \mathbb{F}_q[x]$, that is, on $\mathbb{F}_q(\beta) = \mathbb{F}_{q^m}$. Hence $\psi = \Phi^k$. ∎

17.2. Separable extensions

Remark 17.3 (Recall). Let F be a field. We write $E \mid F$ for a field extension $F \subseteq E$.

1) $[E : F]$ is the dimension of E over F, and $E \mid F$ is *finite* if it is finite. For $F \subseteq K \subseteq E$, $[E : F] = [E : K][K : F]$ (tower law).
2) $\alpha \in E$ is *algebraic* over F if $g(\alpha) = 0$ for some $0 \neq g \in F[x]$. Then its minimal polynomial $m_{\alpha,F}$ is irreducible, divides every $g \in F[x]$ with $g(\alpha) = 0$, and $[F(\alpha) : F] = \deg m_{\alpha,F}$. Every element of a finite extension is algebraic; $E \mid F$ is *algebraic* if every element of E is.
3) If $f \in F[x]$ is irreducible, then $F[x]/(f)$ is an extension of degree $\deg f$ containing a root of f. If α is a root of f in an extension, then $g(x) + (f) \mapsto g(\alpha)$ is an isomorphism $F[x]/(f) \to F(\alpha)$.
4) Every nonzero $f \in F[x]$ has a *splitting field*, an extension in which f is a product of linear factors and which is generated by the roots of f; it is unique up to F-isomorphism.
5) Every field F has an *algebraic closure* $\overline{F}$, an algebraic extension in which every nonconstant polynomial has a root. $\mathbb{C}$ is algebraically closed.

6) The monic greatest common divisor of $f, g \in F[x]$ is the same in $K[x]$ for every extension $K \mid F$, since the Euclidean algorithm runs in $F[x]$.

7) If S is an infinite subset of a field L, a nonzero $P \in L[c_1, \ldots, c_k]$ does not vanish on all of S^k.

A ring homomorphism between fields is injective; it is called an *embedding* or *monomorphism*, and an *F-monomorphism* if it fixes F pointwise.

Definition 17.4 (Separable polynomial). A nonzero polynomial $f \in F[x]$ is *separable* if every root of f in its splitting field is simple.

Example 17.5. $x^2 + 1 \in \mathbb{R}[x]$ has splitting field $\mathbb{C}$ and simple roots $\pm i$, so it is separable. $(x-1)^2(x-2) \in \mathbb{Q}[x]$ is not separable, since 1 is a double root.

Proposition 17.6. A nonzero $f \in F[x]$ is not separable if and only if $\deg \gcd(f, f') \geq 1$.

Proof. Let E be the splitting field of f; by Remark 17.3 6) the gcd may be computed in $E[x]$.

($\Rightarrow$) Let $\alpha \in E$ be a multiple root. Then $f(x) = (x-\alpha)^r h(x)$ with $r \geq 2$ and $h \in E[x]$, so

$$f'(x) = r(x-\alpha)^{r-1} h(x) + (x-\alpha)^r h'(x).$$

Hence $x - \alpha$ is a common divisor of f and f', and $\deg \gcd(f, f') \geq 1$.

($\Leftarrow$) Let $g = \gcd(f, f')$ with $\deg g \geq 1$. Since $g \mid f$, g splits in E; let $\alpha \in E$ be a root of g. Then $x - \alpha$ divides f and f'. Write $f(x) = (x-\alpha)^r h(x)$ with $r \geq 1$ and $h(\alpha) \neq 0$, and suppose $r = 1$. Then $f'(x) = h(x) + (x-\alpha)h'(x)$, and $f'(\alpha) = 0$ forces $h(\alpha) = 0$, a contradiction. So α is a multiple root of f. ∎

Lemma 17.7. An irreducible $f \in F[x]$ is separable if and only if $f' \neq 0$. In particular, if $\operatorname{char} F = 0$, every irreducible polynomial is separable.

Proof. Since f is irreducible, $\gcd(f, f')$ is 1 or f up to a unit. If $f' \neq 0$, then $\deg f' < \deg f$ and $f \nmid f'$, so the gcd is 1 and f is separable by Proposition 17.6. If $f' = 0$, the gcd is f, of degree at least 1. In characteristic 0, $f' \neq 0$ whenever $\deg f \geq 1$. ∎

Theorem 17.8. Over a finite field every irreducible polynomial is separable.

Proof. Let F be finite, $f \in F[x]$ irreducible, and suppose that f is not separable. By Lemma 17.7, $f' = 0$. Let $\operatorname{char} F = p$ (Theorem 14.2). Since $(x^p)' = p x^{p-1} = 0$, the condition $f' = \sum_i i a_i x^{i-1} = 0$ holds exactly when $a_i = 0$ for $p \nmid i$, so

$$f(x) = a_0 + a_1 x^p + \cdots + a_n (x^p)^n.$$

By Theorem 17.2 with $q = p$, the map $b \mapsto b^p$ is surjective on F, so $a_i = b_i^p$ with $b_i \in F$. The binomial argument of Theorem 14.4 holds in every commutative ring of characteristic p, in particular in $F[x]$, hence

$$f(x) = b_0^p + b_1^p x^p + \cdots + b_n^p x^{np} = (b_0 + b_1 x + \cdots + b_n x^n)^p.$$

As $\deg f \geq 1$ and $p \geq 2$, this contradicts the irreducibility of f. ■

Definition 17.9 (Separable extension). Let $E \mid F$ be an extension. An element $\alpha \in E$ is *separable* over F if it is algebraic over F and its minimal polynomial is separable. The extension is *separable* if every element of E is separable over F.

Theorem 17.10. Let $F \subseteq K \subseteq E$ be fields with E separable over F. Then E is separable over K, and K is separable over F.

Proof. 1) $K \mid F$ is separable: every $\beta \in K$ lies in E, and is therefore separable over F.

2) $E \mid K$ is separable: let $\alpha \in E$. Then α is separable over F, in particular algebraic over F, hence over K. Since $m_{\alpha,F} \in F[x] \subseteq K[x]$ vanishes at α, Remark 17.3 2) gives $m_{\alpha,K} \mid m_{\alpha,F}$. So every root of $m_{\alpha,K}$ in a splitting field of $m_{\alpha,F}$ is a root of $m_{\alpha,F}$ of at least the same multiplicity. As $m_{\alpha,F}$ has only simple roots, so does $m_{\alpha,K}$, and α is separable over K. ■

Example 17.11 (A non-separable extension). Fix a prime p, let t be an indeterminate and $F = \mathbb{F}_p(t)$, the field of rational functions over $\mathbb{F}_p$. Let E be obtained by adjoining to F a root s of $x^p - t$. Then $x^p - t$ is irreducible over F, and $E \mid F$ is not separable: in $E[x]$,

$$x^p - t = x^p - s^p = (x - s)^p.$$

Irreducibility of $x^p - t$. First, $s \notin F$: if $s = g/h$ with $g, h \in \mathbb{F}_p[t]$, $h \neq 0$, then $g^p = th^p$, and comparing degrees in t gives $p \deg g = 1 + p \deg h$, which is impossible. A monic factor of $x^p - t$ in $F[x]$ divides $(x - s)^p$ in $E[x]$, so it equals $(x - s)^k$ with $0 \leq k \leq p$. Its coefficient of x^{k-1} is $-ks$, which lies in F. For $1 \leq k \leq p - 1$, k is invertible in $\mathbb{F}_p$, so $s \in F$, a contradiction. Hence the only monic factors are 1 and $x^p - t$. ■

Theorem 17.12. Let $E \mid F$ be an extension, $\sigma : E \to E$ an F-monomorphism, and $f \in F[x]$ a polynomial that splits over E. If $\alpha \in E$ is a root of f, then so is $\sigma(\alpha)$. Hence σ permutes the roots of f in E.

Proof. Let $f(x) = b_0 + b_1 x + \cdots + b_n x^n \in F[x]$ with $f(\alpha) = b_0 + b_1\alpha + \cdots + b_n\alpha^n = 0$. Then

$$\begin{aligned} 0 = \sigma(0) &= \sigma(b_0 + b_1\alpha + \cdots + b_n\alpha^n) = \sigma(b_0) + \sigma(b_1\alpha) + \cdots + \sigma(b_n\alpha^n) \\ &= b_0 + b_1\sigma(\alpha) + \cdots + b_n\sigma(\alpha)^n = f(\sigma(\alpha)), \end{aligned}$$

since σ is an F-monomorphism and fixes the coefficients. So σ maps the finite set of roots of f in E into itself, and it is injective, hence it permutes this set. ■

17.3. Embeddings and primitive elements

Theorem 17.13. Let $E \mid F$ be a finite separable extension with $[E:F]=n$, let L be an algebraically closed field, and let $\sigma : F \to L$ be an embedding. Then σ has exactly n extensions to embeddings $E \to L$. In particular this holds for $L = \overline{F}$.

Proof. We proceed by induction on n. For $n=1$, $E=F$ and σ is its own unique extension.

Let $n>1$, choose $\alpha \in E \setminus F$, and let $r = \deg m_{\alpha,F} > 1$. In the tower $F \subseteq F(\alpha) \subseteq E$,

$$n = [E:F] = [E:F(\alpha)][F(\alpha):F] = \frac{n}{r}\cdot r.$$

Let $g = \sigma\big(m_{\alpha,F}\big) \in \sigma(F)[x]$ be the polynomial obtained by applying σ to the coefficients. Since $\sigma : F \to \sigma(F)$ is an isomorphism, it induces a ring isomorphism $F[x] \to \sigma(F)[x]$ compatible with derivatives and gcds, so g is irreducible and $\deg \gcd(g,g') = \deg \gcd\big(m_{\alpha,F}, m'_{\alpha,F}\big)$, which is 0 by Proposition 17.6 because α is separable over F. By Proposition 17.6, g is separable, so it has exactly r distinct roots in the algebraically closed field L.

For each root $\beta \in L$ of g define

$$\psi : F(\alpha) \to \sigma(F)(\beta), \quad \psi\big(b_0 + b_1\alpha + \cdots + b_k\alpha^k\big) = \sigma(b_0) + \sigma(b_1)\beta + \cdots + \sigma(b_k)\beta^k.$$

This is well defined and an isomorphism, being the composite $F(\alpha) \cong F[x]/\big(m_{\alpha,F}\big) \cong \sigma(F)[x]/(g) \cong \sigma(F)(\beta)$ of the isomorphisms of Remark 17.3 3). Its restriction to F is σ. Conversely, every extension ψ of σ to $F(\alpha)$ satisfies $g(\psi(\alpha)) = \psi\big(m_{\alpha,F}(\alpha)\big) = 0$, by the computation in the proof of Theorem 17.12, and is determined by $\psi(\alpha)$. So σ has exactly r extensions to $F(\alpha)$, one for each root β of g.

$E \mid F(\alpha)$ is separable by Theorem 17.10, and $[E:F(\alpha)] = n/r < n$. By the induction hypothesis each of the r embeddings ψ has exactly n/r extensions to E. Distinct ψ have no common extension, and every extension of σ to E restricts to some ψ. Hence σ has exactly $\frac{n}{r}\cdot r = n$ extensions. ∎

Theorem 17.14 (Primitive element theorem). If $E \mid F$ is a finite separable extension, then $E = F(\gamma)$ for some $\gamma \in E$. Such a γ is called a *primitive element* of $E \mid F$.

Proof. If F is finite, so is E, and a generator γ of $E^\times$ (Theorem 14.2) satisfies $E = F(\gamma)$.

Let F be infinite and $n = [E:F]$. Write $E = F(\alpha_1, \ldots, \alpha_k)$, for instance with a basis. By Theorem 17.13 applied to the inclusion $F \to \overline{F}$, there are exactly n distinct F-embeddings $\sigma_1, \ldots, \sigma_n : E \to \overline{F}$. An F-embedding of E is determined by its values on $\alpha_1, \ldots, \alpha_k$, so for $i \neq j$ some l has $\sigma_i(\alpha_l) \neq \sigma_j(\alpha_l)$. Hence

$$P(c_1, \ldots, c_k) = \prod_{i<j} \sum_{l=1}^{k} c_l\big(\sigma_i(\alpha_l) - \sigma_j(\alpha_l)\big) \in \overline{F}[c_1, \ldots, c_k]$$

is a product of nonzero linear forms, so $P \neq 0$. By Remark 17.3 7) there is $(c_1, \ldots, c_k) \in F^k$ with $P(c_1, \ldots, c_k) \neq 0$. Put $\gamma = \sum_l c_l\alpha_l \in E$. Then the elements $\sigma_i(\gamma) = \sum_l c_l\sigma_i(\alpha_l)$ are pairwise

distinct, and each is a root of $m_{\gamma,F}$ by the computation in the proof of Theorem 17.12. So $[F(\gamma) : F] = \deg m_{\gamma,F} \geq n = [E : F]$, and since $F(\gamma) \subseteq E$, $F(\gamma) = E$. ∎

Remark 17.15. For a finite field E, a generator of $E^\times$, called a primitive element of E in Theorem 14.2, is a primitive element of $E \mid F$ for every subfield F. The converse fails: $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$, but i has order 4 in $\mathbb{F}_9^\times$, which has order 8.

17.4. Normal extensions

Definition 17.16 (Normal extension). An algebraic extension $E \mid F$ is *normal* if every irreducible polynomial in $F[x]$ that has at least one root in E splits into linear factors in $E[x]$.

Example 17.17. $\mathbb{Q}\left(\sqrt[3]{2}\right) \mid \mathbb{Q}$ is not normal. The polynomial $x^3 - 2$ is irreducible over $\mathbb{Q}$: it has degree 3 and no rational root, since $z^3 = 2$ would give $3v_2(z) = 1$. Its roots are $\sqrt[3]{2}$, $\varepsilon\sqrt[3]{2}$, $\varepsilon^2\sqrt[3]{2}$, where $\varepsilon = e^{2\pi i/3}$ is a primitive third root of unity. The last two are not real, while $\mathbb{Q}\left(\sqrt[3]{2}\right) \subseteq \mathbb{R}$, so they do not lie in $\mathbb{Q}\left(\sqrt[3]{2}\right)$.

$\mathbb{Q}\left(\sqrt{2}\right) \mid \mathbb{Q}$ is normal; the roots of $x^2 - 2$ are $\pm\sqrt{2}$. More generally, every extension $E \mid F$ with $[E : F] = 2$ is normal. If $f \in F[x]$ is irreducible with a root $\alpha \in E$, then f is a scalar multiple of $m_{\alpha,F}$ and $\deg f = [F(\alpha) : F]$ divides 2. If $\deg f = 1$, f is linear. If $\deg f = 2$, then $f = c(x - \alpha)(x - \alpha')$ in $E[x]$, where $x - \alpha'$ is the quotient of f/c by $x - \alpha$ in $E[x]$, so $\alpha' \in E$.

Theorem 17.18. Let $F \subseteq E \subseteq \mathbb{C}$ be fields with $[E : F]$ finite. Then $E \mid F$ is normal if and only if every F-monomorphism $\tau : E \to \mathbb{C}$ is an F-automorphism of E, that is, $\tau(E) = E$.

Proof. ($\Rightarrow$) Let $E \mid F$ be normal and $\tau : E \to \mathbb{C}$ an F-monomorphism. For $\gamma \in E$, the irreducible polynomial $m_{\gamma,F}$ has the root $\gamma \in E$, so it splits into linear factors in $E[x]$. By the computation in the proof of Theorem 17.12, $\tau(\gamma)$ is a root of $m_{\gamma,F}$, hence $\tau(\gamma) \in E$. So $\tau(E) \subseteq E$. Since τ is an injective F-linear map of the finite-dimensional F-space E into itself, it is surjective, and τ is an F-automorphism of E.

($\Leftarrow$) Let $f \in F[x]$ be irreducible with a root $\alpha \in E$, and let $\beta \in \mathbb{C}$ be any root of f, that is, a conjugate of α over F. As in the proof of Theorem 17.13, $g(\alpha) \mapsto g(\beta)$ defines an F-embedding $\psi : F(\alpha) \to \mathbb{C}$. Since $\operatorname{char} F = 0$, $E \mid F(\alpha)$ is separable by Lemma 17.7, and $\mathbb{C}$ is algebraically closed, so by Theorem 17.13 ψ extends to an F-monomorphism $\tau : E \to \mathbb{C}$, with $\tau(\alpha) = \beta$. By hypothesis $\tau(E) = E$, so $\beta \in E$. Thus every root of f in $\mathbb{C}$ lies in E; since f splits into linear factors over $\mathbb{C}$, it splits into linear factors in $E[x]$. Hence $E \mid F$ is normal. ∎

Lemma 17.19. Let $E \mid F$ be a finite extension, L an algebraically closed field and $\sigma : F \to L$ an embedding. Then σ extends to an embedding $E \to L$.

Proof. We follow the proof of Theorem 17.13, without separability. We proceed by induction on $[E : F]$; for $[E : F] = 1$ there is nothing to prove. Otherwise choose $\alpha \in E \setminus F$. The polynomial $g = \sigma\left(m_{\alpha,F}\right)$ has a root β in L, since L is algebraically closed, and $g(\alpha) \mapsto g(\beta)$ extends σ to an

embedding $\psi : F(\alpha) \to L$, as in that proof. Since $[E : F(\alpha)] < [E : F]$, the induction hypothesis extends ψ to E. ∎

Proposition 17.20. Let $E \mid F$ be a finite extension and Ω an algebraically closed field containing E. Then $E \mid F$ is normal if and only if $\tau(E) = E$ for every F-monomorphism $\tau : E \to \Omega$.

Proof. This is the proof of Theorem 17.18 with $\mathbb{C}$ replaced by Ω and Theorem 17.13 replaced by Lemma 17.19, which needs no separability. In ($\Leftarrow$), the irreducible f splits into linear factors over the algebraically closed field Ω, and every root of f in Ω lies in E. ∎

Theorem 17.21. A finite extension $E \mid F$ is normal if and only if E is a splitting field over F of some $f \in F[x]$.

Proof. ($\Rightarrow$) Let $[E : F] = n$, let $\alpha_1, ..., \alpha_n \in E$ be a basis of E over F, and let $f_i \in F[x]$ be the minimal polynomial of α_i. Since $E \mid F$ is normal and f_i is irreducible with the root $\alpha_i \in E$, f_i splits over E. Hence $f = f_1 \cdots f_n$ splits over E. It remains to show that E is generated by the roots of f. If $F \subseteq K \subseteq E$ and f splits over K, then $\alpha_i \in K$ for every i, and since the α_i span E, $K = E$.

($\Leftarrow$) Let E be a splitting field of $f \in F[x]$, let $\alpha_1, ..., \alpha_n$ be the roots of f in E, so that $E = F(\alpha_1, ..., \alpha_n)$, and let Ω be an algebraic closure of E (Remark 17.3). Let $\tau : E \to \Omega$ be an F-monomorphism. By the computation in the proof of Theorem 17.12, each $\tau(\alpha_i)$ is a root of f in Ω. Since f is a product of linear factors over E, its roots in Ω are $\alpha_1, ..., \alpha_n$, so $\tau(\alpha_i) = \alpha_j$ for some j. Hence $\tau(E) = F(\tau(\alpha_1), ..., \tau(\alpha_n)) \subseteq E$. As τ is F-linear and injective, $\dim_F \tau(E) = n = \dim_F E$, so $\tau(E) = E$. By Proposition 17.20, $E \mid F$ is normal. ∎

Corollary 17.22. Let $F \subseteq K \subseteq E$ be fields with $[E : F]$ finite. If $E \mid F$ is normal, then so is $E \mid K$.

Proof. By Theorem 17.21, E is a splitting field over F of some $f \in F[x]$, so $E = F(\alpha_1, ..., \alpha_n)$, where $\alpha_1, ..., \alpha_n$ are the roots of f. Since $f \in F[x] \subseteq K[x]$ and $F \subseteq K \subseteq E$, also $E = K(\alpha_1, ..., \alpha_n)$. Hence E is a splitting field of f over K, and $E \mid K$ is normal by Theorem 17.21. ∎

18. Galois theory

Definition 18.1 (Galois extension). An algebraic extension $E \mid F$ is a *Galois extension* if it is normal and separable.

Definition 18.2 (Galois group). Let $E \mid F$ be a field extension. The *Galois group* $\mathrm{Gal}(E \mid F)$ is the set of F-automorphisms of E. It is a group under composition, since compositions and inverses of F-automorphisms are F-automorphisms.

Proposition 18.3. Let $E \mid F$ be a finite Galois extension with $[E : F] = n$. Then $|\mathrm{Gal}(E \mid F)| = n$.

Proof. Let Ω be an algebraic closure of E and $\iota : E \to \Omega$ the inclusion. By Theorem 17.13, applied to the inclusion $F \to \Omega$, there are exactly n F-monomorphisms $E \to \Omega$. Since $E \mid F$ is normal, each of them has image E by Proposition 17.20, so it equals $\iota \circ \sigma$ for a unique $\sigma \in \mathrm{Gal}(E \mid F)$. Conversely $\iota \circ \sigma$ is an F-monomorphism for every $\sigma \in \mathrm{Gal}(E \mid F)$. Hence $\sigma \mapsto \iota \circ \sigma$ is a bijection from $\mathrm{Gal}(E \mid F)$ onto the F-monomorphisms $E \to \Omega$. ∎

Example 18.4. $\mathrm{Gal}\left(\mathbb{Q}\left(\sqrt[3]{2}\right) \mid \mathbb{Q}\right) = \{\mathrm{id}\}$. Let σ be a $\mathbb{Q}$-automorphism of $E = \mathbb{Q}\left(\sqrt[3]{2}\right)$. By Theorem 17.12, $\sigma\left(\sqrt[3]{2}\right)$ is a root of $x^3 - 2$ in $E \subseteq \mathbb{R}$. Since $t \mapsto t^3$ is strictly increasing on $\mathbb{R}$, the only real root is $\sqrt[3]{2}$, so $\sigma\left(\sqrt[3]{2}\right) = \sqrt[3]{2}$, and σ is the identity on $\mathbb{Q}\left(\sqrt[3]{2}\right)$. Here $|\mathrm{Gal}(E \mid \mathbb{Q})| = 1 < 3 = [E : \mathbb{Q}]$; Proposition 18.3 does not apply, since $E \mid \mathbb{Q}$ is not normal (Example 17.17).

Definition 18.5 (Fixed field). Let $E \mid F$ be an extension, $G = \mathrm{Gal}(E \mid F)$ and $H \leq G$ a subgroup. The *fixed field* of H is

$$\mathcal{F}(H) = \{x \in E : \sigma(x) = x \text{ for all } \sigma \in H\}.$$

It is a subfield of E, since each $\sigma \in H$ is a ring homomorphism, and it contains F, since each $\sigma \in H$ fixes F.

Definition 18.6. Let $E \mid F$ be an extension, $G = \mathrm{Gal}(E \mid F)$, and let K be an *intermediate field*, that is, $F \subseteq K \subseteq E$. Put

$$\mathcal{G}(K) = \mathrm{Gal}(E \mid K) = \{\sigma \in G : \sigma(x) = x \text{ for all } x \in K\},$$

the subgroup of G fixing K pointwise.

Example 18.7. $\mathbb{C} \mid \mathbb{R}$ is a Galois extension with $\mathrm{Gal}(\mathbb{C} \mid \mathbb{R}) = \{\mathrm{id}, \text{complex conjugation}\} \cong \mathbb{Z}/2\mathbb{Z}$. Indeed, $[\mathbb{C} : \mathbb{R}] = 2$, so $\mathbb{C} \mid \mathbb{R}$ is normal (Example 17.17), and separable by Lemma 17.7, since $\mathrm{char}\ \mathbb{R} = 0$. By Proposition 18.3, $|\mathrm{Gal}(\mathbb{C} \mid \mathbb{R})| = 2$, and complex conjugation is an $\mathbb{R}$-automorphism different from the identity.

Theorem 18.8. Let $E \mid F$ be a finite Galois extension with $G = \mathrm{Gal}(E \mid F)$.

1) The fixed field of G is F.

2) If H is a proper subgroup of G, then $\mathcal{F}(H) \supsetneq F$.

Proof. 1) Let $F_0 = \mathcal{F}(G)$, so $F \subseteq F_0 \subseteq E$. Every $\sigma \in G$ fixes F_0 pointwise, so $G \subseteq \mathrm{Gal}(E \mid F_0)$. Conversely every F_0-automorphism of E fixes $F \subseteq F_0$, so $\mathrm{Gal}(E \mid F_0) \subseteq G$, and $G = \mathrm{Gal}(E \mid F_0)$. The extension $E \mid F_0$ is Galois: it is normal by Corollary 17.22 and separable by Theorem 17.10. By Proposition 18.3, $[E : F_0] = |\mathrm{Gal}(E \mid F_0)| = |G| = [E : F]$. By the tower law $[E : F] = [E : F_0][F_0 : F]$, so $[F_0 : F] = 1$ and $F_0 = F$.

2) Suppose $F = \mathcal{F}(H)$. Since $E \mid F$ is finite and separable, Theorem 17.14 gives $\alpha \in E$ with $E = F(\alpha)$. Consider

$$f(x) = \prod_{\sigma \in H} (x - \sigma(\alpha)) \in E[x].$$

We show $f \in F[x]$. For $\tau \in H$ the map $\sigma \mapsto \tau\sigma$ is a bijection of H, so

$$\prod_{\sigma\in H}(x - \tau(\sigma(\alpha))) = \prod_{\sigma\in H}(x - \sigma(\alpha)).$$

Applying τ to the coefficients of f therefore leaves f unchanged, so every coefficient lies in $\mathcal{F}(H) = F$, and $f \in F[x]$. Moreover $f(\alpha) = 0$, since $\mathrm{id} \in H$. Then $\deg f = |H| < |G| = [E : F]$ by Proposition 18.3. On the other hand $m_{\alpha,F}$ divides f, so $\deg f \geq [F(\alpha) : F] = [E : F]$, a contradiction. Hence $F \neq \mathcal{F}(H)$, and since $F \subseteq \mathcal{F}(H)$, $\mathcal{F}(H) \supsetneq F$. ■

Remark 18.9. For a finite Galois extension $E \mid F$ with $G = \mathrm{Gal}(E \mid F)$ there are two maps between the intermediate fields and the subgroups of G:

$$\mathcal{G} : \{\text{intermediate fields}\} \to \{\text{subgroups of } G\}, \quad K \mapsto \mathcal{G}(K) = \mathrm{Gal}(E \mid K),$$

$$\mathcal{F} : \{\text{subgroups of } G\} \to \{\text{intermediate fields}\}, \quad H \mapsto \mathcal{F}(H).$$

By Theorem 18.11 they are mutually inverse bijections.

Example 18.10. Let $E = \mathbb{Q}\left(i, \sqrt{5}\right)$ and $F = \mathbb{Q}$. Determine $G = \mathrm{Gal}(E \mid \mathbb{Q})$, its subgroups and the corresponding intermediate fields.

Solution. E is the splitting field of $(x^2 + 1)(x^2 - 5)$ over $\mathbb{Q}$, so $E \mid \mathbb{Q}$ is normal by Theorem 17.21, and separable since $\mathrm{char}\ \mathbb{Q} = 0$. We have $[\mathbb{Q}(i) : \mathbb{Q}] = 2$, and $\sqrt{5} \notin \mathbb{Q}(i)$: an element $a + bi$ with $a, b \in \mathbb{Q}$ is real only if $b = 0$, and $\sqrt{5} \notin \mathbb{Q}$. Hence $[E : \mathbb{Q}(i)] = 2$ and $[E : \mathbb{Q}] = 4$, so $|G| = 4$ by Proposition 18.3. Each $\sigma \in G$ is determined by $\sigma(i) \in \{i, -i\}$ and $\sigma\left(\sqrt{5}\right) \in \left\{\sqrt{5}, -\sqrt{5}\right\}$, by Theorem 17.12. These are four possibilities, so all of them occur:

$$\sigma_1 : \left(i, \sqrt{5}\right) \mapsto \left(i, \sqrt{5}\right), \quad \sigma_2 : \left(i, \sqrt{5}\right) \mapsto \left(i, -\sqrt{5}\right),$$
$$\sigma_3 : \left(i, \sqrt{5}\right) \mapsto \left(-i, \sqrt{5}\right), \quad \sigma_4 : \left(i, \sqrt{5}\right) \mapsto \left(-i, -\sqrt{5}\right).$$

Each σ_j satisfies $\sigma_j^2 = \sigma_1 = \mathrm{id}$. The proper subgroups are $\{\sigma_1\}$, $\{\sigma_1, \sigma_2\}$, $\{\sigma_1, \sigma_3\}$ and $\{\sigma_1, \sigma_4\}$. Since σ_2 fixes i, σ_3 fixes $\sqrt{5}$, and $\sigma_4\left(i\sqrt{5}\right) = (-i)\left(-\sqrt{5}\right) = i\sqrt{5}$, each of the three subgroups of order 2 fixes a field of degree 2 over $\mathbb{Q}$:

$$\mathcal{F}(\{\sigma_1, \sigma_2\}) = \mathbb{Q}(i), \quad \mathcal{F}(\{\sigma_1, \sigma_3\}) = \mathbb{Q}\left(\sqrt{5}\right), \quad \mathcal{F}(\{\sigma_1, \sigma_4\}) = \mathbb{Q}\left(i\sqrt{5}\right).$$

Equality holds in each case. For instance $K = \mathcal{F}(\{\sigma_1, \sigma_2\})$ satisfies $\mathbb{Q}(i) \subseteq K \subsetneq E$, since $\sigma_2\left(\sqrt{5}\right) \neq \sqrt{5}$. As $[E : \mathbb{Q}(i)] = 2$, the tower law forces $K = \mathbb{Q}(i)$. The same argument applies to $\mathbb{Q}\left(\sqrt{5}\right)$ and $\mathbb{Q}\left(i\sqrt{5}\right)$, both of degree 2 over $\mathbb{Q}$ because $\sqrt{5}, i\sqrt{5} \notin \mathbb{Q}$ are roots of $x^2 - 5$ and $x^2 + 5$. Finally $\mathcal{F}(\{\sigma_1\}) = E$ and $\mathcal{F}(G) = \mathbb{Q}$ by Theorem 18.8.

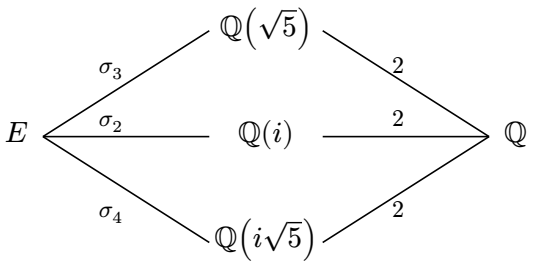


Figure 18.1. *Intermediate fields of $\mathbb{Q}(i, \sqrt{5}) \mid \mathbb{Q}$. An edge from E is labelled with the nontrivial element of $\mathrm{Gal}(E \mid K)$, an edge to $\mathbb{Q}$ with $[K : \mathbb{Q}]$.*

◊

Theorem 18.11 (Fundamental theorem of Galois theory, part I). Let $E \mid F$ be a finite Galois extension with $G = \mathrm{Gal}(E \mid F)$.

1) The map $\mathcal{F}$ is a bijection from the subgroups of G onto the intermediate fields, with inverse $\mathcal{G}$.
2) Let $H \leq G$, $K = \mathcal{F}(H)$ and $\sigma \in G$. Then $\sigma(K) = \{\sigma(x) : x \in K\}$ is the intermediate field fixed by $\sigma H \sigma^{-1}$, that is, $\sigma(K) = \mathcal{F}(\sigma H \sigma^{-1})$.

Proof. 1) We show $\mathcal{G}(\mathcal{F}(H)) = H$ for every subgroup H and $\mathcal{F}(\mathcal{G}(K)) = K$ for every intermediate field K.

Let $H \leq G$. Every $\sigma \in H$ fixes $\mathcal{F}(H)$, so $\sigma \in \mathrm{Gal}(E \mid \mathcal{F}(H)) = \mathcal{G}(\mathcal{F}(H))$, and $H \subseteq \mathcal{G}(\mathcal{F}(H))$. The extension $E \mid \mathcal{F}(H)$ is Galois, by Corollary 17.22 and Theorem 17.10, with Galois group $\mathcal{G}(\mathcal{F}(H))$. If H were a proper subgroup of it, Theorem 18.8 2), applied to $E \mid \mathcal{F}(H)$, would give that the fixed field of H strictly contains $\mathcal{F}(H)$. But the fixed field of H is $\mathcal{F}(H)$ by definition. Hence $H = \mathcal{G}(\mathcal{F}(H))$.

Let K be an intermediate field. Then $\mathcal{F}(\mathcal{G}(K))$ is the fixed field of $\mathrm{Gal}(E \mid K)$. Since $E \mid K$ is Galois, by Corollary 17.22 and Theorem 17.10, this fixed field is K by Theorem 18.8 1).

2) Since σ is an automorphism of E,

$$\begin{aligned}\mathcal{F}(\sigma H \sigma^{-1}) &= \{x \in E : (\sigma\tau\sigma^{-1})(x) = x \text{ for all } \tau \in H\} \\ &= \{x \in E : \tau(\sigma^{-1}(x)) = \sigma^{-1}(x) \text{ for all } \tau \in H\} \\ &= \{x \in E : \sigma^{-1}(x) \in \mathcal{F}(H)\} = \sigma(\mathcal{F}(H)) = \sigma(K).\end{aligned}$$

■

Theorem 18.12 (Fundamental theorem of Galois theory, part II). Let $E \mid F$ be a finite Galois extension with $G = \mathrm{Gal}(E \mid F)$, let $H \leq G$, and let $K = \mathcal{F}(H)$ be the intermediate field fixed by H. Then:

1) $E \mid K$ is a Galois extension.
2) $K \mid F$ is normal if and only if H is a normal subgroup of G.
3) If H is normal in G, then $\mathrm{Gal}(K \mid F) \cong G/H = \mathrm{Gal}(E \mid F)/\mathrm{Gal}(E \mid K)$.
4) $[K : F] = |G : H|$, whether or not $K \mid F$ is normal.

Proof. 1) $E \mid K$ is normal by Corollary 17.22 and separable by Theorem 17.10.

2) Let Ω be an algebraic closure of E. We first show that the F-monomorphisms $K \to \Omega$ are exactly the restrictions $\sigma|_K$ with $\sigma \in G$. Each $\sigma|_K$ is one. Conversely, let $\tau : K \to \Omega$ be an F-monomorphism. By Lemma 17.19, applied to $E \mid K$, τ extends to an F-monomorphism $\tau' : E \to \Omega$. Since $E \mid F$ is normal, $\tau'(E) = E$ by Proposition 17.20, so $\tau' \in G$ and $\tau = \tau'|_K$.

By Proposition 17.20 applied to $K \mid F$, it follows that $K \mid F$ is normal if and only if $\sigma(K) = K$ for every $\sigma \in G$. Since $K = \mathcal{F}(H)$ and $\sigma(\mathcal{F}(H)) = \mathcal{F}(\sigma H \sigma^{-1})$ by Theorem 18.11 2), this holds if and only if $\mathcal{F}(\sigma H \sigma^{-1}) = \mathcal{F}(H)$ for every $\sigma \in G$. As $\mathcal{F}$ is injective by Theorem 18.11 1), this is equivalent to $\sigma H \sigma^{-1} = H$ for every $\sigma \in G$, that is, to H being normal in G.

3) Let H be normal in G. By 2), $\sigma(K) = K$ for every $\sigma \in G$, so $\sigma|_K \in \mathrm{Gal}(K \mid F)$, and

$$G \to \mathrm{Gal}(K \mid F), \quad \sigma \mapsto \sigma|_K,$$

is a group homomorphism. Its kernel is $\{\sigma \in G : \sigma|_K = \mathrm{id}\} = \mathrm{Gal}(E \mid K) = \mathcal{G}(\mathcal{F}(H)) = H$, by Theorem 18.11 1). It is surjective: every $\rho \in \mathrm{Gal}(K \mid F)$, composed with the inclusion $K \to \Omega$, is an F-monomorphism $K \to \Omega$, hence the restriction of some $\sigma \in G$ by the first paragraph of 2). By the first isomorphism theorem, $G/H \cong \mathrm{Gal}(K \mid F)$.

4) By Theorem 18.11 1), $H = \mathrm{Gal}(E \mid K)$. Since $E \mid F$ and $E \mid K$ are Galois, Proposition 18.3 gives $|G| = [E : F]$ and $|H| = [E : K]$. By the tower law $[E : F] = [E : K][K : F]$, so $|G| = |H|\,[K : F]$, and by Lagrange's theorem $|G : H| = |G| \,/\, |H| = [K : F]$. ∎

19. Worked exercises on fields

Throughout the first two exercises, q is a prime power, $m \in \mathbb{N}$, $L = \mathbb{F}_{q^m}$ and $K = \mathbb{F}_q$; we write $\mathrm{N} = \mathrm{N}_{L \mid K}$ and $\mathrm{tr} = \mathrm{tr}_{L \mid K}$. By Theorem 14.2, $c^q = c$ for every $c \in K$.

Example 19.1. Show that for $\alpha, \beta \in L$ and $a, c \in K$:

1) $\mathrm{N}(\alpha\beta) = \mathrm{N}(\alpha)\,\mathrm{N}(\beta)$;

2) $\mathrm{N}(c\alpha) = c^m\,\mathrm{N}(\alpha)$;

3) $\mathrm{N}(a) = a^m$.

Solution. 1) Rearranging the factors,

$$\mathrm{N}(\alpha)\,\mathrm{N}(\beta) = \alpha\alpha^q \cdots \alpha^{q^{m-1}} \cdot \beta\beta^q \cdots \beta^{q^{m-1}} = (\alpha\beta)(\alpha\beta)^q \cdots (\alpha\beta)^{q^{m-1}} = \mathrm{N}(\alpha\beta).$$

2) Since $c^{q^i} = c$ for every i,

$$\mathrm{N}(c\alpha) = c\alpha \cdot (c\alpha)^q \cdots (c\alpha)^{q^{m-1}} = \left(cc^q \cdots c^{q^{m-1}}\right)\left(\alpha\alpha^q \cdots \alpha^{q^{m-1}}\right) = c^m\,\mathrm{N}(\alpha).$$

3) $\mathrm{N}(a) = aa^q \cdots a^{q^{m-1}} = a^m$, since $a^{q^i} = a$. ◊

Example 19.2. Show that for $\alpha, \beta \in L$ and $a, c \in K$:

1) $\mathrm{tr}(\alpha + \beta) = \mathrm{tr}(\alpha) + \mathrm{tr}(\beta)$;

2) $\mathrm{tr}(c\alpha) = c\,\mathrm{tr}(\alpha)$;

3) $\operatorname{tr}(a) = ma$.

Solution. 1) $\alpha \mapsto \alpha^{q^i}$ is a ring homomorphism (Theorem 17.2), so

$$\begin{aligned}\operatorname{tr}(\alpha+\beta) &= (\alpha+\beta) + (\alpha+\beta)^q + \cdots + (\alpha+\beta)^{q^{m-1}} \\ &= \alpha + \beta + \alpha^q + \beta^q + \cdots + \alpha^{q^{m-1}} + \beta^{q^{m-1}} = \operatorname{tr}(\alpha) + \operatorname{tr}(\beta).\end{aligned}$$

2) Using $c^{q^i} = c$,

$$\operatorname{tr}(c\alpha) = c\alpha + (c\alpha)^q + \cdots + (c\alpha)^{q^{m-1}} = c\alpha + c\alpha^q + \cdots + c\alpha^{q^{m-1}} = c\operatorname{tr}(\alpha).$$

3) $\operatorname{tr}(a) = a + a^q + \cdots + a^{q^{m-1}} = ma$, since $a^{q^i} = a$. ◊

Definition 19.3 (Perfect field). A field F is *perfect* if every irreducible polynomial in $F[x]$ is separable.

By Lemma 17.7 and Theorem 17.8, fields of characteristic 0 and finite fields are perfect.

Example 19.4. Show that an algebraic extension of a perfect field is perfect.

Solution. Let K be perfect and $L \mid K$ algebraic. Let $f \in L[x]$ be irreducible; we show that f is separable. Let E be a splitting field of f over L and $\alpha_1 \in E$ a root of f. Since f is irreducible, it is a scalar multiple of $m_{\alpha_1, L}$.

The element α_1 is algebraic over K. Indeed, the coefficients of f are algebraic over K, so they generate a finite extension K' of K (tower law). As α_1 is algebraic over K', $[K'(\alpha_1) : K]$ is finite, and α_1 is algebraic over K.

Let $g = m_{\alpha_1, K}$. Then $g \in K[x] \subseteq L[x]$ and $g(\alpha_1) = 0$, so $f \mid g$ in $L[x]$. Since K is perfect, g is separable: in a splitting field E' of g over E, $g = (x-\alpha_1)\cdots(x-\alpha_r)$ with pairwise distinct α_i. As $f \mid g$, f splits in $E'[x]$ into pairwise distinct linear factors. Hence f is separable, and L is perfect. ◊

Example 19.5. Let p be a prime and $a \in \mathbb{F}_p$. Show that $f(x) = x^p - x + a$ is separable over $\mathbb{F}_p$.

Solution. $f'(x) = px^{p-1} - 1 = -1$ in $\mathbb{F}_p[x]$. Hence $\gcd(f, f') = 1$, and f is separable by Proposition 17.6. ◊

Example 19.6. Is $x^4 - 7$ irreducible over $\mathbb{F}_5$? Determine the irreducible factors of $x^4 + 1$ over $\mathbb{F}_3$.

Solution. Over $\mathbb{F}_5$, $x^4 - 7 = x^4 - 2$. By Theorem 14.2, $x^4 \in \{0, 1\}$ for $x \in \mathbb{F}_5$, so $x^4 - 2$ has no root and no linear factor. If it were reducible, it would therefore be a product of two monic quadratics:

$$x^4 - 2 = (x^2 + ax + b)(x^2 + cx + d) = x^4 + (a+c)x^3 + (b + d + ac)x^2 + (ad + bc)x + bd.$$

Comparing coefficients, $a + c = 0$, $b + d + ac = 0$, $ad + bc = 0$, $bd = -2$. So $c = -a$, $b + d = a^2$, $a(d - b) = 0$ and $bd = -2$.

- If $a = 0$, then $b + d = 0$ and $-b^2 = -2$, so $b^2 = 2$. But $\left(\frac{2}{5}\right) = -1$, since $5 \equiv -3 \pmod 8$.
- If $a \neq 0$, then $b = d$ and $b^2 = -2$. But $\left(\frac{-2}{5}\right) = \left(\frac{-1}{5}\right)\left(\frac{2}{5}\right) = 1 \cdot (-1) = -1$.

So $x^4 - 7$ is irreducible over $\mathbb{F}_5$.

Over $\mathbb{F}_3$, $x^4 + 1 \in \{1, 2\}$ for every $x \in \mathbb{F}_3$, again since $x^4 \in \{0, 1\}$, so there is no linear factor. With the same ansatz, $c = -a$, $b + d = a^2$, $a(d - b) = 0$ and $bd = 1$. In $\mathbb{F}_3$ the units ± 1 are their own inverses, so $bd = 1$ forces $b = d = 1$ or $b = d = -1$.
- If $b = d = 1$, then $a^2 = 2$, which has no solution in $\mathbb{F}_3$.
- If $b = d = -1$, then $a^2 = -2 = 1$, so $a = 1$, $c = -1$ is a solution.

Hence

$$x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1) \quad \text{in } \mathbb{F}_3[x],$$

which can be checked directly: the product is $(x^2 - 1)^2 - x^2 = x^4 - 3x^2 + 1 = x^4 + 1$. Both factors are irreducible, since neither has a root in $\mathbb{F}_3$: $x^2 + x - 1$ takes the values $-1, 1, 2$ and $x^2 - x - 1$ the values $-1, -1, 1$ at $x = 0, 1, 2$. ◊

Definition 19.7 (Purely inseparable polynomial). Let F be a field with algebraic closure $\overline{F}$. A nonconstant $f \in F[x]$ is *purely inseparable* if it has exactly one root in $\overline{F}$.

Example 19.8. Let F be a finite field and $f \in F[x]$ purely inseparable, with root $\alpha \in \overline{F}$. Prove that $f = c \cdot m_{\alpha,F}^k$ for some $k \in \mathbb{N}$ and $c \in F^\times$.

Solution. We proceed by induction on $n = \deg f \geq 1$. Since $m_{\alpha,F}$ divides f in $F[x]$, write $f = m_{\alpha,F} g$ with $g \in F[x]$ and $\deg g < n$. If $\deg g = 0$, then $g = c \in F^\times$ and $f = c \cdot m_{\alpha,F}$. Otherwise g has a root in $\overline{F}$, and every root of g is a root of f, so g is purely inseparable with the root α. By the induction hypothesis $g = a_0 \cdot m_{\alpha,F}^{k_0}$ with $k_0 \in \mathbb{N}$ and $a_0 \in F^\times$, hence

$$f = m_{\alpha,F} \cdot g = a_0 \cdot m_{\alpha,F}^{k_0+1}.$$

The case $n = 1$ is the case $\deg g = 0$. The argument uses only that F is a field. For finite F more is true: $m_{\alpha,F}$ is separable by Theorem 17.8, and its only root is α, so $m_{\alpha,F} = x - \alpha$ and $\alpha \in F$. ◊

Example 19.9. Show that no irreducible polynomial of degree at least 2 over $\mathbb{F}_2$ has an even number of nonzero terms.

Solution. Let $f = x^k + \varepsilon_{k-1}x^{k-1} + \cdots + \varepsilon_1 x + \varepsilon_0$ with $k \geq 2$ and $\varepsilon_i \in \{0, 1\}$. Then $f(1)$ is the number of nonzero terms modulo 2. If this number is even, then $f(1) = 0$, so $x - 1$ divides f, and since $\deg f \geq 2$, f is reducible. The degree condition is needed: $x + 1$ is irreducible with two nonzero terms. ◊

Example 19.10. Let q be a prime power and $d \in \mathbb{N}$. Prove that $x^{q^d} - x$ is the product of all monic irreducible $f \in \mathbb{F}_q[x]$ with $\deg f \mid d$, each with multiplicity 1. Design an algorithm that tests whether a polynomial over $\mathbb{F}_q$ is irreducible.

Solution. Let $P = x^{q^d} - x$. Since $P' = q^d x^{q^d-1} - 1 = -1$, $\gcd(P, P') = 1$, so by Proposition 17.6 P has no multiple roots and no irreducible factor of multiplicity greater than 1. As P is monic, it is the product of its distinct monic irreducible factors. It remains to show that a monic irreducible $f \in \mathbb{F}_q[x]$ divides P if and only if $\deg f \mid d$. Let $r = \deg f$.

Suppose $r \mid d$. In the field $\mathbb{F}_q[x]/(f)$, which has q^r elements (Remark 14.5), the class α of x is a root of f and satisfies $\alpha^{q^r} = \alpha$ by Theorem 14.2. Iterating d/r times gives $\alpha^{q^d} = \alpha$, so $P(\alpha) = 0$, and $f = m_{\alpha,\mathbb{F}_q}$ divides P.

Conversely, suppose $f \mid P$. Let E be a splitting field of P over $\mathbb{F}_q$ and $R \subseteq E$ the set of roots of P; it has exactly q^d elements, since the roots are simple. R is a subfield, as in the proof of Lemma 14.6, and it contains $\mathbb{F}_q$. A root $\alpha \in E$ of f lies in R, so $\mathbb{F}_q(\alpha) \subseteq R$. Then R is a vector space over $\mathbb{F}_q(\alpha)$, which has q^r elements, so $q^d = (q^r)^s$ for some $s \in \mathbb{N}$, and $r \mid d$.

Algorithm. Let $f \in \mathbb{F}_q[x]$ be monic of degree $n \geq 1$.

1) If $x^{q^n} \not\equiv x \pmod f$, answer NO.
2) If $\gcd\left(f, x^{q^{n/r}} - x\right) \neq 1$ for some prime $r \mid n$, answer NO.
3) Otherwise answer YES.

The powers $x^{q^k} \bmod f$ are computed by repeated q-th powering modulo f.

Correctness. Let f be irreducible. By the first part $f \mid x^{q^n} - x$, so step 1 passes. For a prime $r \mid n$, $\gcd\left(f, x^{q^{n/r}} - x\right)$ is 1 or f, and it is not f, since $n \nmid n/r$. So the answer is YES. Conversely, suppose the answer is YES. By step 1, $f \mid x^{q^n} - x$, so by the first part f is a product of distinct monic irreducible factors whose degrees divide n. If f were reducible, it would have a factor g of degree $e < n$ with $e \mid n$. Choose a prime r dividing n/e; then $e \mid n/r$, so $g \mid x^{q^{n/r}} - x$ and step 2 answers NO. Hence f is irreducible.

Step 2 cannot be replaced by the congruence test $x^{q^{n/r}} \equiv x \pmod f$. Over $\mathbb{F}_2$, $f = x(x^2 + x + 1)(x^3 + x + 1)$ has degree 6 and passes step 1, since all factor degrees divide 6. Neither $x^{2^3} \equiv x$ nor $x^{2^2} \equiv x \pmod f$ holds, because $2 \nmid 3$ and $3 \nmid 2$; yet f is reducible. ◊

Example 19.11. Give fields $F \subseteq K \subseteq L$ such that $L \mid K$ and $K \mid F$ are normal, but $L \mid F$ is not normal.

Solution. Take $\mathbb{Q} \subseteq \mathbb{Q}\left(\sqrt{2}\right) \subseteq \mathbb{Q}\left(\sqrt[4]{2}\right)$, where $\sqrt[4]{2}$ is the positive real fourth root of 2 and $\sqrt{2} = \sqrt[4]{2}^2$.

First, $x^4 - 2$ is irreducible over $\mathbb{Q}$. Its complex roots are $\pm a$ and $\pm ia$ with $a = \sqrt[4]{2}$, and none is rational, since $v_2(z^4) = 4v_2(z) \neq 1$. A monic quadratic factor over $\mathbb{Q}$ would be $(x - r)(x - r')$ for two of these roots, with constant term $rr' \in \left\{-\sqrt{2}, \sqrt{2}, \pm i\sqrt{2}\right\}$, none of which is rational. Hence $\left[\mathbb{Q}\left(\sqrt[4]{2}\right) : \mathbb{Q}\right] = 4$, and by the tower law $\left[\mathbb{Q}\left(\sqrt[4]{2}\right) : \mathbb{Q}\left(\sqrt{2}\right)\right] = 2$.

$\mathbb{Q}\left(\sqrt{2}\right) \mid \mathbb{Q}$ and $\mathbb{Q}\left(\sqrt[4]{2}\right) \mid \mathbb{Q}\left(\sqrt{2}\right)$ have degree 2, so both are normal (Example 17.17); the relevant polynomials are $x^2 - 2$ and $x^2 - \sqrt{2}$. But $\mathbb{Q}\left(\sqrt[4]{2}\right) \mid \mathbb{Q}$ is not normal: the irreducible $x^4 - 2$ has the root $\sqrt[4]{2} \in \mathbb{Q}\left(\sqrt[4]{2}\right)$, while its roots $\pm i\sqrt[4]{2}$ are not real and so do not lie in $\mathbb{Q}\left(\sqrt[4]{2}\right) \subseteq \mathbb{R}$. ◊

Example 19.12. Prove that $\mathbb{Q}\left(\sqrt[3]{3}\right) \mid \mathbb{Q}$ is not normal.

Solution. $x^3 - 3$ is irreducible over $\mathbb{Q}$, since it has degree 3 and no rational root ($z^3 = 3$ would give $3v_3(z) = 1$). Its roots are $\sqrt[3]{3}$, $\varepsilon\sqrt[3]{3}$, $\varepsilon^2\sqrt[3]{3}$, where ε is a primitive third root of unity. The root $\varepsilon\sqrt[3]{3}$ is not real, so $\varepsilon\sqrt[3]{3} \notin \mathbb{Q}\left(\sqrt[3]{3}\right) \subseteq \mathbb{R}$. ◊

20. Cyclotomic fields

Definition 20.1 (Cyclotomic extension). Let F be a field and $n \in \mathbb{N}$ with $\operatorname{char} F \nmid n$. A *cyclotomic extension* of F is a splitting field E of $x^n - 1$ over F.

Proposition 20.2. Let E be a cyclotomic extension of F for $x^n - 1$. The roots of $x^n - 1$ in E, the *n-th roots of unity*, form a cyclic subgroup μ_n of order n of $E^\times$. A generator of μ_n is a *primitive n-th root of unity*, denoted ζ_n, and $E = F(\zeta_n)$.

Proof. The derivative of $x^n - 1$ is nx^{n-1}, which is nonzero because $\operatorname{char} F \nmid n$ and vanishes only at 0. Hence $\gcd(x^n - 1, nx^{n-1}) = 1$, and $x^n - 1$ has n distinct roots in E by Proposition 17.6. They form a subgroup of $E^\times$, since $(\alpha\beta^{-1})^n = \alpha^n\beta^{-n} = 1$. A finite subgroup of the multiplicative group of a field is cyclic: the second paragraph of the proof of Theorem 14.2 uses only that a polynomial $x^d - 1$ has at most d roots in a field. So $\mu_n = \{1, \zeta_n, \ldots, \zeta_n^{n-1}\}$, and E, being generated over F by μ_n, equals $F(\zeta_n)$. ■

Example 20.3. For a prime p, $\mathbb{Q}(\zeta_p)$ is a cyclotomic extension of $\mathbb{Q}$.

Proposition 20.4. Let $E = F(\zeta_n)$ be a cyclotomic extension. For $\sigma \in \operatorname{Gal}(E \mid F)$ there is $r_\sigma \in \mathbb{Z}$ with $\gcd(r_\sigma, n) = 1$ and $\sigma(\zeta_n) = \zeta_n^{r_\sigma}$, and $\sigma \mapsto r_\sigma \bmod n$ is an injective group homomorphism $\operatorname{Gal}(E \mid F) \to U_n$, where $U_n = (\mathbb{Z}/n\mathbb{Z})^\times$. In particular $\operatorname{Gal}(E \mid F)$ is abelian. If $\operatorname{char} F = 0$ or F is finite, then $E \mid F$ is Galois and $[E : F] = |\operatorname{Gal}(E \mid F)|$ divides $\varphi(n)$.

Proof. By Theorem 17.12, σ maps μ_n into itself, and it preserves orders of elements, since it is an injective homomorphism of $E^\times$. So $\sigma(\zeta_n)$ is a primitive n-th root of unity, $\sigma(\zeta_n) = \zeta_n^{r_\sigma}$ with $\gcd(r_\sigma, n) = 1$, and r_σ is determined modulo n. For $\sigma, \tau \in \operatorname{Gal}(E \mid F)$,

$$(\tau\sigma)(\zeta_n) = \tau(\zeta_n^{r_\sigma}) = \tau(\zeta_n)^{r_\sigma} = \zeta_n^{r_\tau r_\sigma},$$

so the map is a homomorphism. It is injective, since σ is determined by $\sigma(\zeta_n)$ on $E = F(\zeta_n)$.

If $\operatorname{char} F = 0$ or F is finite, every irreducible polynomial over F is separable by Lemma 17.7 and Theorem 17.8, so $E \mid F$ is separable. It is normal by Theorem 17.21, being a splitting field. Hence it is Galois, and the last statement follows from Proposition 18.3 and Lagrange's theorem, as $|U_n| = \varphi(n)$. ■

From now on $F = \mathbb{Q}$ and $\zeta_n = e^{2\pi i/n} \in \mathbb{C}$. The primitive n-th roots of unity in $\mathbb{C}$ are ζ_n^r with $1 \le r \le n$ and $\gcd(r, n) = 1$.

Definition 20.5 (Cyclotomic polynomial). The *n-th cyclotomic polynomial* is

$$\Phi_n(x) = \prod_{1 \le r \le n,\ \gcd(r,n)=1} (x - \zeta_n^r), \quad \deg \Phi_n = \varphi(n).$$

Proposition 20.6. For every $n \in \mathbb{N}$, $x^n - 1 = \prod_{d \mid n} \Phi_d(x)$.

Proof. Both sides are monic with simple roots, so it suffices to compare roots. If $\zeta^n = 1$ and ζ has order d in $\mathbb{C}^\times$, then $d \mid n$ and ζ is a primitive d-th root of unity, that is, a root of Φ_d. Conversely, if $d \mid n$, every root ζ of Φ_d satisfies $\zeta^n = \left(\zeta^d\right)^{n/d} = 1$. Since every root of unity has exactly one order, the roots of the Φ_d, $d \mid n$, are pairwise distinct. ∎

Example 20.7. If $n = p$ is prime, then $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + x^{p-2} + \cdots + x + 1$, by Proposition 20.6 with $\Phi_1(x) = x - 1$.

Proposition 20.8. $\Phi_n \in \mathbb{Z}[x]$ for every $n \in \mathbb{N}$.

Proof. Induction on n. For $n = 1$, $\Phi_1(x) = x - 1$. Let $n > 1$ and assume $\Phi_k \in \mathbb{Z}[x]$ for every $k \le n - 1$. By Proposition 20.6,

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x) = \Phi_n(x) \prod_{d \mid n,\ d \ne n} \Phi_d(x), \quad \text{so} \quad \Phi_n(x) = \frac{x^n - 1}{\prod_{d \mid n,\ d \ne n} \Phi_d(x)}.$$

The denominator is monic and lies in $\mathbb{Z}[x]$ by the induction hypothesis. Division with remainder by a monic polynomial in $\mathbb{Z}[x]$ stays in $\mathbb{Z}[x]$, and the remainder is 0, so $\Phi_n \in \mathbb{Z}[x]$. ∎

Remark 20.9 (Recall). *Gauss's lemma.* If $h \in \mathbb{Z}[x]$ is monic and $h = fg$ with monic $f, g \in \mathbb{Q}[x]$, then $f, g \in \mathbb{Z}[x]$.

Theorem 20.10. Φ_n is irreducible over $\mathbb{Q}$. Consequently $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \varphi(n)$ and $\mathrm{Gal}(\mathbb{Q}(\zeta_n) \mid \mathbb{Q}) \cong U_n$.

Proof. Let f be the minimal polynomial over $\mathbb{Q}$ of a primitive n-th root of unity θ. Then f is monic and $f \mid x^n - 1$, so $x^n - 1 = fg$ with $f, g \in \mathbb{Z}[x]$ monic, by Remark 20.9. We show that θ^p is a root of f for every prime $p \nmid n$.

Suppose $f(\theta^p) \ne 0$. Since $(\theta^p)^n - 1 = 0 = f(\theta^p) g(\theta^p)$, we get $g(\theta^p) = 0$. So θ is a root of $g(x^p)$, and $f(x) \mid g(x^p)$ in $\mathbb{Q}[x]$, hence $g(x^p) = f(x)h(x)$ with $h \in \mathbb{Z}[x]$ by Remark 20.9. Reduce modulo p and work in $\mathbb{F}_p[x]$; for $q = \sum a_i x^i \in \mathbb{Z}[x]$ write $\overline{q} = \sum (a_i \bmod p) x^i$. By Theorem 14.4 and $a^p = a$ for $a \in \mathbb{F}_p$,

$$\overline{g}(x^p) = \overline{g}(x)^p = \overline{f}(x)\overline{h}(x) \in \mathbb{F}_p[x].$$

Hence any irreducible factor of $\overline{f}$ divides $\overline{g}^p$, hence $\overline{g}$, so $\overline{f}$ and $\overline{g}$ have a common factor. Reducing $x^n - 1 = fg$ gives $x^n - 1 = \overline{f}\,\overline{g}$ in $\mathbb{F}_p[x]$, so $x^n - 1$ has a multiple root over $\mathbb{F}_p$. But its derivative nx^{n-1} is nonzero in $\mathbb{F}_p[x]$, since $p \nmid n$, and $\gcd(x^n - 1, nx^{n-1}) = 1$, contradicting Proposition 17.6.

Since θ^p is again a primitive n-th root of unity and f, being monic and irreducible with the root θ^p, is also its minimal polynomial, the argument applies to θ^p in place of θ. Every primitive n-th root of unity is ζ_n^r with $\gcd(r,n)=1$, and writing $r = p_1 \cdots p_k$ as a product of primes not dividing n, it is obtained from ζ_n by successive prime powers. Hence all primitive n-th roots of unity are roots of $f = m_{\zeta_n,\mathbb{Q}}$, so Φ_n, whose roots are simple, divides f in $\mathbb{Q}[x]$. Since f is irreducible and Φ_n is monic and nonconstant, $f = \Phi_n$. The last statement follows from Proposition 20.4, since $|\operatorname{Gal}(\mathbb{Q}(\zeta_n) \mid \mathbb{Q})| = \varphi(n) = |U_n|$. ∎

21. Ruler and compass constructions

The two admissible operations in the plane $\mathbb{R}^2$ are:

1) *ruler*: draw the straight line through two given points;

2) *compass*: draw the circle with a given point as centre passing through a given point.

Definition 21.1 (Constructible point). Let $P_0 \subseteq \mathbb{R}^2$. A point $r \in \mathbb{R}^2$ is *constructible* from P_0 if there is a sequence $r_1, \dots, r_n = r$ of points such that for each $i = 1, \dots, n$, the point r_i is an intersection point of two distinct lines or circles drawn by operations 1 or 2 from points of $P_{i-1} = P_0 \cup \{r_1, \dots, r_{i-1}\}$. A real number x is *constructible* if $(x, 0)$ is constructible from $\{(0,0), (1,0)\}$.

With each stage we associate a subfield of $\mathbb{R}$. Let $\mathbb{K}_0$ be the subfield of $\mathbb{R}$ generated by the coordinates of the points of P_0. If $r_i = (x_i, y_i)$, put inductively $\mathbb{K}_i = \mathbb{K}_{i-1}(x_i, y_i)$. Then $\mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_n \subseteq \mathbb{R}$.

Lemma 21.2. For each i, the coordinates x_i and y_i are zeros of polynomials of degree at most 2 over $\mathbb{K}_{i-1}$.

Proof. All points of P_{i-1} have coordinates in $\mathbb{K}_{i-1}$. The line through $A = (p, q)$ and $B = (r, s)$ is

$$(s - q)(x - p) = (r - p)(y - q),$$

a linear equation with coefficients in $\mathbb{K}_{i-1}$. The circle with centre $C = (t, u)$ through $D = (v, w)$ is

$$(x - t)^2 + (y - u)^2 = \rho^2, \quad \rho^2 = (v - t)^2 + (w - u)^2 \in \mathbb{K}_{i-1},$$

that is, $x^2 + y^2 + ax + by + c = 0$ with $a, b, c \in \mathbb{K}_{i-1}$.

Two lines. The intersection point solves a 2×2 linear system over $\mathbb{K}_{i-1}$ with nonzero determinant, so $x_i, y_i \in \mathbb{K}_{i-1}$ by Cramer's rule.

A line and a circle. If the line is not vertical, it is $y = mx + k$ with $m, k \in \mathbb{K}_{i-1}$. Substituting into the circle gives a polynomial of degree 2 over $\mathbb{K}_{i-1}$ vanishing at x_i, and $y_i = mx_i + k$ satisfies the polynomial obtained by substituting $x = (y - k)/m$ if $m \neq 0$, or $y_i = k$ if $m = 0$. A vertical line $x = k$ gives $x_i = k$ and a quadratic equation for y_i.

Two circles. Subtracting the equations of two distinct circles $x^2 + y^2 + ax + by + c = 0$ and $x^2 + y^2 + a'x + b'y + c' = 0$ gives the linear equation $(a - a')x + (b - b')y + (c - c') = 0$. If $a = a'$

and $b = b'$, the circles are concentric and distinct and do not meet. Otherwise this is a line over $\mathbb{K}_{i-1}$ through every common point, and we are in the previous case. ∎

Theorem 21.3. If $r = (x, y)$ is constructible from $P_0 \subseteq \mathbb{R}^2$ and $\mathbb{K}_0$ is the subfield of $\mathbb{R}$ generated by the coordinates of the points of P_0, then $[\mathbb{K}_0(x) : \mathbb{K}_0]$ and $[\mathbb{K}_0(y) : \mathbb{K}_0]$ are powers of 2.

Proof. By Lemma 21.2, $[\mathbb{K}_{i-1}(x_i) : \mathbb{K}_{i-1}] \in \{1, 2\}$, and y_i is a zero of a polynomial of degree at most 2 over $\mathbb{K}_{i-1} \subseteq \mathbb{K}_{i-1}(x_i)$, so $[\mathbb{K}_{i-1}(x_i, y_i) : \mathbb{K}_{i-1}(x_i)] \in \{1, 2\}$. By the tower law

$$[\mathbb{K}_{i-1}(x_i, y_i) : \mathbb{K}_{i-1}] = [\mathbb{K}_{i-1}(x_i, y_i) : \mathbb{K}_{i-1}(x_i)][\mathbb{K}_{i-1}(x_i) : \mathbb{K}_{i-1}] \in \{1, 2, 4\}.$$

Since $\mathbb{K}_i = \mathbb{K}_{i-1}(x_i, y_i)$, the tower law gives that $[\mathbb{K}_n : \mathbb{K}_0]$ is a power of 2. As $x = x_n \in \mathbb{K}_n$, $[\mathbb{K}_n : \mathbb{K}_0] = [\mathbb{K}_n : \mathbb{K}_0(x)][\mathbb{K}_0(x) : \mathbb{K}_0]$, so $[\mathbb{K}_0(x) : \mathbb{K}_0]$ divides a power of 2 and is itself a power of 2. The same holds for y. ∎

Below, $P_0 = \{(0, 0), (1, 0)\}$, so $\mathbb{K}_0 = \mathbb{Q}$. We use the elementary fact that if a segment of length ℓ has been constructed, so has $(\ell, 0)$: transfer the segment to the origin and intersect the circle of radius ℓ about $(0, 0)$ with the x-axis.

Theorem 21.4 (Duplication of the cube). The cube cannot be duplicated with ruler and compass: there is no construction from P_0 of the side of a cube whose volume is twice the volume of the unit cube.

Proof. The unit cube has volume 1. A cube of volume 2 has side $\alpha = \sqrt[3]{2}$, so a duplication would make $(\alpha, 0)$ constructible from P_0. By Theorem 21.3, $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ would be a power of 2. But α is a root of $x^3 - 2$, which is irreducible over $\mathbb{Q}$, since it has degree 3 and no rational root ($3v_2(z) = 1$ is impossible). So $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 3$, which is not a power of 2. ∎

Theorem 21.5 (Trisection of the angle). The angle $60°$ cannot be trisected with ruler and compass.

Proof. The angle $60°$ is constructible from P_0: the circles of radius 1 about $(0, 0)$ and $(1, 0)$ meet at $\left(1/2, \sqrt{3}/2\right)$. A trisection would make the point $(\cos 20°, \sin 20°)$, and hence $\alpha = \cos 20°$, constructible from P_0. Recall $\cos(3\beta) = 4\cos^3\beta - 3\cos\beta$. With $\beta = 20°$,

$$\frac{1}{2} = \cos 60° = 4\cos^3 20° - 3\cos 20° \iff 8\cos^3 20° - 6\cos 20° - 1 = 0,$$

so α is a root of $f(x) = 8x^3 - 6x - 1$. Putting $y = 2x$, $f(x) = y^3 - 3y - 1$. A rational root of the monic integer polynomial $y^3 - 3y - 1$ would be an integer dividing 1, but $1 - 3 - 1 = -3$ and $-1 + 3 - 1 = 1$. So $y^3 - 3y - 1$ has no rational root, and being cubic, it is irreducible over $\mathbb{Q}$; hence so is f. Therefore $[\mathbb{Q}(\cos 20°) : \mathbb{Q}] = 3$, not a power of 2, contradicting Theorem 21.3. ∎

Theorem 21.6 (Squaring the circle). The circle cannot be squared with ruler and compass: there is no construction of a square whose area equals that of a given circle.

Proof. Take the unit circle with centre $(0, 0)$, so P_0 is as above. A square of area π has side $\sqrt{\pi}$, so squaring the circle would make $(\sqrt{\pi}, 0)$ constructible. By Theorem 21.3, $[\mathbb{Q}(\sqrt{\pi}) : \mathbb{Q}]$ would be

a power of 2, in particular finite, so $\sqrt{\pi}$ and hence $\pi = (\sqrt{\pi})^2$ would be algebraic over $\mathbb{Q}$. This contradicts the following theorem. ∎

Theorem 21.7 (Lindemann) (cited)**.** π is transcendental over $\mathbb{Q}$.

Lemma 21.8. Let $t \in \mathbb{N}$. If $2^t + 1$ is prime, then t is a power of 2.

Proof. If t is odd, then $2 \equiv -1 \pmod 3$ gives $2^t \equiv (-1)^t = -1 \pmod 3$, so $3 \mid 2^t + 1$; as $2^t + 1$ is prime, $2^t + 1 = 3$ and $t = 1 = 2^0$. In general write $t = 2^k r$ with $k \geq 0$ and r odd. Then

$$2^t + 1 = \left(2^{2^k}\right)^r + 1 \equiv (-1)^r + 1 = 0 \pmod{2^{2^k} + 1},$$

so $2^{2^k} + 1$ divides the prime $2^t + 1$. Since $2^{2^k} + 1 > 1$, it equals $2^t + 1$, so $r = 1$ and $t = 2^k$. ∎

Definition 21.9 (Fermat numbers). The *Fermat numbers* are $F_n = 2^{2^n} + 1$, $n \geq 0$. A prime of this form is a *Fermat prime*. The first five, $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$, $F_4 = 65537$, are prime.

Definition 21.10. Let $n \geq 3$. The regular n-gon is *constructible* by ruler and compass if the point $(\cos(2\pi/n), \sin(2\pi/n))$ is constructible from $P_0 = \{(0,0), (1,0)\}$.

Theorem 21.11 (Regular n-gon). Let $n \geq 3$. The regular n-gon is constructible by ruler and compass if and only if $n = 2^k p_1 \cdots p_r$ with $k, r \geq 0$ and $p_1, \ldots, p_r$ pairwise distinct Fermat primes.

Proof. We prove necessity; sufficiency is cited below. Suppose the regular n-gon is constructible. Then $c = \cos(2\pi/n)$ is a coordinate of a constructible point, so $[\mathbb{Q}(c) : \mathbb{Q}]$ is a power of 2 by Theorem 21.3. Let $s = \sin(2\pi/n)$ and $\varepsilon_n = c + is = \zeta_n$. Since

$$(is)^2 = -s^2 = c^2 - 1 \in \mathbb{Q}(c),$$

the degree $[\mathbb{Q}(c)(is) : \mathbb{Q}(c)]$ is 1 or 2, so $[\mathbb{Q}(c, is) : \mathbb{Q}]$ is a power of 2. As $\varepsilon_n \in \mathbb{Q}(c, is)$, the tower law and Theorem 20.10 show that $\varphi(n) = [\mathbb{Q}(\varepsilon_n) : \mathbb{Q}]$ divides a power of 2, so $\varphi(n) = 2^m$ for some m.

Write $n = 2^k p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ with distinct odd primes p_i and $\alpha_i \geq 1$. Then

$$\varphi(n) = \varphi\big(2^k\big)\big(p_1^{\alpha_1} - p_1^{\alpha_1 - 1}\big) \cdots \big(p_r^{\alpha_r} - p_r^{\alpha_r - 1}\big) = \varphi\big(2^k\big) p_1^{\alpha_1 - 1} \cdots p_r^{\alpha_r - 1} (p_1 - 1) \cdots (p_r - 1) = 2^m.$$

As the p_i are odd, $\alpha_i = 1$ and $p_i - 1 = 2^{\beta_i}$ for every i. By Lemma 21.8 each $p_i = 2^{\beta_i} + 1$ is a Fermat prime. ∎

Theorem 21.12 (Gauss) (cited)**.** If $n = 2^k p_1 \cdots p_r$ with pairwise distinct Fermat primes p_i, then the regular n-gon is constructible.

22. Worked exercises on Galois groups

Remark 22.1 (Recall). *Schönemann–Eisenstein criterion.* Let $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ and p a prime with $p \nmid a_n$, $p \mid a_i$ for $i < n$ and $p^2 \nmid a_0$. Then f is irreducible over $\mathbb{Q}$.

Example 22.2. Determine $G = \mathrm{Gal}\big(\mathbb{Q}\big(\sqrt[4]{2}, i\big) \mid \mathbb{Q}\big)$.

Solution. Let $a = \sqrt[4]{2} > 0$. Over $\mathbb{C}$,

$$x^4 - 2 = \big(x^2 + \sqrt{2}\big)\big(x^2 - \sqrt{2}\big) = (x-a)(x+a)(x-ia)(x+ia),$$

so $E = \mathbb{Q}(a, i)$ is the splitting field of $x^4 - 2$, and $E \mid \mathbb{Q}$ is normal by Theorem 17.21 and Galois since $\mathrm{char}\ \mathbb{Q} = 0$. By Remark 22.1 with $p = 2$, $x^4 - 2$ is irreducible over $\mathbb{Q}$, so $[\mathbb{Q}(a) : \mathbb{Q}] = 4$. Since $i \notin \mathbb{Q}(a) \subseteq \mathbb{R}$, $[E : \mathbb{Q}(a)] = 2$, and $|G| = [E : \mathbb{Q}] = 8$ by Proposition 18.3. By the tower law $[E : \mathbb{Q}(i)] = 4$, so $x^4 - 2$ is the minimal polynomial of a over $\mathbb{Q}(i)$.

Let τ be complex conjugation restricted to E; it maps E onto E by Proposition 17.20, so $\tau \in G$, with $\tau(a) = a$, $\tau(i) = -i$. By Lemma 17.19, the $\mathbb{Q}(i)$-embedding $\mathbb{Q}(i)(a) \to \mathbb{C}$ with $a \mapsto ia$ exists, and it maps E onto E by Proposition 17.20; call it σ. So $\sigma(a) = ia$, $\sigma(i) = i$, and

$$\sigma^2(a) = \sigma(ia) = -a, \quad \sigma^3(a) = \sigma(-a) = -ia, \quad \sigma^4(a) = a.$$

Hence σ has order 4, τ has order 2, and the eight elements $1, \sigma, \sigma^2, \sigma^3, \tau, \sigma\tau, \sigma^2\tau, \sigma^3\tau$ are pairwise distinct, since they differ on a or on i. So

$$G = \{1, \sigma, \sigma^2, \sigma^3, \tau, \sigma\tau, \sigma^2\tau, \sigma^3\tau\}.$$

Checking on the generators, $\tau^{-1}\sigma\tau(a) = \tau\sigma(a) = \tau(ia) = -ia = \sigma^3(a)$ and $\tau^{-1}\sigma\tau(i) = i = \sigma^3(i)$, so $\tau^{-1}\sigma\tau = \sigma^3 = \sigma^{-1}$. Therefore $G \cong D_4 = \langle \sigma, \tau \mid \sigma^4 = \tau^2 = 1, \tau\sigma\tau = \sigma^{-1} \rangle$, the dihedral group of order 8. ◊

Example 22.3. Determine $G = \mathrm{Gal}\Big(\mathbb{Q}\Big(\sqrt{2+\sqrt{2}}\Big) \mid \mathbb{Q}\Big)$.

Solution. Let $\alpha = \sqrt{2+\sqrt{2}}$. Then

$$\alpha^2 = 2 + \sqrt{2} \iff \alpha^2 - 2 = \sqrt{2} \implies (\alpha^2 - 2)^2 = 2 \iff \alpha^4 - 4\alpha^2 + 2 = 0.$$

By Remark 22.1 with $p = 2$, $x^4 - 4x^2 + 2$ is irreducible, so $m_{\alpha,\mathbb{Q}} = x^4 - 4x^2 + 2$ and $[\mathbb{Q}(\alpha) : \mathbb{Q}] = 4$. Its roots are $\pm\sqrt{2 \pm \sqrt{2}}$. Let $\beta = \sqrt{2 - \sqrt{2}}$; since $\alpha\beta = \sqrt{4-2} = \sqrt{2}$, we get $\beta = \sqrt{2}/\alpha = (\alpha^2 - 2)/\alpha \in \mathbb{Q}(\alpha)$. So all roots lie in $\mathbb{Q}(\alpha)$, which is their splitting field; $\mathbb{Q}(\alpha) \mid \mathbb{Q}$ is normal and separable, hence Galois, and $|G| = 4$.

Let $\sigma \in G$ with $\sigma(\alpha) = \beta$; it exists by Theorem 17.13 and Proposition 17.20, since β is a root of $m_{\alpha,\mathbb{Q}}$. Then

$$2 + \sigma\big(\sqrt{2}\big) = \sigma\big(2 + \sqrt{2}\big) = \sigma(\alpha^2) = \sigma(\alpha)^2 = \beta^2 = 2 - \sqrt{2},$$

so $\sigma\big(\sqrt{2}\big) = -\sqrt{2}$, and

$$\sigma^2(\alpha) = \sigma(\beta) = \sigma\left(\frac{\sqrt{2}}{\alpha}\right) = \frac{\sigma\left(\sqrt{2}\right)}{\sigma(\alpha)} = \frac{-\sqrt{2}}{\beta} = -\alpha.$$

Hence $\sigma^2 \neq \mathrm{id}$, σ has order 4, and $G = \langle\sigma\rangle \cong \mathbb{Z}_4$. ◊

Example 22.4. Determine the Galois groups of $x^3 + 2$ and $x^4 - 7x^2 + 10$ over $\mathbb{Q}$.

Solution. $x^3 + 2$. Since $x^3 + 2 = x^3 + \left(\sqrt[3]{2}\right)^3 = \left(x + \sqrt[3]{2}\right)\left(x^2 - \sqrt[3]{2}x + \sqrt[3]{4}\right)$, the roots are

$$x_1 = -\sqrt[3]{2}, \quad x_2 = \frac{\sqrt[3]{2}}{2} + \frac{i\sqrt{3}\sqrt[3]{2}}{2}, \quad x_3 = \frac{\sqrt[3]{2}}{2} - \frac{i\sqrt{3}\sqrt[3]{2}}{2}.$$

The splitting field is $E = \mathbb{Q}\left(\sqrt[3]{2}, i\sqrt{3}\right)$. By Remark 22.1 with $p = 2$, $x^3 + 2$ is irreducible, so $\left[\mathbb{Q}\left(\sqrt[3]{2}\right) : \mathbb{Q}\right] = 3$, and $i\sqrt{3} \notin \mathbb{Q}\left(\sqrt[3]{2}\right) \subseteq \mathbb{R}$ gives $[E : \mathbb{Q}] = 6 = |G|$ by Proposition 18.3. Each $\sigma \in G$ permutes $\{x_1, x_2, x_3\}$ by Theorem 17.12 and is determined by this permutation, since E is generated by the roots. This gives an injective homomorphism $G \to S_3$, and as $|S_3| = 3! = 6 = |G|$, $G \cong S_3$.

$x^4 - 7x^2 + 10 = (x^2 - 5)(x^2 - 2)$. The roots are $\pm\sqrt{5}, \pm\sqrt{2}$, and the splitting field is $E = \mathbb{Q}\left(\sqrt{2}, \sqrt{5}\right)$. Here $\sqrt{5} \notin \mathbb{Q}\left(\sqrt{2}\right)$: if $\sqrt{5} = u + v\sqrt{2}$ with $u, v \in \mathbb{Q}$, then $5 = u^2 + 2v^2 + 2uv\sqrt{2}$ forces $uv = 0$, and $v = 0$ or $u = 0$ would make $\sqrt{5}$ or $\sqrt{5/2}$ rational. Hence $[E : \mathbb{Q}] = 4 = |G|$. As in the example on $\mathbb{Q}\left(i, \sqrt{5}\right)$ in Section 18, the elements of G are given by sign changes:

$$\sigma : \sqrt{5} \mapsto -\sqrt{5},\ \sqrt{2} \mapsto \sqrt{2}; \quad \tau : \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{5} \mapsto \sqrt{5}; \quad \sigma\tau : \sqrt{2} \mapsto -\sqrt{2},\ \sqrt{5} \mapsto -\sqrt{5}.$$

Each satisfies $\sigma^2 = \tau^2 = (\sigma\tau)^2 = \mathrm{id}$, so $G = \{\mathrm{id}, \sigma, \tau, \sigma\tau\} \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group. ◊

Definition 22.5 (Galois group of a polynomial). Let F be a field and $f \in F[x]$ nonconstant. The *Galois group of f over F* is $\mathrm{Gal}(E \mid F)$, where E is a splitting field of f over F.

Example 22.6. Let F be a field, $f \in F[x]$ and $c \in F$. Show that the Galois groups of $f(x)$ and $f(x + c)$ over F are isomorphic.

Solution. Let $\alpha_1, \ldots, \alpha_n$ be the roots of f in an algebraic closure $\overline{F}$. Then $f(\beta + c) = 0$ if and only if $\beta + c = \alpha_i$ for some i, so the roots of $f(x + c)$ are $\alpha_1 - c, \ldots, \alpha_n - c$. Their splitting fields in $\overline{F}$ are

$$L = F(\alpha_1, \ldots, \alpha_n), \quad L' = F(\alpha_1 - c, \ldots, \alpha_n - c),$$

and $L = L'$, since $c \in F$. So both Galois groups equal $\mathrm{Gal}(L \mid F)$. ◊

Example 22.7. Let F be a finite field with $\mathrm{char}\ F = p$ and $|F| = p^n$. Show that $F \mid \mathbb{F}_p$ is a Galois extension whose Galois group is generated by the Frobenius automorphism $\sigma(x) = x^p$.

Solution. Let $f(x) = x^{p^n} - x$. Then $f'(x) = p^n x^{p^n - 1} - 1 = -1$, so f is separable by Proposition 17.6. By Theorem 14.2 every element of F is a root of f, and f has at most p^n roots, so F is exactly the set of roots of f and is a splitting field of f over $\mathbb{F}_p$. Hence $F \mid \mathbb{F}_p$ is normal (Theorem 17.21) and separable (Theorem 17.8), so it is a Galois extension.

Let $G = \mathrm{Gal}(F \mid \mathbb{F}_p)$. By Theorem 17.2 with $q = p$, $\sigma \in G$, so $\mathbb{F}_p \subseteq \mathcal{F}(\langle\sigma\rangle)$. The fixed field of $\langle\sigma\rangle$ consists of the roots of $x^p - x$ in F, so it has at most p elements; hence $\mathcal{F}(\langle\sigma\rangle) = \mathbb{F}_p$. By Theorem 18.11, $\langle\sigma\rangle = \mathcal{G}(\mathbb{F}_p) = G$. This reproves part of Theorem 17.2 for $q = p$. ◊

Remark 22.8 (Recall). For angles, measured in degrees: an angle α is *constructible* if $(\cos\alpha, \sin\alpha)$ is constructible from $P_0 = \{(0,0),(1,0)\}$. Sums and differences of constructible angles are constructible, and so is half of a constructible angle.

Example 22.9. For which $n \in \mathbb{N}$ is the angle $n°$ constructible?

Solution. We show that $n°$ is constructible if and only if $3 \mid n$. By Theorem 21.11 the regular 5-gon is constructible, 5 being a Fermat prime, so $72°$ is constructible. The angle $60°$ is constructible, hence so are $12° = 72° - 60°$, $3°$ (bisecting twice) and every multiple $3k°$.

Conversely, if $(3k+1)°$ were constructible, so would be $1° = (3k+1)° - 3k°$. If $(3k+2)°$ were constructible, so would be $1° = 3(k+1)° - (3k+2)°$. Then $20°$, a sum of twenty copies of $1°$, would be constructible, and the angle $60°$ could be trisected, contradicting the trisection theorem of Section 21. ◊

Example 22.10. [p4: statement of the exercise not recorded; the solution concerns an isosceles triangle with legs a, base $2b$, height f_a and inscribed circle of radius r]

Solution.

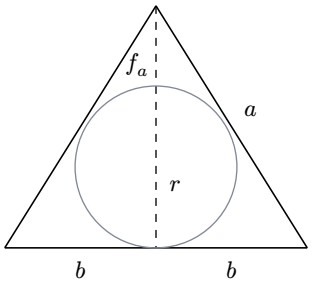


Figure 22.1. *Isosceles triangle with legs a, base $2b$, height f_a and inradius r.*

The height is $f_a = \sqrt{a^2 - b^2}$ and the semiperimeter is $S = (2a + 2b)/2 = a + b$. Computing the area in two ways,

$$\frac{2b \cdot f_a}{2} = b f_a, \quad rS = r(a+b),$$

where r is the radius of the inscribed circle. Hence $b\sqrt{a^2 - b^2} = r(a+b)$, so $b^2(a^2 - b^2) = r^2(a+b)^2$, and dividing by $a + b > 0$,

$$b^2(a-b) = r^2(a+b) \iff b^3 - ab^2 + r^2 a + r^2 b = 0.$$

With $r = 1$, b is a root of $f(x) = x^3 - ax^2 + x + a$.

Whether f is irreducible over $\mathbb{Q}$ depends on a: for $a = 10/3$, $f(2) = 8 - 40/3 + 2 + 10/3 = 0$. Take $a = 5$. Then $f(x) = x^3 - 5x^2 + x + 5$ has no rational root, since $f(1) = 2$, $f(-1) = -2$, $f(5) =$

10, $f(-5) = -150$, so it is irreducible over $\mathbb{Q}$. It has roots in $(1,2)$ and $(4,5)$, since $f(1) = 2 > 0 > -5 = f(2)$ and $f(4) = -7 < 0 < 10 = f(5)$; each gives an isosceles triangle with leg 5 and inradius 1. For both, $[\mathbb{Q}(b) : \mathbb{Q}] = 3$, which is not a power of 2, so b is not constructible from the data $a = 5$, $r = 1$ by Theorem 21.3. ◊

23. Solvable groups

Remark 23.1 (Recall). For a group G, the *commutator subgroup* $G' = \langle g^{-1}h^{-1}gh : g, h \in G \rangle$ is a normal subgroup of G. For $N \trianglelefteq G$, the quotient G/N is abelian if and only if $G' \leq N$. If $\varphi : G \to H$ is a surjective homomorphism, then $\varphi(G') = H'$.

Definition 23.2 (Derived series, solvable group). The *derived series* (commutator series) of a group G is $G^{(0)} = G$, $G^{(n+1)} = \left(G^{(n)}\right)'$ for $n \geq 0$. The group G is *solvable* if $G^{(k)} = 1$ for some $k \in \mathbb{N}$.

Definition 23.3 (Normal series). A *normal series* of G is a chain $1 = H_0 \trianglelefteq H_1 \trianglelefteq \cdots \trianglelefteq H_n = G$ of subgroups, each normal in the next. Its *factors* are the groups H_{i+1}/H_i.

Proposition 23.4. A group G is solvable if and only if it has a normal series with abelian factors.

Proof. ($\Rightarrow$) If $G^{(k)} = 1$, then $1 = G^{(k)} \trianglelefteq G^{(k-1)} \trianglelefteq \cdots \trianglelefteq G^{(0)} = G$, and $G^{(i)}/G^{(i+1)} = G^{(i)}/\left(G^{(i)}\right)'$ is abelian.

($\Leftarrow$) Let $1 = H_0 \trianglelefteq \cdots \trianglelefteq H_n = G$ have abelian factors. We show $G^{(i)} \leq H_{n-i}$ by induction on i. For $i = 0$ this is $G = H_n$. If $G^{(i)} \leq H_{n-i}$, then $G^{(i+1)} = \left(G^{(i)}\right)' \leq (H_{n-i})' \leq H_{n-i-1}$, the last inclusion because H_{n-i}/H_{n-i-1} is abelian. For $i = n$ we get $G^{(n)} \leq H_0 = 1$. ■

Proposition 23.5. If G and H are solvable groups, then so is $G \times H$.

Proof. Commutators in $G \times H$ are formed componentwise, so $(G \times H)' = G' \times H'$, and by induction $(G \times H)^{(i)} = G^{(i)} \times H^{(i)}$. If $G^{(k)} = 1$ and $H^{(l)} = 1$, then $(G \times H)^{(\max(k,l))} = 1$. ■

Proposition 23.6. A finite group G is solvable if and only if it has a normal series whose factors are cyclic.

Proof. Cyclic groups are abelian, so one direction is Proposition 23.4. Conversely, take a normal series with abelian factors. It suffices to refine each step $H_i \trianglelefteq H_{i+1}$ with $A = H_{i+1}/H_i$ finite abelian. By induction on $|A|$, A has a chain $1 = A_0 \leq A_1 \leq \cdots \leq A_s = A$ with cyclic factors: for $A \neq 1$ choose $1 \neq g \in A$, put $A_1 = \langle g \rangle$, and lift a chain of A/A_1, which is smaller. Every subgroup of the abelian group A is normal in it, and the preimages of the A_j in H_{i+1} form a chain from H_i to H_{i+1}, each normal in the next, with factors A_{j+1}/A_j. ■

Definition 23.7 (Alternating group). A_n is the group of even permutations of $\{1, \dots, n\}$; $|A_n| = n!/2$ for $n \geq 2$. A group is *simple* if its only normal subgroups are 1 and itself.

Theorem 23.8 (cited)**.** A_n is simple for $n \geq 5$.

Theorem 23.9. S_n is solvable if and only if $n \leq 4$.

Proof. S_1 and S_2 are abelian. $1 \trianglelefteq A_3 \trianglelefteq S_3$ has factors of orders 3 and 2. For S_4, let $V = \{\mathrm{id}, (12)(34), (13)(24), (14)(23)\}$, which is normal in S_4, being the union of id and a conjugacy class. Then $1 \trianglelefteq V \trianglelefteq A_4 \trianglelefteq S_4$ has factors $V \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, A_4/V of order 3, and S_4/A_4 of order 2, all abelian. By Proposition 23.4 these groups are solvable.

Let $n \geq 5$. Since $(123)(345) \neq (345)(123)$, A_n is not abelian, so $A_{n'} \neq 1$. As $A_{n'}$ is normal in the simple group A_n, $A_{n'} = A_n$, and the derived series of A_n is constant. Since $A_n \leq S_n$, induction gives $A_n = A_n^{(i)} \leq S_n^{(i)}$ for every i, so the derived series of S_n never reaches 1, and S_n is not solvable. ∎

24. Solvability by radicals

Throughout, all fields considered over a field $\mathbb{K}$ lie in a fixed algebraic closure Ω of $\mathbb{K}$.

Definition 24.1 (Radical extension). An extension $\mathbb{K} \subseteq \mathbb{K}(b)$ with $b^n = a \in \mathbb{K}$ for some $n \in \mathbb{N}$ is a *radical extension*; we write $\mathbb{K}(b) = \mathbb{K}(\sqrt[n]{a})$.

Remark 24.2. The notation $\sqrt[n]{a} = b$ only says $b^n = a$, $b \in \Omega$, and is ambiguous: $x^n - a$ has several roots. When $\mathbb{K}$ contains the n-th roots of unity, $\mathbb{K}(\sqrt[n]{a})$ contains all roots of $x^n - a$, namely $\sqrt[n]{a} \cdot \zeta$ with $\zeta^n = 1$, so the field does not depend on the choice.

Definition 24.3 (Radical tower, root extension). A tower $\mathbb{K} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_\ell$ in which every $\mathbb{K}_i \mid \mathbb{K}_{i-1}$ is a radical extension is a *radical tower*. Then $\mathbb{K}_\ell \mid \mathbb{K}$ is a *root extension*, and the elements of $\mathbb{K}_\ell$ are said to be *expressible by radicals* over $\mathbb{K}$.

Definition 24.4 (Solvable by radicals). A polynomial $f \in \mathbb{K}[x]$ is *solvable by radicals* over $\mathbb{K}$ if all its roots in Ω lie in a root extension of $\mathbb{K}$.

Definition 24.5 (Cyclic extension). A *cyclic extension* is a Galois extension $\mathbb{F} \mid \mathbb{K}$ whose Galois group $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is cyclic.

Lemma 24.6 (Dedekind). Let $\mathbb{F}$ be a field and $\sigma_1, \dots, \sigma_n$ pairwise distinct automorphisms of $\mathbb{F}$. If $c_1, \dots, c_n \in \mathbb{F}$ satisfy $c_1\sigma_1(x) + \cdots + c_n\sigma_n(x) = 0$ for all $x \in \mathbb{F}$, then $c_1 = \cdots = c_n = 0$.

Proof. Suppose not, and take a nontrivial relation with the least number of nonzero coefficients; renumbering, $c_1\sigma_1 + \cdots + c_k\sigma_k = 0$ with all $c_i \neq 0$. Then $k \geq 2$, since $\sigma_1(1) = 1 \neq 0$. Choose $y \in$

$\mathbb{F}$ with $\sigma_1(y) \neq \sigma_k(y)$. For all x, $\sum_i c_i\sigma_i(yx) = \sum_i c_i\sigma_i(y)\sigma_i(x) = 0$, and also $\sigma_k(y)\sum_i c_i\sigma_i(x) = 0$. Subtracting,

$$\sum_{i=1}^{k-1} c_i(\sigma_i(y) - \sigma_k(y))\sigma_i(x) = 0 \quad \text{for all } x \in \mathbb{F},$$

a relation with fewer nonzero terms whose first coefficient $c_1(\sigma_1(y) - \sigma_k(y))$ is nonzero. This contradicts minimality. ■

Theorem 24.7 (Kummer). Let $n \in \mathbb{N}$, let $\operatorname{char} \mathbb{K} = 0$, and assume that $\mathbb{K}$ contains the n-th roots of unity.

1) For $a \in \mathbb{K}$ let $\mathbb{F} = \mathbb{K}(\sqrt[n]{a})$. Then $\mathbb{F} \mid \mathbb{K}$ is cyclic of degree d for some $d \mid n$.

2) If $\mathbb{F} \mid \mathbb{K}$ is cyclic of degree n, then $\mathbb{F} = \mathbb{K}(\sqrt[n]{a})$ for some $a \in \mathbb{K}$.

Proof. By Proposition 20.2, $\mathbb{K}$ contains a primitive n-th root of unity ξ, and the n-th roots of unity are ξ^i, $0 \leq i \leq n-1$, pairwise distinct.

1) If $a = 0$, then $\mathbb{F} = \mathbb{K}$ and the claim holds with $d = 1$. Let $a \neq 0$ and $b = \sqrt[n]{a}$, so $b \neq 0$. The roots of $x^n - a$ are the n distinct elements $b\xi^i$, $0 \leq i \leq n-1$, all in $\mathbb{F}$. So $\mathbb{F}$ is the splitting field of $x^n - a$ over $\mathbb{K}$, hence normal (Theorem 17.21). Since $\operatorname{char} \mathbb{K} = 0$, $\mathbb{F} \mid \mathbb{K}$ is separable by Lemma 17.7, so it is Galois, and $|\operatorname{Gal}(\mathbb{F} \mid \mathbb{K})| = [\mathbb{F} : \mathbb{K}]$ by Proposition 18.3. Each $\sigma \in \operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$ is determined by $\sigma(b) = b\xi^{i_\sigma}$, with $i_\sigma \in \mathbb{Z}_n$, by Theorem 17.12. The map $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K}) \to \mathbb{Z}_n$, $\sigma \mapsto i_\sigma$, is injective, and it is a homomorphism: since $\xi \in \mathbb{K}$ is fixed,

$$(\tau\sigma)(b) = \tau(b\xi^{i_\sigma}) = \tau(b)\xi^{i_\sigma} = b\xi^{i_\tau + i_\sigma}.$$

The map need not be surjective. Hence $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$ is isomorphic to a subgroup of $\mathbb{Z}_n$, that is, cyclic of some order $d \mid n$, and $[\mathbb{F} : \mathbb{K}] = d$.

2) Let $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K}) = \langle\sigma\rangle$ with $|\sigma| = n$. For $\alpha \in \mathbb{F}$ define the *Lagrange resolvent*

$$(\alpha, \xi) = \alpha + \xi\sigma(\alpha) + \xi^2\sigma^2(\alpha) + \cdots + \xi^{n-1}\sigma^{n-1}(\alpha).$$

The automorphisms $1, \sigma, \ldots, \sigma^{n-1}$ are pairwise distinct, so by Lemma 24.6 the map $1 + \xi\sigma + \cdots + \xi^{n-1}\sigma^{n-1}$ is not identically 0; choose $\alpha \in \mathbb{F}$ with $\rho = (\alpha, \xi) \neq 0$. Since $\xi \in \mathbb{K}$ and $\sigma^n = 1$,

$$\sigma(\rho) = \sigma(\alpha) + \xi\sigma^2(\alpha) + \cdots + \xi^{n-1}\sigma^n(\alpha) = \xi^{-1}(\xi\sigma(\alpha) + \cdots + \xi^{n-1}\sigma^{n-1}(\alpha) + \xi^n\alpha) = \xi^{-1}\rho.$$

Hence $\sigma(\rho^n) = \xi^{-n}\rho^n = \rho^n$, so ρ^n lies in the fixed field of $\langle\sigma\rangle = \operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$, which is $\mathbb{K}$ by Theorem 18.8. For $0 < i < n$, $\sigma^i(\rho) = \xi^{-i}\rho \neq \rho$, since ξ is primitive and $\rho \neq 0$; so $\sigma^i \notin \operatorname{Gal}(\mathbb{F} \mid \mathbb{K}(\rho))$. Thus $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K}(\rho)) = \{\mathrm{id}\}$, and by Theorem 18.11, $\mathbb{K}(\rho) = \mathcal{F}(\{\mathrm{id}\}) = \mathbb{F}$. With $a = \rho^n \in \mathbb{K}$, $\mathbb{F} = \mathbb{K}(\rho) = \mathbb{K}(\sqrt[n]{a})$. ■

Remark 24.8. Let $\operatorname{char} \mathbb{K} = 0$ and ζ_n a primitive n-th root of unity. Then $\mathbb{K} \subseteq \mathbb{K}(\zeta_n)$ is a Galois extension with abelian Galois group, by Proposition 20.4.

Definition 24.9 (Compositum, normal closure). For fields $\mathbb{F}, \mathbb{E} \subseteq \Omega$, the *compositum* $\mathbb{F}\mathbb{E}$ is the smallest subfield of Ω containing $\mathbb{F} \cup \mathbb{E}$. For a finite extension $\mathbb{F} = \mathbb{K}(\gamma_1, ..., \gamma_s)$, the *normal closure* of $\mathbb{F} \mid \mathbb{K}$ is the splitting field in Ω over $\mathbb{K}$ of $\prod_j m_{\gamma_j, \mathbb{K}}$.

Lemma 24.10. Let $\operatorname{char} \mathbb{K} = 0$.

1) If $\mathbb{E} \mid \mathbb{K}$ is a root extension and $\mathbb{F} \supseteq \mathbb{K}$ is any field, then $\mathbb{F}\mathbb{E} \mid \mathbb{F}$ is a root extension.

2) If $\mathbb{F} \mid \mathbb{K}$ and $\mathbb{E} \mid \mathbb{K}$ are root extensions, then so is $\mathbb{F}\mathbb{E} \mid \mathbb{K}$.

3) If $\mathbb{F} \mid \mathbb{K}$ is a root extension, then so is its normal closure $\mathbb{L} \mid \mathbb{K}$.

Proof. 1) Let $\mathbb{K} = \mathbb{E}_0 \subseteq \cdots \subseteq \mathbb{E}_\ell = \mathbb{E}$ with $\mathbb{E}_i = \mathbb{E}_{i-1}(b_i)$ and $b_i^{n_i} \in \mathbb{E}_{i-1}$. Then $\mathbb{F}\mathbb{E} = \mathbb{F}(b_1, ..., b_\ell)$, and $b_i^{n_i} \in \mathbb{E}_{i-1} \subseteq \mathbb{F}(b_1, ..., b_{i-1})$. So $\mathbb{F} \subseteq \mathbb{F}(b_1) \subseteq \cdots \subseteq \mathbb{F}(b_1, ..., b_\ell)$ is a radical tower. For instance, if $\mathbb{E} = \mathbb{K}(\sqrt[n]{a})$, then $\mathbb{F}\mathbb{E} = \mathbb{F}(\sqrt[n]{a})$.

2) By 1), $\mathbb{F}\mathbb{E} \mid \mathbb{F}$ is a root extension. Appending its radical tower to one of $\mathbb{F} \mid \mathbb{K}$ gives a radical tower from $\mathbb{K}$ to $\mathbb{F}\mathbb{E}$.

3) $\mathbb{F} \mid \mathbb{K}$ is finite, so $\mathbb{F} = \mathbb{K}(\gamma_1, ..., \gamma_s)$ with γ_j algebraic, and $\mathbb{L}$ is finite over $\mathbb{K}$, being generated by the finitely many roots of $\prod_j m_{\gamma_j, \mathbb{K}}$. We show that $\mathbb{L}$ is the compositum of the fields $\tau(\mathbb{F})$, where τ runs over the finitely many $\mathbb{K}$-monomorphisms $\mathbb{F} \to \Omega$. Each $\tau(\gamma_j)$ is a root of $m_{\gamma_j, \mathbb{K}}$ by Theorem 17.12, so $\tau(\mathbb{F}) \subseteq \mathbb{L}$. Conversely, for a root β of $m_{\gamma_j, \mathbb{K}}$ there is a $\mathbb{K}$-embedding $\mathbb{K}(\gamma_j) \to \Omega$ with $\gamma_j \mapsto \beta$, as in the proof of Theorem 17.13, and it extends to $\mathbb{F}$ by Lemma 17.19; so $\beta \in \tau(\mathbb{F})$ for some τ. If $\mathbb{K} = \mathbb{F}_0 \subseteq \cdots \subseteq \mathbb{F}_\ell = \mathbb{F}$ is a radical tower with $\mathbb{F}_i = \mathbb{F}_{i-1}(b_i)$, then $\tau(\mathbb{F}_i) = \tau(\mathbb{F}_{i-1})(\tau(b_i))$ and $\tau(b_i)^{n_i} = \tau(b_i^{n_i}) \in \tau(\mathbb{F}_{i-1})$, so each $\tau(\mathbb{F}) \mid \mathbb{K}$ is a root extension. By 2), applied finitely many times, $\mathbb{L} \mid \mathbb{K}$ is a root extension. ∎

Theorem 24.11 (Galois). Let $\operatorname{char} \mathbb{K} = 0$, $f \in \mathbb{K}[x]$, and let $\mathbb{F}$ be the splitting field of f over $\mathbb{K}$. Then f is solvable by radicals if and only if $\operatorname{Gal}(\mathbb{F} \mid \mathbb{K})$ is solvable.

Proof. ($\Rightarrow$) Assume that f is solvable by radicals. Each root of f lies in a root extension of $\mathbb{K}$; by Lemma 24.10 2), their compositum is a root extension containing all roots of f. So there is a radical tower $\mathbb{K} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_n$ with $\mathbb{K}_{i+1} = \mathbb{K}_i(b_i)$, $b_i^{m_i} = a_i \in \mathbb{K}_i$, and $\mathbb{F} \subseteq \mathbb{K}_n$. Let $m = \operatorname{lcm}(m_0, ..., m_{n-1})$, ξ a primitive m-th root of unity, $\mathbb{E}_0 = \mathbb{K}(\xi)$ and $\mathbb{E}_{i+1} = \mathbb{E}_i(b_i)$. Then $\mathbb{E}_0 \mid \mathbb{K}$ is abelian Galois by Proposition 20.4, and each $\mathbb{E}_{i+1} \mid \mathbb{E}_i$ is cyclic by Theorem 24.7 1), since $m_i \mid m$ and $\mathbb{E}_i$ contains the m_i-th roots of unity $\xi^{m/m_i \cdot j}$.

Let $\mathbb{L}$ be the normal closure of $\mathbb{E}_n \mid \mathbb{K}$. By the proof of Lemma 24.10 3), $\mathbb{L}$ is the compositum of the fields $\tau(\mathbb{E}_n)$ for the $\mathbb{K}$-monomorphisms $\tau : \mathbb{E}_n \to \Omega$. Since $\tau(\xi)$ is a primitive m-th root of unity, $\mathbb{K}(\tau(\xi)) = \mathbb{K}(\xi)$, both being the splitting field of $x^m - 1$. So $\tau(\mathbb{E}_n) = \mathbb{K}(\xi)(\tau(b_0), ..., \tau(b_{n-1}))$ with $\tau(b_i)^{m_i} = \tau(a_i) \in \tau(\mathbb{E}_i)$. Concatenating these towers for the finitely many τ gives

$$\mathbb{K} \subseteq \mathbb{K}(\xi) = \mathbb{L}_0 \subseteq \mathbb{L}_1 \subseteq \cdots \subseteq \mathbb{L}_N = \mathbb{L}, \quad \mathbb{L}_{j+1} = \mathbb{L}_j(c_j), \quad c_j^{e_j} \in \mathbb{L}_j, \quad e_j \mid m.$$

As above, each $\mathbb{L}_{j+1} \mid \mathbb{L}_j$ is cyclic by Theorem 24.7 1), and $\mathbb{L}_0 \mid \mathbb{K}$ is abelian Galois. $\mathbb{L} \mid \mathbb{K}$ is Galois, being normal (Theorem 17.21) in characteristic 0, and $\mathbb{F} \subseteq \mathbb{L}$, as $\mathbb{L}$ contains all roots of f.

Let $G = \mathrm{Gal}(\mathbb{L} \mid \mathbb{K})$ and $G_j = \mathrm{Gal}(\mathbb{L} \mid \mathbb{L}_j)$, so $G \geq G_0 \geq G_1 \geq \cdots \geq G_N = 1$. Since $\mathbb{L}_{j+1} \mid \mathbb{L}_j$ is normal, Theorem 18.12 applied to the Galois extension $\mathbb{L} \mid \mathbb{L}_j$ (Corollary 17.22) gives $G_{j+1} \trianglelefteq G_j$ and

$$G_j/G_{j+1} = \mathrm{Gal}(\mathbb{L} \mid \mathbb{L}_j)/\mathrm{Gal}(\mathbb{L} \mid \mathbb{L}_{j+1}) \cong \mathrm{Gal}(\mathbb{L}_{j+1} \mid \mathbb{L}_j),$$

which is abelian. Likewise $G_0 \trianglelefteq G$ with $G/G_0 \cong \mathrm{Gal}(\mathbb{L}_0 \mid \mathbb{K})$ abelian. By Proposition 23.4, G is solvable. Finally $\mathbb{F} \mid \mathbb{K}$ is normal and $\mathbb{K} \subseteq \mathbb{F} \subseteq \mathbb{L}$, so by Theorem 18.12,

$$\mathrm{Gal}(\mathbb{F} \mid \mathbb{K}) \cong \mathrm{Gal}(\mathbb{L} \mid \mathbb{K})/\mathrm{Gal}(\mathbb{L} \mid \mathbb{F}),$$

The projection $\pi : \mathrm{Gal}(\mathbb{L} \mid \mathbb{K}) \to \mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is surjective, so $\pi(H') = \pi(H)'$ for every subgroup H, and by induction $\pi\big(G^{(i)}\big) = \mathrm{Gal}(\mathbb{F} \mid \mathbb{K})^{(i)}$. As $G^{(k)} = 1$ for some k, $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})^{(k)} = 1$, and $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is solvable.

($\Leftarrow$) Assume $\mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$ is solvable. By Proposition 23.6 there is a normal series

$$1 = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = \mathrm{Gal}(\mathbb{F} \mid \mathbb{K})$$

with G_{i-1}/G_i cyclic of order m_i. Let $\mathbb{K}_i = \mathcal{F}(G_i)$. Then $\mathbb{K} = \mathbb{K}_0 \subseteq \mathbb{K}_1 \subseteq \cdots \subseteq \mathbb{K}_n = \mathbb{F}$, by Theorem 18.8 and Theorem 18.11. By Theorem 18.11, $G_{i-1} = \mathrm{Gal}(\mathbb{F} \mid \mathbb{K}_{i-1})$, and since $G_i \trianglelefteq G_{i-1}$, Theorem 18.12 applied to $\mathbb{F} \mid \mathbb{K}_{i-1}$ shows that $\mathbb{K}_i \mid \mathbb{K}_{i-1}$ is Galois with $\mathrm{Gal}(\mathbb{K}_i \mid \mathbb{K}_{i-1}) \cong G_{i-1}/G_i$, cyclic of order m_i.

Let $m = \mathrm{lcm}(m_1, \ldots, m_n)$, $\mathbb{E}_0 = \mathbb{K}(\xi_m)$ and $\mathbb{E}_i = \mathbb{E}_0\mathbb{K}_i$, so $\mathbb{E}_i = \mathbb{E}_{i-1}\mathbb{K}_i$. By Theorem 17.21, $\mathbb{K}_i$ is the splitting field over $\mathbb{K}_{i-1}$ of some $g \in \mathbb{K}_{i-1}[x]$, so $\mathbb{E}_i = \mathbb{E}_{i-1}(\text{roots of } g)$ is the splitting field of g over $\mathbb{E}_{i-1}$, and $\mathbb{E}_i \mid \mathbb{E}_{i-1}$ is Galois. For $\sigma \in \mathrm{Gal}(\mathbb{E}_i \mid \mathbb{E}_{i-1})$, σ permutes the roots of g (Theorem 17.12), so $\sigma(\mathbb{K}_i) = \mathbb{K}_i$, and $\sigma|_{\mathbb{K}_i}$ fixes $\mathbb{K}_{i-1} \subseteq \mathbb{E}_{i-1}$. The restriction $\mathrm{Gal}(\mathbb{E}_i \mid \mathbb{E}_{i-1}) \to \mathrm{Gal}(\mathbb{K}_i \mid \mathbb{K}_{i-1})$ is an injective homomorphism, since $\mathbb{E}_i$ is generated over $\mathbb{E}_{i-1}$ by the roots of g. Hence $\mathrm{Gal}(\mathbb{E}_i \mid \mathbb{E}_{i-1})$ is isomorphic to a subgroup of a cyclic group of order m_i, so it is cyclic of some order $d_i \mid m_i$. The field $\mathbb{E}_{i-1} \supseteq \mathbb{E}_0$ contains the primitive d_i-th root of unity ξ_m^{m/d_i}, so by Theorem 24.7 2), $\mathbb{E}_i = \mathbb{E}_{i-1}\big(\sqrt[d_i]{c_i}\big)$ with $c_i \in \mathbb{E}_{i-1}$. Together with $\mathbb{E}_0 = \mathbb{K}\big(\sqrt[m]{1}\big)$, this shows that $\mathbb{E}_n \mid \mathbb{K}$ is a root extension. It contains $\mathbb{K}_n = \mathbb{F}$, hence all roots of f, so f is solvable by radicals. ■

Remark 24.12 (Recall). *Cauchy's theorem.* If G is a finite group and p is a prime dividing $|G|$, then G has an element of order p.

Lemma 24.13. Let p be a prime and $G \leq S_p$. If G contains a p-cycle and a transposition, then $G = S_p$.

Proof. Let $(a\ b) \in G$ be a transposition and $c \in G$ a p-cycle. Some power c^k with $1 \leq k < p$ maps a to b; since p is prime, c^k is again a p-cycle. Relabelling the points, $a = 1$, $b = 2$ and $c^k = (1\ 2\ \ldots\ p)$. Conjugating $(1\ 2)$ by $(c^k)^j$ gives $(j+1\ j+2)$, so G contains all transpositions $(i\ i+1)$, $1 \leq i \leq p-1$, which generate S_p. ■

Example 24.14. Show that $f(x) = x^5 - 20x + 6 \in \mathbb{Q}[x]$ is not solvable by radicals.

Solution. By Remark 22.1 with $p = 2$, f is irreducible over $\mathbb{Q}$. Let $\mathbb{E} \subseteq \mathbb{C}$ be its splitting field. Since $\mathrm{char}\ \mathbb{Q} = 0$, f is separable (Lemma 17.7) and has five distinct roots $\alpha_1, \ldots, \alpha_5$. $G = \mathrm{Gal}(\mathbb{E} \mid \mathbb{Q})$

permutes the roots and is determined by this action, so $G \leq S_5$. As $[\mathbb{Q}(\alpha_i) : \mathbb{Q}] = 5$, the tower law gives $5 \mid [\mathbb{E} : \mathbb{Q}] = |G|$. By Cauchy's theorem G has an element of order 5; an element of S_5 of order 5 has cycle lengths with least common multiple 5, so it is a 5-cycle.

Now $f'(x) = 5x^4 - 20$ and $f''(x) = 20x^3$. As f'' has a single real zero, f' has at most 2 real zeros by Rolle's theorem, and so f has at most 3 real zeros. From the values

x	-3	0	1	3
$f(x)$	-177	6	-13	189

and the intermediate value theorem, f has at least 3 real zeros. So f has exactly three real roots $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ and two nonreal roots $\alpha_4, \alpha_5 \in \mathbb{C}$ with $\overline{\alpha_4} = \alpha_5$. Complex conjugation maps $\mathbb{E}$ onto $\mathbb{E}$ by Proposition 17.20, so its restriction lies in G; it fixes $\alpha_1, \alpha_2, \alpha_3$ and swaps α_4, α_5, so $(4\ 5) \in G$. By Lemma 24.13, $G = S_5$, which is not solvable by Theorem 23.9. By Theorem 24.11, f is not solvable by radicals. ◊

Example 24.15. Is $f(x) = x^5 - 2x^3 - 8x + 2 \in \mathbb{Q}[x]$ solvable by radicals?

Solution. No. By Remark 22.1 with $p = 2$, f is irreducible over $\mathbb{Q}$. We have

$$f'(x) = 5x^4 - 6x^2 - 8 = (x^2 - 2)(5x^2 + 4),$$

whose real zeros are $\pm\sqrt{2}$. By Rolle's theorem f has at most 3 real zeros. From

x	-3	0	1	3
$f(x)$	-163	2	-7	167

and the intermediate value theorem, f has zeros in $(-3, 0)$, $(0, 1)$ and $(1, 3)$. So its roots are $a_1, a_2, a_3 \in \mathbb{R}$ and $a_4, a_5 \in \mathbb{C} \setminus \mathbb{R}$ with $\overline{a_4} = a_5$.

Let $\mathbb{K}$ be the splitting field of f over $\mathbb{Q}$ and $G = \mathrm{Gal}(\mathbb{K} \mid \mathbb{Q}) \leq S_5$. Complex conjugation restricts to an element $\sigma \in G$ (Proposition 17.20) with $\sigma(a_4) = a_5$, fixing a_1, a_2, a_3, so G contains a transposition. Since f is irreducible, $5 = [\mathbb{Q}(a_1) : \mathbb{Q}]$ divides $[\mathbb{K} : \mathbb{Q}] = |G|$, and by Cauchy's theorem G contains an element of order 5, that is, a 5-cycle. By Lemma 24.13, $G = S_5$; for instance $S_5 = \langle (1\ 2\ 3\ 4\ 5), (4\ 5) \rangle$. S_5 is not solvable (Theorem 23.9), since A_5 is simple, so f is not solvable by radicals by Theorem 24.11. ◊

25. Review exercises

Example 25.1. Solve $x^3 - 2x^2 + 3x + 2 \equiv 0 \pmod{25}$.

Solution. Let $f(x) = x^3 - 2x^2 + 3x + 2$. Modulo 5, the values $f(0), \ldots, f(4)$ are $2, 4, 3, 0, 1$, so $x \equiv 3 \pmod 5$ is the only solution. Since $f'(x) = 3x^2 - 4x + 3$ and $f'(3) = 27 - 12 + 3 = 18 \not\equiv 0 \pmod 5$, Theorem 6.2 gives exactly one solution modulo 25.

Write $x = 5t + 3$. By Lemma 6.3,

$$f(5t + 3) \equiv f(3) + 5tf'(3) = 20 + 90t \pmod{25},$$

so $90t + 20 \equiv 0 \pmod{25}$, that is, $18t + 4 \equiv 0 \pmod{5}$, or $3t + 4 \equiv 0 \pmod{5}$. Multiplying by 2 gives $6t + 8 \equiv t + 3 \equiv 0 \pmod{5}$, so $t \equiv 2 \pmod{5}$ and

$$x \equiv 13 \pmod{25}.$$

Indeed $f(13) = 1900 = 76 \cdot 25$. ◊

Example 25.2. Let $p > 3$ be a prime. Compute $\left(\frac{3}{p}\right)$.

Solution. Since $p \neq 2, 3$, the symbol is defined and Theorem 4.6 applies with $3 \equiv -1 \pmod{4}$. We use Proposition 4.3 1) and $\left(\frac{1}{3}\right) = 1$, $\left(\frac{2}{3}\right) = -1$.

Case I: $p \equiv -1 \pmod{4}$. By Theorem 4.6,

$$\left(\frac{3}{p}\right) = -\left(\frac{p}{3}\right) = \begin{cases} -\left(\frac{1}{3}\right) = -1 & \text{if } p \equiv 1 \pmod{3} \\ -\left(\frac{2}{3}\right) = -(-1) = 1 & \text{if } p \equiv -1 \pmod{3}. \end{cases}$$

Case II: $p \equiv 1 \pmod{4}$. By Theorem 4.6,

$$\left(\frac{3}{p}\right) = \left(\frac{p}{3}\right) = \begin{cases} \left(\frac{1}{3}\right) = 1 & \text{if } p \equiv 1 \pmod{3} \\ \left(\frac{2}{3}\right) = -1 & \text{if } p \equiv -1 \pmod{3}. \end{cases}$$

Combining the conditions modulo 4 and 3 by the Chinese remainder theorem:

- $p \equiv -1 \pmod{4}$, $p \equiv 1 \pmod{3}$: $p = 4t - 1$ with $4t - 1 \equiv 1 \pmod{3}$, so $t \equiv 2 \pmod{3}$ and $p \equiv 7 \pmod{12}$; $\left(\frac{3}{p}\right) = -1$.
- $p \equiv -1 \pmod{4}$, $p \equiv -1 \pmod{3}$: $4t - 1 \equiv -1 \pmod{3}$, so $t \equiv 0 \pmod{3}$ and $p \equiv -1 \pmod{12}$; $\left(\frac{3}{p}\right) = 1$.
- $p \equiv 1 \pmod{4}$, $p \equiv 1 \pmod{3}$: $p = 4t + 1$ with $4t + 1 \equiv 1 \pmod{3}$, so $t \equiv 0 \pmod{3}$ and $p \equiv 1 \pmod{12}$; $\left(\frac{3}{p}\right) = 1$.
- $p \equiv 1 \pmod{4}$, $p \equiv -1 \pmod{3}$: $4t + 1 \equiv -1 \pmod{3}$, that is, $4t \equiv 1 \pmod{3}$, so $t \equiv 1 \pmod{3}$ and $p \equiv 5 \pmod{12}$; $\left(\frac{3}{p}\right) = -1$.

Hence

$$\left(\frac{3}{p}\right) = \begin{cases} 1 & \text{if } p \equiv \pm 1 \pmod{12} \\ -1 & \text{if } p \equiv \pm 5 \pmod{12}. \end{cases}$$

◊

Example 25.3. Prove that the field extension $\mathbb{Q} \subseteq \mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$ is Galois and determine its Galois group.

Solution. We need that $\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right) \mid \mathbb{Q}$ is normal and separable. It is separable since $\operatorname{char} \mathbb{Q} = 0$ (Lemma 17.7). The minimal polynomials are $m_{\sqrt{3}}(x) = x^2 - 3$ and $m_{\sqrt{5}}(x) = x^2 - 5$, and

$$(x^2 - 3)(x^2 - 5) = \left(x - \sqrt{3}\right)\left(x + \sqrt{3}\right)\left(x - \sqrt{5}\right)\left(x + \sqrt{5}\right),$$

so $\mathbb{Q}\left(\sqrt{3}, \sqrt{5}\right)$ is the splitting field of $(x^2 - 3)(x^2 - 5)$, hence normal (Theorem 17.21), and the extension is Galois.

$\sqrt{5} \notin \mathbb{Q}(\sqrt{3})$: if $\sqrt{5} = u + v\sqrt{3}$ with $u, v \in \mathbb{Q}$, then $5 = u^2 + 3v^2 + 2uv\sqrt{3}$ forces $uv = 0$, and either case makes $\sqrt{5}$ or $\sqrt{5/3}$ rational. So

$$\left[\mathbb{Q}(\sqrt{3}, \sqrt{5}) : \mathbb{Q}\right] = \left[\mathbb{Q}(\sqrt{3}, \sqrt{5}) : \mathbb{Q}(\sqrt{3})\right]\left[\mathbb{Q}(\sqrt{3}) : \mathbb{Q}\right] = 2 \cdot 2 = 4,$$

and $G = \mathrm{Gal}\left(\mathbb{Q}(\sqrt{3}, \sqrt{5}) \mid \mathbb{Q}\right)$ has order 4 (Proposition 18.3). For $\sigma \in G$, $\sigma(3) = 3$ gives $\sigma(\sqrt{3})^2 = \sigma(3) = 3$, so $\sigma(\sqrt{3}) = \pm\sqrt{3}$, and similarly $\sigma(\sqrt{5}) = \pm\sqrt{5}$. These are four possibilities, and $|G| = 4$, so all occur:

$$\mathrm{id} : \sqrt{3} \mapsto \sqrt{3},\ \sqrt{5} \mapsto \sqrt{5}, \quad \alpha : \sqrt{3} \mapsto -\sqrt{3},\ \sqrt{5} \mapsto \sqrt{5},$$
$$\beta : \sqrt{3} \mapsto \sqrt{3},\ \sqrt{5} \mapsto -\sqrt{5}, \quad \gamma : \sqrt{3} \mapsto -\sqrt{3},\ \sqrt{5} \mapsto -\sqrt{5}.$$

Then $(\alpha\beta)(\sqrt{3}) = -\sqrt{3} = \gamma(\sqrt{3})$ and $(\alpha\beta)(\sqrt{5}) = -\sqrt{5} = \gamma(\sqrt{5})$, so $\alpha\beta = \gamma$. Likewise $(\alpha\gamma)(\sqrt{3}) = \sqrt{3}$, $(\alpha\gamma)(\sqrt{5}) = -\sqrt{5}$, so $\alpha\gamma = \beta$, and similarly $\gamma\beta = \alpha$. Every element has order at most 2, so $G \cong \mathbb{Z}_2 \times \mathbb{Z}_2$, the Klein four-group. ◊

Example 25.4. Find the smallest positive primitive root modulo 13. Make an index table and solve $3x^7 \equiv 8 \pmod{13}$.

Solution. $\varphi(13) = 12$, so $O_{13}(2) \mid 12$ by Proposition 1.3, that is, $O_{13}(2) \in \{1, 2, 3, 4, 6, 12\}$. By Procedure 1.12 it suffices to check $2^{12/2} = 2^6$ and $2^{12/3} = 2^4$: $2^4 = 16 \equiv 3$ and $2^6 = 64 \equiv -1 \pmod{13}$. Hence $O_{13}(2) = 12$, and 2 is a primitive root modulo 13. Since $O_{13}(1) = 1$, 2 is the smallest positive primitive root. The powers of 2 modulo 13 are:

α	0	1	2	3	4	5	6	7	8	9	10	11	12
2^α	1	2	4	8	3	6	12	11	9	5	10	7	1

From the table, $\mathrm{ind}_2\, 3 = 4$ and $\mathrm{ind}_2\, 8 = 3$. Taking indices in $3x^7 \equiv 8 \pmod{13}$, Theorem 2.3 gives

$$\mathrm{ind}_2\, 3 + 7\,\mathrm{ind}_2\, x \equiv \mathrm{ind}_2\, 8 \pmod{12}, \quad \text{that is,} \quad 4 + 7\,\mathrm{ind}_2\, x \equiv 3 \pmod{12},$$

so $7\,\mathrm{ind}_2\, x \equiv -1 \pmod{12}$. Put $y = \mathrm{ind}_2\, x$ and multiply by 5: $35y \equiv -5 \pmod{12}$. Since $35 \equiv -1 \pmod{12}$, $-y \equiv -5$ and $y \equiv 5 \pmod{12}$. Hence $x \equiv 2^5 = 32 \equiv 6 \pmod{13}$. Indeed $6^7 \equiv 7 \pmod{13}$ and $3 \cdot 7 = 21 \equiv 8 \pmod{13}$. ◊

Example 25.5. For $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_r^{\alpha_r}$, define the arithmetic function

$$f(n) = \begin{cases} (-1)^{p_1 + p_2 + \cdots + p_r} & \text{if } n > 1 \\ 1 & \text{if } n = 1. \end{cases}$$

Prove that f is multiplicative and determine its summation function.

Solution. Here the p_i are distinct primes and $\alpha_i \geq 1$. Let $F(n) = \sum_{d \mid n} f(d)$ be the summation function.

f is multiplicative. Let $m = q_1^{\beta_1} \cdots q_s^{\beta_s}$ with distinct primes q_j, and $\gcd(n, m) = 1$, so $p_i \neq q_j$ for all i, j. Then the distinct prime divisors of mn are the p_i and the q_j, and

$$f(mn) = f\left(p_1^{\alpha_1} \cdots p_r^{\alpha_r} q_1^{\beta_1} \cdots q_s^{\beta_s}\right) = (-1)^{p_1 + \cdots + p_r + q_1 + \cdots + q_s} = (-1)^{\sum_i p_i} (-1)^{\sum_j q_j} = f(n) f(m).$$

If $n = 1$ or $m = 1$, this holds since $f(1) = 1$.

F. By Theorem 7.8, F is multiplicative, so it suffices to compute it on prime powers. For $\alpha \geq 1$,

$$F(p^\alpha) = \sum_{d|p^\alpha} f(d) = f(1) + f(p) + \cdots + f(p^\alpha) = 1 + \alpha(-1)^p = \begin{cases} \alpha + 1 \text{ if } p = 2 \\ 1 - \alpha \text{ if } p > 2. \end{cases}$$

Hence, for $n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$,

$$F(n) = \prod_{i=1}^{r} F\left(p_i^{\alpha_i}\right) = \begin{cases} \prod_{i=1}^{r} (1 - \alpha_i) & \text{if } 2 \nmid n \\ (1 + \alpha_1) \prod_{i=2}^{r} (1 - \alpha_i) & \text{if } 2 \mid n \text{ and } p_1 = 2. \end{cases}$$

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